

APPLIED CHEMISTRY, MATERIALS AND SUSTAINABLE ENGINEERING HANDBOOK

Chemistry for Engineering Practices — complete theory, laboratory and examination guide in the **Molecular Engineering Laboratory** theme.

Course code	2003A
Course type	Practicum
PDF semester field	Semester I
Catalogue placement	Semester II
Teaching scheme	3:0:3:0
Contact hours	90
Credits	4.5
Summary CIE	40
Summary ESE	60

Course-outcome summary: Apply atomic, solution, analytical, electrochemical, corrosion, material and green-chemistry concepts to engineering practice.

OFFICIAL SOURCE DECLARATION

Source identity and verified inconsistencies

This handbook is an original educational expansion of the supplied official SITTTR Kerala Diploma Curriculum Revision 2026 syllabus for **Course 2003A — Chemistry for Engineering Practices**. The suffix **A** is part of the course code and is preserved everywhere.

Semester-source inconsistency: the supplied course-description PDF prints “**Semester I**”, while the current Revision 2026 programme catalogue places code **2003A** in **Semester II**. Website navigation follows the verified catalogue placement; the PDF wording is reproduced here without alteration.

Hour-allocation note: the detailed module table totals 45 theory hours and 39 practical hours (84), while the course description states 90 contact hours and the weekly scheme is 3:0:3:0. These official values are retained in their own contexts.

Assessment-table inconsistency: the course-description summary shows CIE 40 and ESE 60. The teaching-and-assessment table separately shows theory CIE 20 + ESE 60 and practical CIE 20 + ESE 0, while the detailed practicum methodology presents its own converted internal components and a 60-mark end-semester component. Some assessment values differ between the summary and detailed tables of the official syllabus. The values are reproduced in their respective contexts without alteration.

Official PDF course-description wording: “Chemistry for engineering practices is a foundational course designed to understand the basic concepts of Chemistry and their practical applications in engineering and daily life. This course enables students to develop problem-solving skills and gain knowledge necessary for selecting various engineering materials and addressing real-world industrial challenges.”

COURSE DASHBOARD

Course map, outcomes and assessment

45 h

Theory allocation

39 h

Listed practical allocation

90 h

Contact hours stated

90 h

Self-learning total

Four official objectives

1. Build knowledge of atoms, molecules, solutions and thermodynamics and relate it to engineering.
2. Apply analytical chemistry and water-treatment methods to domestic and industrial problems.
3. Build electrochemistry and corrosion knowledge for engineering systems

4. Select engineering materials and understand green-chemistry principles.

Course 2003A

Prerequisite

Basic knowledge of Chemistry from secondary education.

How to use this handbook

Study each theory chapter, inspect its diagram, attempt solved and practice problems, then complete at least two laboratory experiments from each module. Use the Formula Bank and Practical Record Centre before assessments.

Course outcomes

CO	Outcome	Bloom level
CO1	Apply basic concepts of atoms, molecules, solutions and thermodynamic processes in scientific and engineering contexts.	Apply
CO2	Apply analytical chemistry, water-treatment methods and acid-base concepts to solve engineering problems.	Apply
CO3	Understand and apply electrochemistry and corrosion principles in engineering systems and industrial challenges.	Apply
CO4	Explore engineering materials for industrial and domestic applications and explain green-chemistry principles promoting sustainable practices.	Apply

CO-PO mapping reproduced from the official course PDF

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8-PO11
CO1	3	2	-	-	-	-	-	-
CO2	3	3	2	3	2	2	2	-
CO3	2	2	2	2	2	-	-	-
CO4	3	2	3	2	2	2	3	-

Major equipment for 30 students

- Conductivity meter, range up to 1000 mS/cm — 1
- Digital/benchtop/portable/pen pH meter — 1
- Bunsen burners — 2
- Digital weighing balance — 1
- Thermometer — 1

Experiment rule

At least two experiments from every module, ten experiments in total, collectively attaining all four course outcomes.

Assessment: continuous evaluation includes preparation, setup, observation, analysis, viva, workmanship, discipline and teamwork. The end-semester theory duration is 2.5 hours for 60 marks.

INTERACTIVE ROADMAP

From atomic structure to sustainable engineering

LABORATORY SAFETY CENTRE

Work safely before working quickly

NEVER PIPETTE BY MOUTH. Use a pipette filler. Formaldehyde, phenol, concentrated acids, concentrated bases and heated vapours require faculty supervision, ventilation and correct PPE. None of the listed experiments is a home activity.

Corrosive

Acids and strong bases can burn skin and eyes. Add acid to water, not water to acid.

Toxic

Avoid inhalation, ingestion and skin contact. Use a fume hood where directed.

Irritant

Wear goggles and gloves; wash exposed skin immediately.

Flammable

Keep away from flames, sparks and hot surfaces.

Oxidising

Keep away from reducing agents and combustible material.

Environmental hazard

Do not discharge metal-ion or resin waste into the sink.

Personal protection

- Lab coat, splash goggles, suitable gloves and closed footwear.
- Tie back long hair; secure loose clothing and jewellery.
- Read chemical labels and Safety Data Sheets before use.

Glassware and instruments

- Inspect burettes, pipettes and flasks for cracks.
- Use a funnel only for filling, then remove it before titration readings.
- Rinse pH and conductivity probes with distilled water; blot gently.
- Never heat a closed system.

Emergency response

- Alert faculty, isolate the area and use the correct spill kit.
- Eyes: rinse at an eyewash for at least 15 minutes and obtain medical help.
- Skin: remove contaminated clothing and rinse with water.

- Broken glass: use brush and pan, never bare hands.

Course 2003A

Waste segregation guide

Waste	Container/action
Dilute neutral aqueous salt solution	Follow laboratory disposal instruction; neutralise only when authorised.
Acid/base waste	Labelled corrosive-waste container.
Nickel, copper, zinc, EDTA or heavy-metal waste	Dedicated hazardous metal-ion container; no drain disposal.
Phenol/formaldehyde/resin waste	Dedicated organic hazardous-waste container in ventilated area.
Broken glass	Rigid labelled glass-sharps container.

MODULE I • CO1

Fundamentals of Chemistry, Solutions and Thermodynamics

Theory 11 h

Practical 9 h

CO1

1.1 Atoms and molecules CO1

An atom is the smallest electrically neutral unit that retains the identity of an element. A molecule contains two or more atoms chemically bonded and behaves as one independent unit. Atomic structure controls bonding, conductivity, corrosion tendency and material behaviour.

Nucleus: compact centre containing protons and neutrons

Electron: negatively charged particle occupying shells around the nucleus

Proton: +1 relative charge and approximately 1 u relative mass

Neutron: zero charge and approximately 1 u relative mass

Atomic number Z equals the number of protons

Mass number A equals protons plus neutrons

For a neutral atom, electrons equal protons

Neutrons = $A - Z$

Malayalam support: ആറ്റത്തിന്റെ കേന്ദ്രമാണ് nucleus. Atomic number എന്നത് proton-കളുടെ എണ്ണമാണ്; neutral atom-ൽ electron-കളുടെ എണ്ണവും അതേ ആയിരിക്കും.

Engineering connection

Atoms and molecules is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

1.2 Solutions and standard solutions **CO1**

A solution is a homogeneous mixture. The component present in larger amount is normally the solvent and the dissolved component is the solute. A standard solution has an accurately known concentration and is prepared using clean glassware, an analytical balance and a volumetric flask.

Aqueous solution: water is the solvent

Concentrated and dilute are comparative, not exact, terms

Primary-standard material should be pure, stable and accurately weighable

Transfer solute quantitatively; rinse the weighing vessel and funnel

Dissolve before making up to the calibration mark

Mix by repeated inversion

Malayalam support: Solute-ന്റെ അളവും final solution volume-വും കൃത്യമായി അറിഞ്ഞാൽ മാത്രമേ standard solution concentration വിശ്വസിക്കാനാകൂ.

Engineering connection

Solutions and standard solutions is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

1.3 Molarity and dilution **CO1**

Molarity is the number of moles of solute in one litre of solution. Convert millilitres to litres before substitution. For solution preparation, required mass equals molarity $\times$ molar mass $\times$ volume in litres. During dilution the amount of solute remains constant, so $C_1V_1 = C_2V_2$.

Mole links particle count to measurable mass

Molar mass is expressed in g mol^{-1}

Molarity changes with temperature because solution volume changes

Use a volumetric pipette for accurate dilution

Never make the final volume while the solution is hot

Malayalam support: Molarity എന്നത് ഒരു litre solution-ലുള്ള solute moles ആണ്. mL നേരിട്ട് formula-യിൽ ഇടത്ത്; litre ആക്കണം.

Engineering connection

Molarity and dilution is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

1.4 Normality and equivalent mass CO1

Normality is the number of gram equivalents per litre of solution. Equivalent mass depends on the reaction: molar mass divided by acidity for an acid, basicity for a base, or the relevant electron/charge factor for a redox or salt reaction.

Normality unit: eq L^{-1} or N

Oxalic acid dihydrate has equivalent mass 63 g eq^{-1} in acid-base reaction

Anhydrous sodium carbonate has equivalent mass 53 g eq^{-1}

Reaction context must be stated

For required mass: $N \times \text{equivalent mass} \times \text{volume in litres}$

Malayalam support: Normality reaction-നെ ആശ്രയിക്കുന്ന concentration ആണ്. Equivalent mass തെറ്റിയാൽ മഴുവൻ calculation തെറ്റും.

Engineering connection

Normality and equivalent mass is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

1.5 Parts per million co1

Parts per million expresses very dilute composition. One ppm means one part solute in one million parts solution by mass. For dilute aqueous solutions with density near 1 kg L^{-1} , 1 mg L^{-1} is approximately 1 ppm.

Exact mass form: $\text{mass solute/mass solution} \times 10^6$

Useful for pollutants, dissolved ions and water quality

Always state whether the basis is mg/L or mass/mass

The dilute-water approximation is not universal

Malayalam support: ppm വളരെ ചെറിയ concentration കാണിക്കാൻ ഉപയോഗിക്കുന്നു. Dilute water-ൽ 1 mg/L ഏകദേശം 1 ppm ആണ്.

Engineering connection

Parts per million is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

1.6 Thermodynamic system and surroundings co1

A thermodynamic system is the chosen quantity of matter or region in space under study. Everything else is the surroundings; the real or imaginary surface separating them is the boundary. Temperature, pressure and volume describe the state.

Open system exchanges matter and energy

Closed system exchanges energy but not matter

Isolated system ideally exchanges neither

A pressure cooker with closed valve approximates a closed system

A running turbine is an open system

A well-insulated sealed flask approximates an isolated system

Malayalam support: പഠനത്തിനായി തിരഞ്ഞെടുത്ത ഭാഗം system; അതിന് പുറത്തുള്ളത് surroundings; ഇടയിലെ പരിധിയാണ് boundary.

Engineering connection

Thermodynamic system and surroundings is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

1.7 Thermodynamic processes CO1

A process is a change from one thermodynamic state to another. The defining condition distinguishes isothermal, isobaric, isochoric and adiabatic paths. Reversible is an ideal quasi-static path; real rapid processes with friction or finite gradients are irreversible.

Isothermal: temperature constant

Isobaric: pressure constant

Isochoric: volume constant

Adiabatic: no heat transfer

Spontaneous: proceeds without continuous external driving under stated conditions

Endothermic: system absorbs heat

Exothermic: system releases heat

Malayalam support: Endothermic process ചൂട് സ്വീകരിക്കുന്നു; exothermic process ചൂട് പുറത്തേക്ക് വിടുന്നു. Temperature change മാത്രം നോക്കാതെ system boundary വ്യക്തമാക്കണം.

Engineering connection

Thermodynamic processes is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

1.8 Internal energy, enthalpy and entropy CO1

Internal energy is the microscopic energy stored in a system. Enthalpy $H = U + PV$ is convenient for constant-pressure heat effects. Entropy describes energy dispersal and the number of accessible microscopic arrangements; at diploma level it is used qualitatively to discuss direction and disorder.

Internal energy is a state function

Enthalpy change is useful for reactions at constant pressure

Positive ΔH indicates endothermic behaviour

Negative ΔH indicates exothermic behaviour

Entropy is not simply visual disorder; it measures energy distribution

Malayalam support: Enthalpy heat effect മനസ്സിലാക്കാൻ സഹായിക്കുന്നു. Entropy energy എത്രവിധം വിതരണം ചെയ്തും എന്ന ആശയം നൽകുന്നു.

Engineering connection

Internal energy, enthalpy and entropy is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.1 Volumetric analysis and titration CO2

Volumetric analysis determines an unknown amount by measuring the volume of a standard solution required for complete reaction. The titrant is delivered from a burette; the analyte is accurately transferred by pipette into a conical flask.

End point: observed indicator colour change

Equivalence point: stoichiometric completion

Titre: final burette reading minus initial reading

Concordant readings agree within the laboratory tolerance

Remove the filling funnel before taking readings

Read the lower meniscus at eye level for colourless solutions

Malayalam support: Known concentration ഉള്ള titrant ഉപയോഗിച്ച് unknown analyte concentration കണ്ടെത്തുന്ന രീതിയാണ് titration.

Engineering connection

Volumetric analysis and titration is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.2 Choice of acid-base indicator CO2

An indicator must change colour within the steep pH change near the equivalence point. Strong acid-strong base titrations can use methyl orange or phenolphthalein. Strong acid-weak base titrations favour methyl orange; weak acid-strong base titrations favour phenolphthalein.

Methyl orange: red in acid, yellow in alkali; transition about pH 3.1–4.4

Phenolphthalein: colourless in acid, pale pink in alkali; transition about pH 8.2–10.0

Use only a few drops

Indicator error increases if too much indicator is added

Malayalam support: Endpoint-നടുത്ത് colour മാറ്റം indicator തിരഞ്ഞെടുക്കണം; ഇഷ്യൂള്ള indicator ഏതും ഉപയോഗിക്കാനാവില്ല.

Engineering connection

Choice of acid-base indicator is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.3 Principle $V_1N_1 = V_2N_2$ CO2

At equivalence in a reaction expressed through normality, gram equivalents of one reactant equal gram equivalents of the other. Therefore $V_1N_1 = V_2N_2$ when volumes use the same unit.

Define the identity of solution 1 and solution 2

Keep both volumes in the same unit

Rearrange before inserting data

Use the average concordant titre

Report concentration with suitable significant figures

Malayalam support: Equivalent-കൾ തുല്യമായതിനാൽ $V_1N_1 = V_2N_2$. V units ഒരേ ആയിരിക്കണം.

Engineering connection

Principle $V_1N_1 = V_2N_2$ is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.4 Water hardness CO2

Hardness is caused mainly by dissolved calcium and magnesium salts. Bicarbonates produce temporary hardness, which can be reduced by boiling. Sulphates and chlorides produce permanent hardness and require chemical softening or ion-exchange/membrane treatment.

Hard water consumes soap and forms scum

Heating produces scale in boilers and heat exchangers

Scale lowers heat-transfer efficiency and can cause overheating

Temporary hardness: $\text{Ca}(\text{HCO}_3)_2$ and $\text{Mg}(\text{HCO}_3)_2$

Permanent hardness: CaSO_4 , MgSO_4 , CaCl_2 and MgCl_2

Malayalam support: Ca^{2+} , Mg^{2+} salts കാരണം soap lather കുറയുകയും scale ഉണ്ടാകുകയും ചെയ്യുന്ന വെള്ളമാണ് hard water.

Engineering connection

Water hardness is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.5 Industrial effluent treatment CO2

Industrial effluent treatment combines physical removal, chemical conversion, biological degradation and advanced polishing according to the pollutant load. It is not identical to municipal drinking-water treatment because the inlet contaminants and discharge objectives differ.

Physical: screening, equalisation, sedimentation and filtration

Chemical: neutralisation, precipitation, coagulation and oxidation

Biological: microorganisms remove biodegradable organics

Advanced: membranes, adsorption, advanced oxidation or ion exchange

Sludge requires separate safe handling

Malayalam support: Industrial effluent-ന്റെ pollutant type അനുസരിച്ച് physical, chemical, biological, advanced stages തിരഞ്ഞെടുക്കുന്നു.

Engineering connection

Industrial effluent treatment is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.6 Potable water and municipal treatment CO2

Potable water is safe and acceptable for drinking. It should have acceptable colour, odour, taste, turbidity and pH, and meet chemical and microbiological safety limits. Municipal treatment commonly uses screening, sedimentation, coagulation, filtration and disinfection.

Screening removes large debris

Coagulation destabilises fine colloids

Sedimentation settles flocs

Filtration removes remaining suspended matter

Chlorine leaves a residual; UV does not

Ozone is a strong oxidant but leaves little residual

Malayalam support: കുടിക്കാൻ സുരക്ഷിതവും രുചി, മണം, നിറം എന്നിവ അംഗീകരിക്കാവുന്നതുമായ വെള്ളമാണ് potable water.

Engineering connection

Potable water and municipal treatment is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.7 Reverse osmosis and nanofiltration CO2

Osmosis moves solvent through a semipermeable membrane toward higher solute concentration. Reverse osmosis applies pressure greater than osmotic pressure to force water in the opposite direction, leaving most

dissolved ions in the reject stream. Nanofiltration operates at lower pressure and preferentially removes multivalent ions and larger solutes.

Feed, permeate and reject streams must be identified

Pretreatment reduces fouling and scaling

RO removes more monovalent salts than nanofiltration

Membranes require pressure, cleaning and periodic replacement

Reject-water management affects sustainability

Malayalam support: RO-യിൽ pressure ഉപയോഗിച്ച് water membrane കടക്കും; dissolved ions കൂടുതലും reject side-ൽ തടയപ്പെടും.

Engineering connection

Reverse osmosis and nanofiltration is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.8 Arrhenius acids, bases and neutralisation CO2

An Arrhenius acid increases H^+ (more accurately H_3O^+) concentration in water; an Arrhenius base increases OH^- concentration. Strong species ionise nearly completely, while weak species ionise partially. Neutralisation combines acid and base to form salt and water.

HCl and HNO_3 are strong acids

CH_3COOH is a weak acid

NaOH and KOH are strong bases

NH_4OH is treated as a weak base in elementary descriptions

Acid-base control is important in water treatment, plating and process chemistry

Malayalam support: Water-ൽ H^+ വർദ്ധിപ്പിക്കുന്നത് Arrhenius acid; OH^- വർദ്ധിപ്പിക്കുന്നത് Arrhenius base.

Engineering connection

Arrhenius acids, bases and neutralisation is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

2.9 pH, pOH and applications CO2

pH is the negative base-10 logarithm of hydrogen-ion concentration; pOH is the corresponding value for hydroxide-ion concentration. At 25 °C for dilute aqueous solutions, $\text{pH} + \text{pOH} = 14$.

pH below 7 is acidic, 7 neutral and above 7 basic at 25 °C

A one-unit pH change represents a tenfold concentration change

pH meters require buffer calibration

Temperature and activity affect precise measurements

Applications include water quality, corrosion control, electroplating and chemical processing

Malayalam support: pH scale logarithmic ആണ്; pH 3 solution, pH 4 solution-നെക്കാൾ ഏകദേശം 10 മടങ്ങ് കൂടുതൽ acidic ആണ്.

Engineering connection

pH, pOH and applications is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

Electrochemistry, Corrosion and Emerging Battery Technologies

Course 2003A

Theory 11 h

Practical 9 h

CO3

3.1 Oxidation, reduction and redox agents CO3

Oxidation is loss of electrons or increase in oxidation number; reduction is gain of electrons or decrease in oxidation number. Both occur together. The oxidising agent accepts electrons and is reduced; the reducing agent donates electrons and is oxidised.

OIL RIG: Oxidation Is Loss,
Reduction Is Gain

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is oxidation

$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ is reduction

Electron balance checks a redox equation

Corrosion, batteries and electroplating are redox systems

Malayalam support: Electron നഷ്ടപ്പെടുന്നത് oxidation; electron നേടുന്നത് reduction. രണ്ടും ഒരേസമയം നടക്കുന്നു.

Engineering connection

Oxidation, reduction and redox agents is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.2 Metallic and electrolytic conduction CO3

Metals conduct through mobile electrons in a solid lattice. Electrolytes conduct through moving ions in solution or molten state. In an electrolyte, cations migrate toward the cathode and anions toward the anode.

Metallic conduction does not require chemical decomposition

Electrolytic conduction usually accompanies electrode reactions

Insulators have very few mobile charge carriers

Conductivity depends on carrier concentration and mobility

Pure water conducts poorly compared with ionic solutions

Malayalam support: Metal-ൽ electrons ആണ് charge carry ചെയ്യുന്നത്; electrolyte-ൽ ions ആണ്.

Engineering connection

Metallic and electrolytic conduction is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.3 Strong, weak and non-electrolytes CO3

Strong electrolytes dissociate almost completely into ions, weak electrolytes only partially ionise, and non-electrolytes form essentially no ions in solution.

NaCl and HCl are strong electrolytes in water

Acetic acid is a weak electrolyte

Sugar solution is a non-electrolyte

Greater ion concentration generally increases conductivity

Dilution affects strong and weak electrolytes differently

Malayalam support: Solution-ൽ mobile ions കൂടുതലായാൽ conductivity സാധാരണയായി കൂടുതലാകും.

Engineering connection

Strong, weak and non-electrolytes is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.4 Electrolysis and electrolytic cells CO3

Electrolysis uses direct-current electrical energy to drive a non-spontaneous chemical change. The anode is positive and supports oxidation; the cathode is negative and supports reduction. Electron flow is through the external circuit, while ions move through the electrolyte.

Cations move to the cathode

Anions move to the anode

Water electrolysis forms hydrogen at the cathode and oxygen at the anode

Gas-volume ratio $H_2:O_2$ is ideally 2:1

Use low-voltage DC and suitable electrodes under supervision

Malayalam support: Electrolysis-ൽ electrical energy ഉപയോഗിച്ച് non-spontaneous reaction നടത്തുന്നു. Cathode-ൽ reduction, anode-ൽ oxidation.

Engineering connection

Electrolysis and electrolytic cells is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.5 Faraday's laws of electrolysis CO3

Faraday's first law states that deposited mass is proportional to charge passed: $m = ZIt$. The second law states that equal charge deposits masses proportional to chemical equivalent masses.

$$Q = It$$

Z is electrochemical equivalent in $kg\ C^{-1}$ or $g\ C^{-1}$

Current in ampere equals coulomb per second

Time must be in seconds

Actual deposition can be lower because current efficiency may be below 100%

Malayalam support: Deposited mass current $\times$ time-ന് proportional ആണ്. Time seconds ആയി ഉപയോഗിക്കുക.

Engineering connection

Faraday's laws of electrolysis is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.6 Nickel electroplating and copper refining

For nickel plating on mild steel, the cleaned steel object is the cathode, nickel is the anode, and a nickel-salt solution supplies Ni^{2+} . In copper refining, impure copper is the anode and a thin pure-copper sheet is the cathode in acidified copper-sulphate solution.

Surface cleaning controls adhesion

Nickel anode: $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$

Cathode: $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$

Copper impurities less noble than Cu enter solution; noble impurities form anode mud

Current density, temperature, pH and agitation influence coating quality

Malayalam support: Electroplating-ൽ coat ചെയ്യേണ്ട object cathode ആണ്; coating metal സാധാരണ anode ആണ്.

Engineering connection

Nickel electroplating and copper refining is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.7 Daniell cell and salt bridge CO3

A Daniell cell converts spontaneous redox energy to electrical energy. Zinc oxidises at the negative anode in ZnSO_4 ; Cu^{2+} reduces at the positive cathode in CuSO_4 . Electrons move through the wire from zinc to copper. The salt bridge maintains electrical neutrality and completes the ionic circuit.

Anode: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$

Cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

Overall: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$

Anions from the bridge migrate toward the anode compartment

Cations from the bridge migrate toward the cathode compartment

Malayalam support: Daniell cell-ൽ electron zinc electrode-ൽ നിന്ന് copper electrode-ലേക്ക് പുറം wire വഴി സഞ്ചരിക്കുന്നു.

Engineering connection

Daniell cell and salt bridge is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.8 Primary, secondary and fuel cells CO3

Primary cells are designed mainly for one discharge; secondary cells can be recharged by reversing the cell reaction; fuel cells continue producing electricity while fuel and oxidant are supplied.

Primary example: alkaline dry cell

Secondary example: lead-acid or lithium-ion battery

Fuel-cell example: hydrogen-oxygen fuel cell

Rechargeability depends on reaction reversibility and cell design

Safe charging requires correct voltage, current and temperature control

Malayalam support: Primary cell സാധാരണ recharge ചെയ്യില്ല; secondary cell recharge ചെയ്യാം; fuel cell-ൽ reactants തുടർച്ചയായി supply ചെയ്യണം.

Engineering connection

Primary, secondary and fuel cells are linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.9 Corrosion and factors affecting rate CO3

Corrosion is deterioration of a material through chemical or electrochemical interaction with its environment. Rusting of iron requires anodic metal dissolution and cathodic oxygen reduction in the presence of moisture.

Moisture and dissolved oxygen promote electrochemical cells	Low pH can accelerate metal dissolution	Salts raise electrolyte conductivity
Crevices create oxygen-concentration cells	Dissimilar-metal contact can produce galvanic corrosion	Temperature, stress and surface condition influence rate

Malayalam support: Metal environment-ുമായി പ്രതികരിച്ചു നശിക്കുന്നതാണ് corrosion. Moisture, salt, dissimilar metals എന്നിവ rate വർദ്ധിപ്പിക്കാം.

Engineering connection

Corrosion and factors affecting rate is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.10 Barrier and cathodic protection CO3

Barrier protection separates the metal from the environment. Metallic anodic coatings such as zinc can protect even when scratched because zinc sacrifices itself; cathodic coatings such as tin protect only while continuous. Cathodic protection forces the structure to act as a cathode.

Paint and polymer coatings require surface preparation

Anodising thickens the protective oxide on aluminium

Anti-rust compounds exclude water and oxygen

Sacrificial zinc or magnesium corrodes instead of steel

Protection systems require inspection and renewal

Malayalam support: Sacrificial anode method-ൽ active metal ആദ്യം corrode ചെയ്യും; steel structure cathode ആയി സംരക്ഷിക്കപ്പെടും.

Engineering connection

Barrier and cathodic protection is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

3.11 Lithium-ion, sodium-ion and solid-state batteries CO3

Lithium-ion batteries shuttle Li^+ between host electrodes through an organic electrolyte. Sodium-ion batteries use more abundant Na^+ but generally have lower energy density. Solid-state batteries replace the flammable liquid electrolyte with a solid ion conductor; they promise safety and energy benefits but face interface and manufacturing challenges.

Charging stores electrical energy as chemical energy

Discharging releases electrical energy

Battery-management systems supervise voltage, current and temperature

Thermal runaway is a key lithium-ion hazard

Deeper electrochemistry is enrichment beyond the basic syllabus

Malayalam support: Battery തിരഞ്ഞെടുപ്പിൽ energy density മാത്രം നോക്കരുത്; safety, cycle life, charging, temperature, cost എന്നിവയും കാണണം.

Engineering connection

Lithium-ion, sodium-ion and solid-state batteries is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

MODULE IV · CO4

Engineering Materials and Green Chemistry

Theory 11 h

Practical 9 h

CO4

4.1 Alloys and purpose of alloying CO4

An alloy is a metallic material containing a base metal and one or more additional elements. Alloying changes crystal structure and microstructure to improve strength, hardness, corrosion resistance, machinability, electrical behaviour or melting characteristics.

Stainless steel: iron with chromium and often nickel; corrosion-resistant equipment

Brass: copper-zinc; terminals, valves and fittings

Bronze: copper-tin; bearings, bushes and sculptures

Solder: low-melting alloy used to join electrical conductors

Composition varies by grade; do not treat one nominal composition as universal

Malayalam support: Base metal-ചേർക്കു മറ്റു elements ചേർത്ത് ആവശ്യമായ properties നേടുന്നതാണ് alloying.

Engineering connection

Alloys and purpose of alloying is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.2 Monomers, polymers and polymerisation CO4

A monomer is a small reactive molecule; polymerisation links many monomers into macromolecular chains or networks. A homopolymer uses one monomer type; a copolymer uses two or more.

Addition polymerisation commonly opens a carbon-carbon double bond

Condensation polymerisation eliminates a small molecule such as water

Chain length and branching affect properties

Cross-linking restricts chain movement

Polymer selection requires electrical, mechanical, thermal and environmental data

Malayalam support: Monomer-ുകൾ ചേർന്ന് നീളമുള്ള polymer chain രൂപപ്പെടുന്ന പ്രക്രിയയാണ് polymerisation.

Engineering connection

Monomers, polymers and polymerisation is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.3 Thermoplastics and thermosets CO4

Thermoplastics soften repeatedly on heating because chains are not permanently cross-linked. Thermosets form a three-dimensional network and do not remelt after curing; excessive heating causes degradation.

Thermoplastics are generally weldable and recyclable by

Thermosets retain shape and heat resistance but are harder

PE and PVC are thermoplastics

remelting
Course 2003A

to recycle

Bakelite is a thermoset

Recycling feasibility also depends on additives, contamination and economics

Malayalam support: Thermoplastic വീണ്ടും soften ചെയ്ത് shape മാറ്റാം; thermoset cure കഴിഞ്ഞാൽ വീണ്ടും melt ചെയ്യാനാവില്ല.

Engineering connection

Thermoplastics and thermosets is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.4 Common engineering polymers CO4

Engineering polymers are selected by balancing dielectric strength, flame behaviour, impact resistance, moisture uptake, chemical resistance and service temperature.

PE: cable insulation, tanks and films

PVC: conduit, cable sheath and pipes; plasticiser controls flexibility

Nylon-66: gears, bushes and fibres; moisture affects dimensions

Bakelite: electrical switches and heat-resistant handles

PET: fibres, bottles and films

PU: foams, coatings and elastomers

PBT: electrical connectors and housings

PC: tough transparent guards and enclosures

PMMA: clear covers and light diffusers

Malayalam support: Polymer name മാത്രം മതിയല്ല; grade, additives, temperature, UV, flame rating എന്നിവ പരിശോധിക്കണം.

Engineering connection

Common engineering polymers is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.5 Natural rubber, vulcanisation and synthetic rubber **CO4**

Natural rubber is mainly cis-1,4-polyisoprene derived from isoprene. Raw rubber is soft, tacky and temperature-sensitive. Vulcanisation creates sulphur cross-links, improving elasticity, tensile strength, wear and temperature resistance.

Cross-links prevent permanent chain slippage

Buna-S uses butadiene and styrene

Buna-N uses butadiene and acrylonitrile

Buna-N resists oils better than natural rubber

Excessive cross-linking produces harder material

Malayalam support: Sulphur cross-links chain-കളെ ബന്ധിപ്പിക്കുന്നതിനാൽ vulcanised rubber കൂടുതൽ ശക്തവും സ്ഥിരവുമാകും.

Engineering connection

Natural rubber, vulcanisation and synthetic rubber is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.6 Nanomaterials, carbon nanotubes, fullerenes and graphene **CO4**

Nanomaterials have structural features roughly in the 1–100 nm range. High surface-area-to-volume ratio and quantum/size effects can change strength, conductivity, optical behaviour and reactivity.

Carbon nanotube: rolled graphene-like cylinder

Fullerene: closed carbon cage such as C₆₀

Graphene: one-atom-thick sp² carbon sheet

Applications include composites, sensors, conductive coatings and energy devices

Health and environmental effects require responsible handling

Malayalam support: Nanoscale-ൽ surface area വളരെ കൂടുതലാകുന്നതിനാൽ bulk material-നേക്കാൾ properties മാറും.

Engineering connection

Nanomaterials, carbon nanotubes, fullerenes and graphene is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.7 Metal-organic frameworks CO4

MOFs are porous crystalline networks built from metal-containing nodes connected by organic linkers. Their pore size and surface chemistry can be tuned for gas storage, separation, catalysis and sensing.

Metal nodes provide coordination sites

Organic linkers define spacing and functionality

High porosity gives large internal surface area

Moisture and thermal stability vary widely

Industrial use depends on cost, shaping, cycling and safety

Malayalam support: Metal nodes-ഉം organic linkers-ഉം ചേർന്ന porous lattice ആണ് MOF.

Engineering connection

Metal-organic frameworks is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.8 Biomaterials and biofuels CO4

A biomaterial interacts with biological systems for diagnosis, repair or replacement. Metallic, ceramic, polymeric and composite biomaterials are selected for biocompatibility, strength, wear, corrosion resistance and sterilisation. Biofuels are fuels made from biomass, including bioethanol, biodiesel and biogas.

Metallic implants: strength and toughness

Ceramics: hardness, wear resistance and bioactivity

Polymers: flexibility and tailored degradation

Composites combine complementary properties

Biofuel sustainability depends on feedstock, land, water, processing and lifecycle emissions

Malayalam support: Biomaterial ശരീരവുമായി സുരക്ഷിതമായി പ്രവർത്തിക്കണം; mechanical strength മാത്രം മതിയല്ല, biocompatibility നിർബന്ധമാണ്.

Engineering connection

Biomaterials and biofuels is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

4.9 Green chemistry CO4

Green chemistry designs chemical products and processes to reduce hazards and waste at the source. It is not merely waste treatment after pollution occurs.

Prevent waste

Improve atom economy

Use less hazardous synthesis and safer products

Choose safer solvents and auxiliaries

Increase energy efficiency

Use renewable feedstocks

Prefer catalytic reagents

Design for degradation

Use real-time analysis and inherently safer conditions

Apply lifecycle thinking in green manufacturing

Malayalam support: Pollution ഉണ്ടാക്കിയ ശേഷം മാത്രം clean ചെയ്യുന്നത് അല്ല; pollution ഉണ്ടാകാതിരിക്കാൻ process design ചെയ്യുന്നതാണ് green chemistry.

Engineering connection

Green chemistry is linked to measurement, material selection, process control, maintenance, safety or sustainability. State the system boundary, units and assumptions before applying the concept.

Self-check

Explain the definition in one sentence, draw the relevant diagram, give one domestic example and one engineering example, and identify one common error.

PRACTICAL RECORD CENTRE

Printable experiment records and selection planner

The records below follow a consistent engineering-laboratory format: experiment number, date, aim, objective, apparatus, chemicals, formula, reaction, theory, diagram/flow chart, procedure, observation, calculations, result, precautions, safety, waste disposal, viva and faculty signature.

Select the official minimum of ten experiments

- ☐ M1 · E1 Preparation of 0.1 N oxalic acid solution
- ☐ M1 · E2 Preparation of 0.1 N sodium carbonate solution
- ☐ M1 · E3 Preparation of 0.1 M zinc sulphate solution
- ☐ M1 · E4 Preparation of 0.1 M magnesium sulphate solution
- ☐ M1 · E5 Endothermic reaction: citric acid and baking soda
- ☐ M1 · E6 Endothermic reaction: barium hydroxide and ammonium chloride
- ☐ M1 · E7 Exothermic neutralisation of HCl and NaOH
- ☐ M1 · E8 Exothermic dissolution of NaOH in water
- ☐ M1 · E9 Exothermic dissolution of calcium chloride

- ☐ M2 · E10 Standardisation of HCl using standard sodium carbonate

- ☐ M2 · E11 Estimation of sodium hydroxide using standard HCl
- ☐ M2 · E12 Estimation of potassium hydroxide using standard oxalic acid
- ☐ M2 · E13 Standardisation of EDTA using ZnSO_4
- ☐ M2 · E14 Estimation of total hardness using standard EDTA
- ☐ M2 · E15 pH measurement using a pH meter
- ☐ M2 · E16 pH using universal indicator
- ☐ M2 · E17 pH using pH test paper
- ☐ M3 · E18 Construction of a salt bridge
- ☐ M3 · E19 Construction of a Daniell cell
- ☐ M3 · E20 Nickel electroplating of a steel spoon
- ☐ M3 · E21 Conductivity measurement of water
- ☐ M3 · E22 Electrolysis of water
- ☐ M4 · E23 Estimation of copper in brass
- ☐ M4 · E24 Estimation of zinc in brass
- ☐ M4 · E25 Estimation of iron in iron ore
- ☐ M4 · E26 Preparation of urea-formaldehyde resin
- ☐ M4 · E27 Preparation of phenol-formaldehyde resin

Experiment 1: Preparation of 0.1 N oxalic acid solution

Date: _____

Mapped outcome: CO1

Aim / objective

Prepare 250 mL of accurately known oxalic acid solution.

Apparatus

Balance, weighing bottle, beaker, funnel, wash bottle, 250 mL volumetric flask

Chemicals

Oxalic acid dihydrate; distilled water

Formula / calculation

$$\text{Mass} = N \times \text{Eq.mass} \times V(\text{L}) = 0.1 \times 63 \times 0.250 = 1.575 \text{ g}$$

Theory and reaction

$\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ provides two replaceable H^+ in acid-base reaction.

Procedure

Course 2003A

1. Weigh 1.575 g
2. dissolve in water
3. transfer quantitatively
4. rinse
5. make to mark
6. stopper and invert.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Mass weighed, final volume, appearance			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** 0.1 N oxalic acid prepared.

Safety: Wear goggles; avoid ingestion. **Waste:** Collect excess acidic solution as instructed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 2: Preparation of 0.1 N sodium carbonate solution

Date: _____

Mapped outcome: CO1

Aim / objective

Prepare 250 mL standard sodium carbonate solution.
Course 2003A

Apparatus

Balance, beaker, funnel, 250 mL volumetric flask

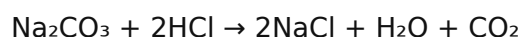
Chemicals

Anhydrous Na_2CO_3 ; distilled water

Formula / calculation

$$\text{Mass} = 0.1 \times 53 \times 0.250 = 1.325 \text{ g}$$

Theory and reaction



Procedure

1. Dry if instructed
2. weigh
3. dissolve
4. transfer
5. make to mark
6. mix.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Mass, volume, clarity			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** 0.1 N Na_2CO_3 prepared.

Safety: Avoid dust; wash spills. **Waste:** Dilute/neutralise only under faculty direction.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 3: Preparation of 0.1 M zinc sulphate solution

Date: _____

Mapped outcome: CO1

Aim / objective

Prepare 250 mL 0.1 M ZnSO_4 solution using the hydrate specified on the reagent label.

Apparatus

Balance, beaker, funnel, volumetric flask

Chemicals

ZnSO_4 reagent; distilled water

Formula / calculation

Mass = M $\times$ molar mass of actual hydrate $\times$ 0.250 L

Theory and reaction

Use the molar mass printed for ZnSO_4 , $\text{ZnSO}_4 \cdot \text{H}_2\text{O}$ or $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ as supplied.

Procedure

1. Calculate for actual hydrate
2. weigh
3. dissolve
4. transfer
5. make to mark.

Observation table

Hydrate form, molar mass, mass, volume

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** 0.1 M ZnSO₄ prepared.

Safety: Zinc salts are harmful; gloves required. **Waste:** Collect zinc-containing waste separately.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 4: Preparation of 0.1 M magnesium sulphate solution

Date: _____

Mapped outcome: CO1

Aim / objective

Prepare 250 mL 0.1 M MgSO₄ using the hydrate stated on the bottle.

Apparatus

Balance and volumetric glassware

Chemicals

MgSO₄ reagent; distilled water

Formula / calculation

Theory and reaction

Hydration changes molar mass and required mass.

Procedure

1. Identify hydrate
2. calculate
3. weigh
4. dissolve
5. transfer
6. make to mark.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Hydrate, mass, volume			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** 0.1 M MgSO₄ prepared.

Safety: Avoid dust and contact. **Waste:** Dispose as directed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 5: Endothermic reaction: citric acid and baking soda

Mapped outcome: C01

Aim / objective

Observe temperature reduction during reaction.

Aim / objective

Observe temperature reduction during reaction.

Apparatus

Cup/beaker, thermometer, spatula

Apparatus

Cup/beaker, thermometer, spatula

Chemicals
Citric acid, NaHCO_3 , water

Chemicals
Citric acid, NaHCO_3 , water

Formula / calculation

Record $\Delta T = \text{final} - \text{initial}$

Formula / calculation

Record $\Delta T = \text{final} - \text{initial}$

Theory and reaction

Acid-bicarbonate reaction absorbs heat overall and releases CO₂.

Theory and reaction

Acid-bicarbonate reaction absorbs heat overall and releases CO₂.

Procedure

1. Measure initial temperature
2. mix measured reactants
3. stir
4. record minimum temperature.

- ### Procedure
1. Measure initial temperature
 2. mix measured reactants
 3. stir
 4. record minimum temperature.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Time and temperature table			

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Time and temperature table			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Temperature decreased; endothermic behaviour observed.

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Temperature decreased; endothermic behaviour observed.

Safety: Do not seal CO₂-generating mixture. **Waste:** Allow gas release; neutral aqueous residue per instruction.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.

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- ## Precautions
- Use clean, correctly rinsed apparatus and labelled reagents.
 - Record readings immediately with units; do not invent concordant values.
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- Repeat until the institution's concordance criterion is met.

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Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 6: Endothermic reaction: barium hydroxide and ammonium chloride

Date: _____

Mapped outcome: CO1

Aim / objective

Demonstrate a strongly endothermic reaction under a fume hood.

Apparatus

Fume hood, beaker, thermometer

Chemicals

Ba(OH)₂ hydrate, NH₄Cl

Formula / calculation

Record ΔT

Theory and reaction

Reaction absorbs heat and may release ammonia; barium compounds are toxic.

Procedure

1. Faculty-supervised microscale setup only
2. mix in hood
3. monitor temperature
4. avoid inhalation.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Temperature and odour observation without direct smelling			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Endothermic cooling observed.

Safety: TOXIC barium; ammonia vapour; goggles, gloves, hood. **Waste:** Dedicated barium hazardous-waste container.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 7: Exothermic neutralisation of HCl and NaOH

Date: _____

Mapped outcome: CO1

Aim / objective

Measure heat release during neutralisation.

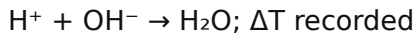
Apparatus

Insulated cup, thermometer, measuring cylinders

Chemicals

Dilute HCl and dilute NaOH

Formula / calculation



Theory and reaction

Strong acid-base neutralisation releases heat.

Procedure

1. Measure equal volumes at similar initial temperature
2. mix
3. stir
4. record peak.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Initial, maximum and ΔT			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Temperature increased.

Safety: Corrosive solutions; splash protection. **Waste:** Neutralise only after verification and faculty approval.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution’s concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Date: _____

Mapped outcome: CO1

Aim / objective

Observe heat released when NaOH dissolves.

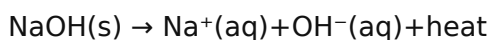
Apparatus

Beaker, thermometer, tongs/spatula

Chemicals

Small quantity NaOH pellets, water

Formula / calculation



Theory and reaction

Hydration of ions releases more energy than lattice separation consumes.

Procedure

1. Add pellets slowly to water while stirring
2. never add water onto a bulk of pellets
3. record temperature.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Mass, water volume, temperature			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Exothermic dissolution observed.

Safety: Highly corrosive; face/eye protection. **Waste:** Collect alkaline waste; do not touch pellets.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.

- Repeat until the institution's concordance criterion is met.

Course 2003A

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 9: Exothermic dissolution of calcium chloride

Date: _____

Mapped outcome: CO1

Aim / objective

Observe temperature rise on dissolving CaCl_2 .

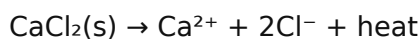
Apparatus

Beaker, thermometer

Chemicals

CaCl_2 , water

Formula / calculation



Theory and reaction

Hydration produces net heat release.

Procedure

1. Record initial temperature
2. add measured salt
3. stir
4. record maximum.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Time-temperature table			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Temperature increased.

Safety: Avoid contact and dust. **Waste:** Dispose as directed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 10: Standardisation of HCl using standard sodium carbonate

Date: _____

Mapped outcome: CO₂

Aim / objective

Determine exact normality of HCl.

Apparatus

Burette, pipette, conical flask, funnel

Chemicals

Standard Na₂CO₃, HCl, methyl orange

Theory and reaction

$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$; methyl orange: yellow to orange/red at endpoint.

Procedure

1. Rinse
2. fill burette with HCl
3. pipette Na_2CO_3
4. add indicator
5. titrate to first permanent endpoint
6. obtain concordant titres.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Initial, final, titre and mean			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Exact HCl normality calculated.

Safety: HCl corrosive; do not blow pipette. **Waste:** Neutral aqueous waste as instructed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 11: Estimation of sodium hydroxide using standard HCl

Date: _____

Mapped outcome: CO2

Aim / objective

Determine NaOH normality.

Apparatus

Titration glassware

Chemicals

NaOH sample, standard HCl, phenolphthalein

Formula / calculation

$$N(\text{NaOH}) = V(\text{HCl}) \times N(\text{HCl}) / V(\text{NaOH})$$

Theory and reaction

$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$; pink to colourless.

Procedure

1. Pipette NaOH
2. add phenolphthalein
3. titrate with HCl to disappearance of pink.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Concordant titre table			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** NaOH normality estimated.

Safety: Corrosive; protect eyes. **Waste:** Neutralise/collect as directed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.

- Repeat until the institution's concordance criterion is met.

Course 2003A

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 12: Estimation of potassium hydroxide using standard oxalic acid

Date: _____

Mapped outcome: CO2

Aim / objective

Determine KOH normality.

Apparatus

Titration glassware

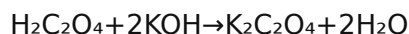
Chemicals

KOH sample, standard oxalic acid, phenolphthalein

Formula / calculation

$N(\text{KOH}) = V(\text{acid}) \times N(\text{acid}) / V(\text{KOH})$

Theory and reaction



Procedure

1. Pipette KOH
2. indicator
3. titrate with oxalic acid to colourless endpoint.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Titre table			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** KOH normality estimated.

Safety: KOH corrosive. **Waste:** Handle neutralised waste as directed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 13: Standardisation of EDTA using ZnSO₄

Date: _____

Mapped outcome: CO2

Aim / objective

Determine exact EDTA concentration.

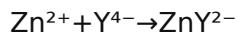
Apparatus

Burette, pipette, conical flask

Chemicals

Standard ZnSO₄, EDTA, buffer, suitable metallochromic indicator

Theory and reaction



Procedure

1. Buffer aliquot
2. add indicator
3. titrate with EDTA to specified colour change.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Readings and endpoint colour			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** EDTA concentration standardised.

Safety: Zinc solutions hazardous. **Waste:** Collect metal-EDTA waste separately.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 14: Estimation of total hardness using standard EDTA

Date: _____

Mapped outcome: CO2

Aim/objective

Determine total hardness of water as CaCO_3 .

Apparatus

Titration glassware

Chemicals

Water sample, standard EDTA, pH 10 buffer, Eriochrome Black T

Formula / calculation

Hardness mg/L as $\text{CaCO}_3 = V_{\text{EDTA}} \times M_{\text{EDTA}} \times 100000 / \text{sample mL}$

Theory and reaction

$\text{Ca}^{2+}/\text{Mg}^{2+}$ form 1:1 complexes with EDTA; wine red to blue endpoint.

Procedure

1. Pipette sample
2. add buffer and indicator
3. titrate to clear blue
4. repeat.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Titre and hardness table			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Total hardness reported in mg/L as CaCO_3 .

Safety: Avoid metal contamination. **Waste:** Collect EDTA/metal waste.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 15: pH measurement using a pH meter

Date: _____

Mapped outcome: CO2

Aim / objective

Measure pH after calibration.

Apparatus

pH meter, beakers, wash bottle, tissue

Chemicals

Buffer pH 4, 7 and/or 9.2; samples

Formula / calculation

$\text{pH} = -\log[\text{H}^+]$

Theory and reaction

Glass electrode potential is converted to pH.

Procedure

1. Warm up
2. calibrate with buffers
3. rinse and blot
4. immerse
5. wait for stable reading
6. rinse between samples
7. store electrode correctly.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Buffer checks, sample temperature and pH			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** pH measured.

Safety: Do not wipe/scratch bulb. **Waste:** Return buffers to waste, not stock bottles.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 16: pH using universal indicator

Date: _____

Mapped outcome: CO2

Aim / objective

Estimate pH by colour comparison.

Apparatus

Droppers, white tile/test tubes

Chemicals

Universal indicator, samples

Formula / calculation

Theory and reaction

Mixed indicators cover a broad pH range.

Procedure

1. Add equal sample volumes
2. add indicator
3. compare colour immediately with chart.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Colour and pH range			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Approximate pH identified.

Safety: Avoid cross-contamination. **Waste:** Small dilute waste as directed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 17: pH using pH test paper

Date: _____

Mapped outcome: CO2

Aim/objective

Estimate sample pH with test paper.

Apparatus

Clean droppers, forceps

Chemicals

pH paper, samples

Formula / calculation

Colour comparison

Theory and reaction

Indicator dyes immobilised on paper change with pH.

Procedure

1. Place drop on strip
2. compare promptly
3. do not dip roll into sample.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Colour and pH			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Approximate pH identified.

Safety: Use clean dropper. **Waste:** Used strips to solid lab waste.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*

3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 18: Construction of a salt bridge

Date: _____

Mapped outcome: CO3

Aim / objective

Prepare a salt bridge for an electrochemical cell.

Apparatus

U-tube, beaker, heater

Chemicals

KCl/KNO₃, agar, water

Formula / calculation

Ions migrate to maintain electroneutrality

Theory and reaction

Inert electrolyte and gel minimise bulk mixing.

Procedure

1. Prepare warm agar-electrolyte solution
2. fill U-tube without bubbles
3. cool
4. keep ends moist.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Continuity and absence of bubbles			

Calculation and result

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Show formula, substitution, unit and significant figures. **Expected record statement:** Salt bridge constructed.

Safety: Heating and glass safety. **Waste:** Dispose gel as directed.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 19: Construction of a Daniell cell

Date: _____

Mapped outcome: CO3

Aim / objective

Construct $\text{Zn}|\text{ZnSO}_4||\text{CuSO}_4|\text{Cu}$ and observe voltage.

Apparatus

Beakers, leads, voltmeter

Chemicals

Zn, Cu, ZnSO_4 , CuSO_4 , salt bridge

Formula / calculation

$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$

Theory and reaction

Procedure

1. Clean electrodes
2. immerse in own electrolytes
3. connect bridge
4. connect Zn to negative and Cu to positive voltmeter terminal
5. read voltage.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Polarity and voltage			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Daniell cell generated electrical energy.

Safety: Metal salts hazardous. **Waste:** Collect Cu/Zn waste separately.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 20: Nickel electroplating of a steel spoon

Date: _____

Mapped outcome: CO3

Aim / objective

Apparatus

Low-voltage DC source, leads, beaker

Chemicals

NiCl₂/NiSO₄ electrolyte, nickel anode, steel spoon

Formula / calculation

$$m = ZIt$$

Theory and reaction

Anode $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$; cathode $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$

Procedure

1. Degrease and clean spoon
2. connect spoon cathode, Ni anode
3. immerse without contact
4. apply controlled current
5. rinse and inspect.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Current, time, before/after mass, appearance			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Nickel coating formed.

Safety: Nickel compounds hazardous; electrical safety. **Waste:** Dedicated nickel waste.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*

2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*

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3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 21: Conductivity measurement of water

Date: _____

Mapped outcome: CO3

Aim / objective

Measure conductivity of a sample.

Apparatus

Conductivity meter, beakers

Chemicals

Calibration standard and water samples

Formula / calculation

Conductivity displayed in $\mu\text{S}/\text{cm}$ or mS/cm

Theory and reaction

Ions carry current; reading depends on temperature.

Procedure

1. Calibrate/check cell
2. rinse
3. immerse
4. remove bubbles
5. wait stable
6. record temperature and reading.

Observation table

Sample, temperature, conductivity

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Conductivity measured.

Safety: Keep connector dry. **Waste:** Samples disposed as classified.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 22: Electrolysis of water

Date: _____

Mapped outcome: CO3

Aim / objective

Demonstrate gas formation at electrodes.

Apparatus

Low-voltage DC source, inert electrodes, gas tubes

Chemicals

Water with approved supporting electrolyte

Theory and reaction

Hydrogen forms at cathode, oxygen at anode; ideal volume ratio 2:1.

Procedure

1. Assemble
2. verify polarity
3. energise
4. collect gases
5. observe volumes
6. do not ignite gases.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Polarity, gas volumes			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Electrolysis demonstrated.

Safety: No flame; low voltage; ventilation. **Waste:** De-energise before dismantling.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 23: Estimation of copper in brass

Date: _____

Mapped outcome: CO4

Aim / objective

Determine copper content using the prescribed laboratory method.

Apparatus

Volumetric glassware and balance

Chemicals

Prepared brass solution and method-specific reagents

Formula / calculation

$$\%Cu = (\text{mass Cu} / \text{mass sample}) \times 100$$

Theory and reaction

Copper is converted to a measurable species by the approved procedure.

Procedure

1. Faculty supplies exact official/local method
2. dissolve weighed sample safely
3. carry out quantitative reaction
4. calculate from standardised reagent.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Sample mass, titres/readings			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Copper percentage calculated.

Safety: Acids and copper salts hazardous. **Waste:** Heavy-metal waste container.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents

- Record readings immediately with units; do not invent concordant values.

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- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 24: Estimation of zinc in brass

Date: _____

Mapped outcome: CO4

Aim / objective

Determine zinc content using the prescribed quantitative method.

Apparatus

Volumetric apparatus

Chemicals

Prepared brass solution and reagents

Formula / calculation

$$\%Zn = (\text{mass Zn} / \text{mass sample}) \times 100$$

Theory and reaction

Zinc is selectively determined or obtained after copper determination according to the approved method.

Procedure

1. Follow institution method
2. maintain quantitative transfer
3. obtain concordant data.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Mass and analytical readings			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Zinc percentage calculated.

Safety: Corrosive reagents; zinc waste. **Waste:** Heavy-metal waste.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 25: Estimation of iron in iron ore

Date: _____

Mapped outcome: CO4

Aim / objective

Estimate iron content by the approved redox/complexometric method.

Apparatus

Burette, pipette, flask

Chemicals

Prepared ore solution and standard reagent

Theory and reaction

Iron oxidation state is controlled before titration.

Procedure

1. Dissolve/prepare sample only under faculty instruction
2. standardise reagent
3. titrate
4. blank-correct if required.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Sample mass, titres			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** Iron percentage calculated.

Safety: Acids and oxidants may be corrosive. **Waste:** Collect metal-containing waste.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 26: Preparation of urea-formaldehyde resin

Date: _____

Mapped outcome: CO4

Aim / objective

Prepare a small resin sample under ventilation.

Apparatus

Fume hood, beaker, water bath, stirrer

Chemicals

Urea, formaldehyde solution, approved catalyst

Formula / calculation

Condensation polymerisation forms methylene-linked network

Theory and reaction

Urea reacts with formaldehyde then condenses during curing.

Procedure

1. FUME HOOD ONLY
2. use microscale quantities
3. control pH and temperature as directed
4. collect resin
5. wash only by approved method.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Time, temperature, appearance			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** UF resin formed.

Safety: FORMALDEHYDE toxic/carcinogenic; vapour and skin hazard; goggles, gloves, hood. **Waste:** All liquid/solid residues to labelled formaldehyde-resin waste.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values

- Repeat until the institution's concordance criterion is met.

Course 2003A

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

Experiment 27: Preparation of phenol-formaldehyde resin

Date: _____

Mapped outcome: CO4

Aim / objective

Prepare a small phenolic resin sample under strict supervision.

Apparatus

Fume hood, water bath, guarded glassware

Chemicals

Phenol, formaldehyde, approved catalyst

Formula / calculation

Phenol and formaldehyde undergo condensation to a cross-linked resin

Theory and reaction

Network structure develops during controlled reaction and curing.

Procedure

1. FUME HOOD ONLY
2. faculty handles concentrated reagents
3. control heating
4. avoid open flame
5. isolate product with approved method.

Observation table

Trial/time/item	Initial reading	Final reading	Difference/result
Time, temperature, colour, texture			

Calculation and result

Show formula, substitution, unit and significant figures. **Expected record statement:** PF resin formed.

Safety: PHENOL is toxic and rapidly absorbed through skin; formaldehyde vapour hazard; corrosive catalyst and heating hazards. **Waste:** Dedicated phenol/formaldehyde hazardous waste; no drain disposal.

Precautions

- Use clean, correctly rinsed apparatus and labelled reagents.
- Record readings immediately with units; do not invent concordant values.
- Repeat until the institution's concordance criterion is met.

Viva questions with answer cues

1. Why is this experiment performed? *State the measured/prepared quantity and engineering relevance.*
2. What is the most important source of error? *Discuss transfer, endpoint, temperature, calibration or surface preparation as applicable.*
3. How is the result verified? *Use repeatability, stoichiometry, polarity, blank or standard check.*

Faculty signature: _____

FORMULA AND EQUATION BANK

Use the correct formula, units and context

1. Protons

$$p = Z$$

Symbols: Z atomic number · **Unit:** count

When to use: Atomic identity

Worked example: For Z=13, p=13

Common mistake: Using mass number instead of atomic number

2. Electrons in a neutral atom

Symbols: Z atomic number · **Unit:** count

When to use: Neutral atoms only

Worked example: For neutral Na, Z=11, e=11

Common mistake: Ignoring ionic charge

3. Neutrons

$$n = A - Z$$

Symbols: A mass number; Z atomic number · **Unit:** count

When to use: Isotope particle count

Worked example: For ^{23}Na , $n=23-11=12$

Common mistake: Subtracting in reverse order

4. Molarity

$$M = n/V = m/(M_r V)$$

Symbols: n mol; V L; m g; M_r g mol $^{-1}$ · **Unit:** mol L $^{-1}$

When to use: Solution concentration

Worked example: 5.85 g NaCl in 1 L: 0.100 M

Common mistake: Using mL as L

5. Required mass

$$m = M M_r V$$

Symbols: M mol L $^{-1}$; M_r g mol $^{-1}$; V L · **Unit:** g

When to use: Preparing molar solution

Worked example: $0.1 \text{ M} \times 58.5 \times 0.25 = 1.4625 \text{ g}$

Common mistake: Wrong hydrate molar mass

6. Normality

$$N = \text{equivalents}/V$$

Symbols: equivalents mol-equivalent; V L · **Unit:** eq L⁻¹

When to use: Reaction-based concentration

Worked example: 6.3 g oxalic acid dihydrate in 1 L = 0.1 N

Common mistake: Ignoring n-factor

7. ppm

$$\text{ppm} = \text{mass solute}/\text{mass solution} \times 10^6$$

Symbols: consistent mass units · **Unit:** ppm

When to use: Trace concentration

Worked example: 2 mg in 2 kg = 1 ppm

Common mistake: Mixing mg/L with exact mass basis without density

8. Dilution

$$C_1V_1=C_2V_2$$

Symbols: C concentration; V volume · **Unit:** matching units

When to use: Dilution without reaction

Worked example: 1 M stock to 0.1 M, 100 mL: V₁=10 mL

Common mistake: Changing amount of solute

9. Titration

$$V_1N_1=V_2N_2$$

Symbols: V matching units; N normality · **Unit:** N

When to use: Equivalence using normality

Worked example: 25×0.1 = 20×N₂; N₂=0.125 N

10. pH

$$\text{pH} = -\log_{10} [\text{H}^+]$$

Symbols: $[\text{H}^+]$ mol L⁻¹ · **Unit:** dimensionless

When to use: Acidity

Worked example: $[\text{H}^+]=10^{-3}$ gives pH 3

Common mistake: Dropping negative sign

11. pOH

$$\text{pOH} = -\log_{10} [\text{OH}^-]$$

Symbols: $[\text{OH}^-]$ mol L⁻¹ · **Unit:** dimensionless

When to use: Basicity

Worked example: $[\text{OH}^-]=10^{-4}$ gives pOH 4

Common mistake: Calling pOH the pH

12. pH-pOH relation

$$\text{pH} + \text{pOH} = 14$$

Symbols: at 25 °C, dilute aqueous solution · **Unit:** dimensionless

When to use: Convert pH and pOH

Worked example: pH 9.5 → pOH 4.5

Common mistake: Applying 14 universally at all temperatures

13. Charge

$$Q = It$$

Symbols: I A; t s · **Unit:** C

When to use: Electrolysis charge

Worked example: 2 A for 300 s gives 600 C
Course 2003A

Common mistake: Using minutes without conversion

14. Faraday first law

$$m = ZIt$$

Symbols: Z g C⁻¹; I A; t s · **Unit:** g

When to use: Metal deposition

Worked example: Z=0.000329 g/C, 2 A, 600 s → 0.395 g

Common mistake: Confusing equivalent mass with Z

15. Faraday second law

$$m_1/m_2 = E_1/E_2$$

Symbols: E equivalent masses · **Unit:** ratio

When to use: Equal charge through different electrolytes

Worked example: Ag/Cu mass ratio = 107.9/31.75

Common mistake: Comparing unequal charges

OFFLINE CALCULATORS

Show the formula, substitution and result

Every calculator validates entries. In the printed handbook, use the worked examples and Formula Bank.

Atomic-particle calculator

Atomic number

Mass number

Enter the values and select Calculate.

Molarity calculator

Solute mass (g)

Molar mass (g/mol)

Course 2003A

Volume (L)

Enter the values and select Calculate.

Required-mass calculator

Molarity (mol/L)

Molar mass (g/mol)

Volume (L)

Enter the values and select Calculate.

Normality calculator

Solute mass (g)

Equivalent mass (g/eq)

Volume (L)

Enter the values and select Calculate.

ppm calculator

Solute mass

Solution mass, same unit

Enter the values and select Calculate.

Dilution calculator

C₁

V₁

C₂

Enter the values and select Calculate.

Titration calculator

V₁

N₁

V₂

Enter the values and select Calculate.

pH calculator

[H⁺] mol/L

Enter the values and select Calculate.

pOH calculator

[OH⁻] mol/L

Enter the values and select Calculate.

Hydrogen-ion calculator

pH

Enter the values and select Calculate.

Faraday charge calculator

Current (A)

Time (s)

Enter the values and select Calculate.

Electrolysis-time calculator

Enter the values and select Calculate.

Metal-deposition calculator

Z (g/C)

Current (A)

Time (s)

Enter the values and select Calculate.

REQUIRED DIAGRAM LIBRARY

Original SVG diagrams for screen and A4 print

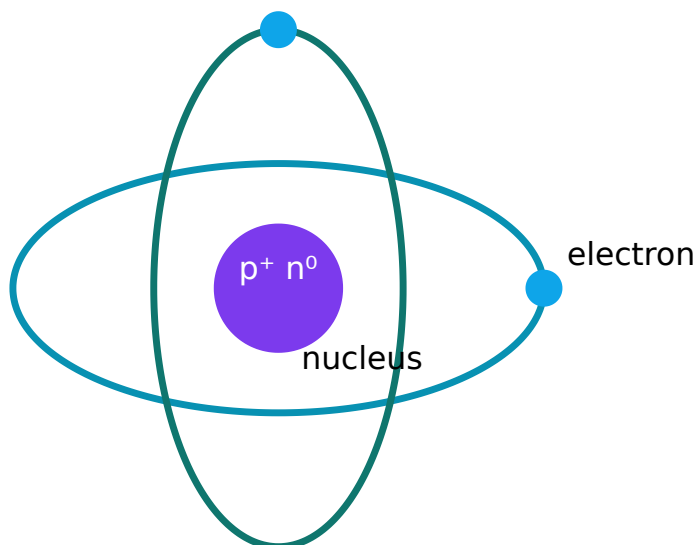


Figure 1. Labelled atomic structure. Labels and arrows identify the principal parts or process direction.

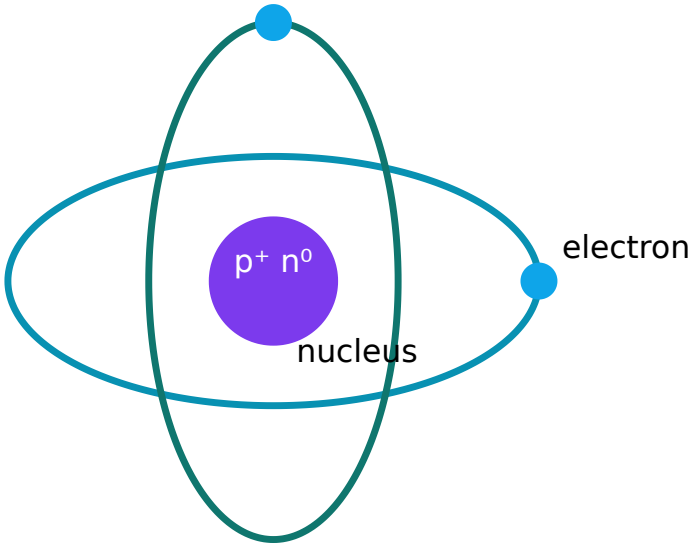


Figure 2. Electron-shell structure. Labels and arrows identify the principal parts or process direction.

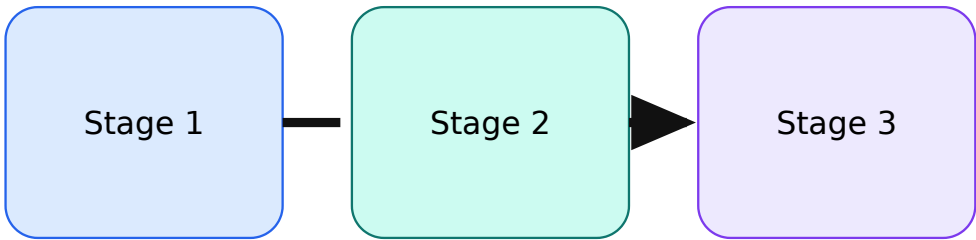


Figure 3. Fundamental-particle comparison. Labels and arrows identify the principal parts or process direction.

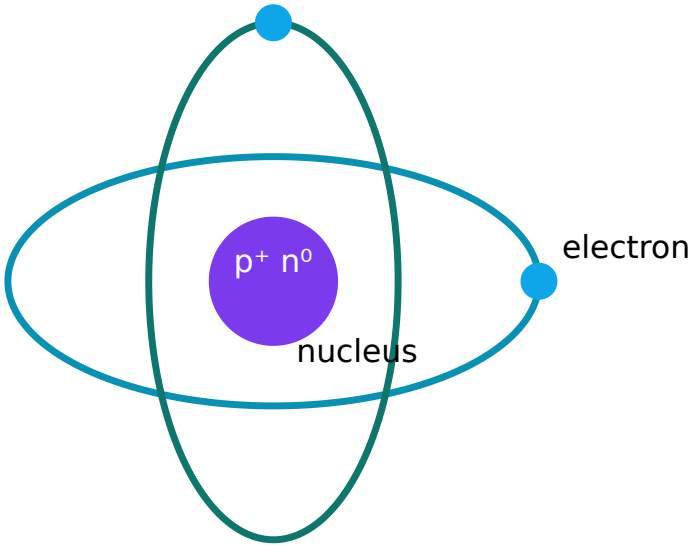


Figure 4. Atomic-number and mass-number notation. Labels and arrows identify the principal parts or process direction.

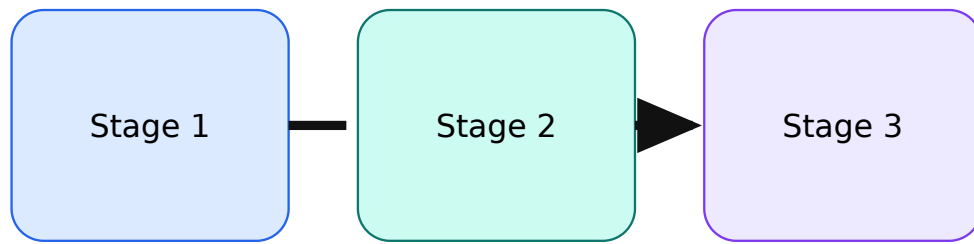


Figure 5. Elements up to Z=20. Labels and arrows identify the principal parts or process direction.

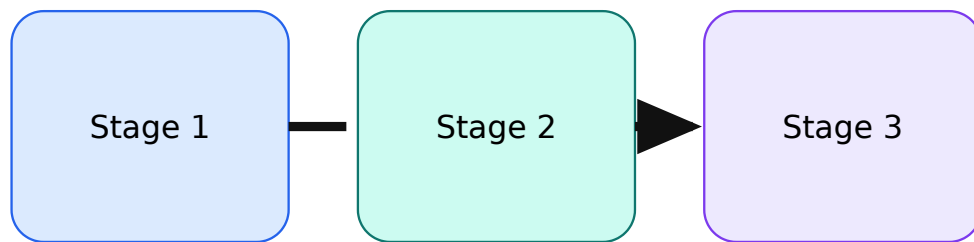


Figure 6. Open thermodynamic system. Labels and arrows identify the principal parts or process direction.

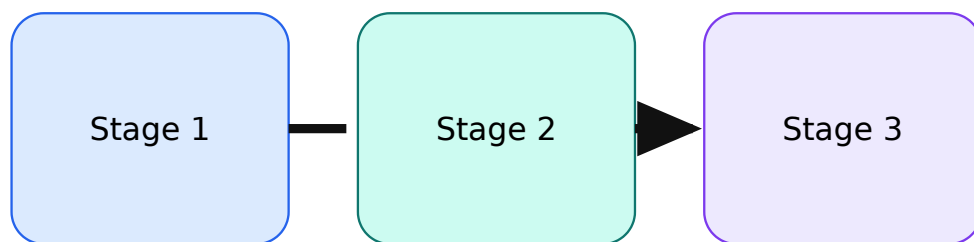


Figure 7. Closed thermodynamic system. Labels and arrows identify the principal parts or process direction.

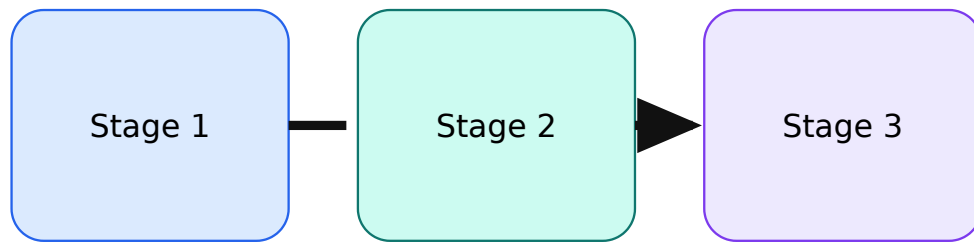


Figure 8. Isolated thermodynamic system. Labels and arrows identify the principal parts or process direction.

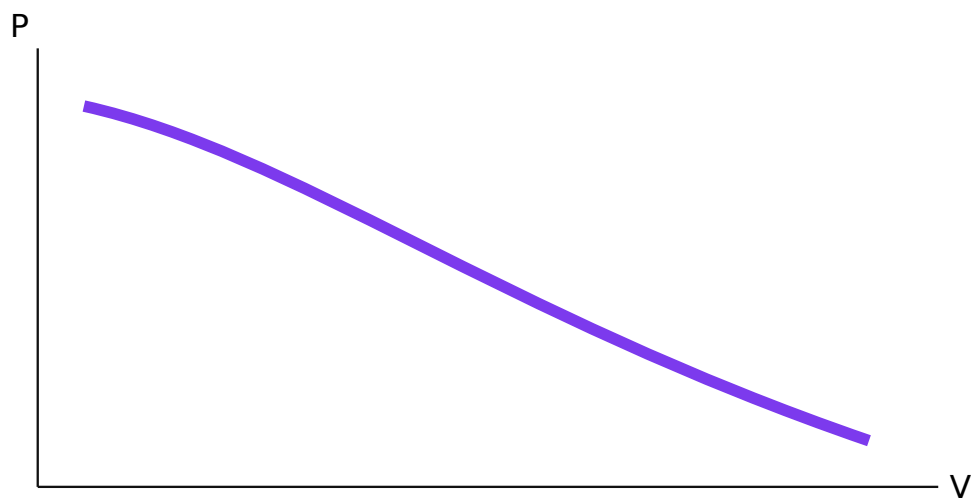


Figure 9. Isothermal P-V graph. Labels and arrows identify the principal parts or process direction.

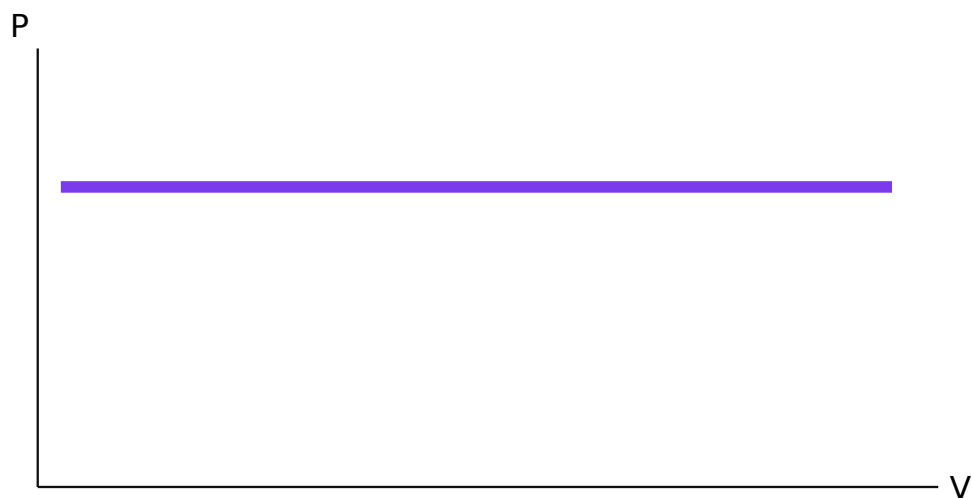


Figure 10. Isobaric P-V graph. Labels and arrows identify the principal parts or process direction.

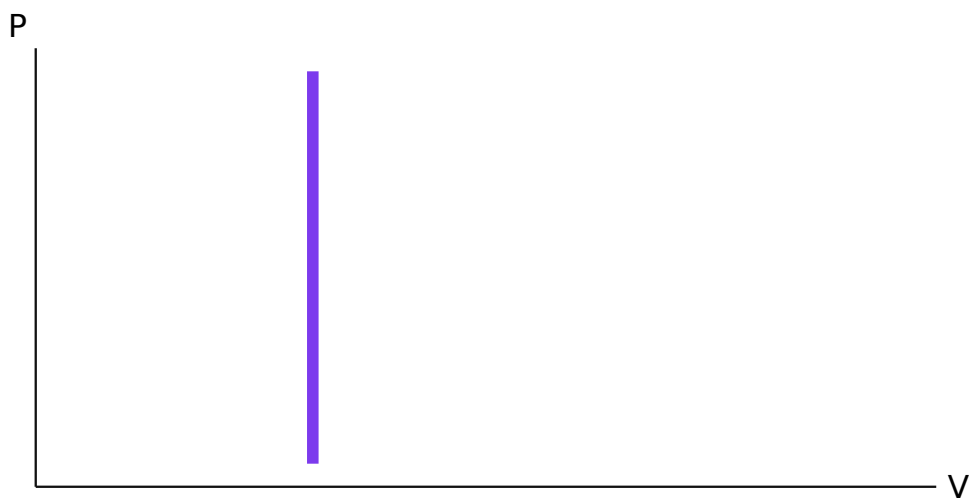


Figure 11. Isochoric P-V graph. Labels and arrows identify the principal parts or process direction.

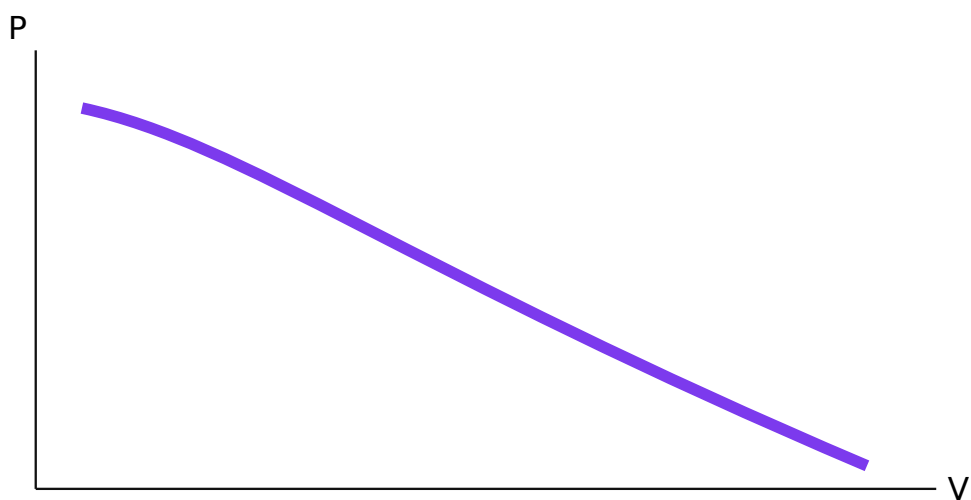


Figure 12. Adiabatic P-V graph. Labels and arrows identify the principal parts or process direction.

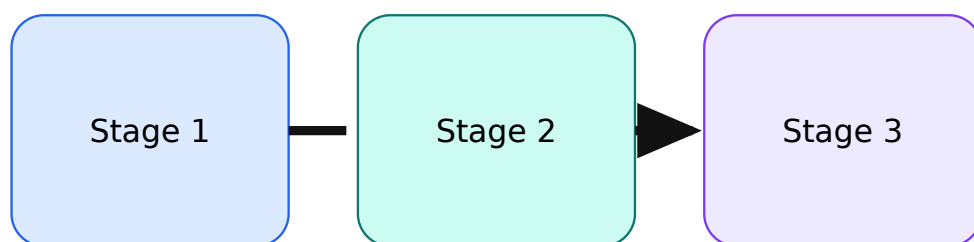


Figure 13. Endothermic energy profile. Labels and arrows identify the principal parts or process direction.

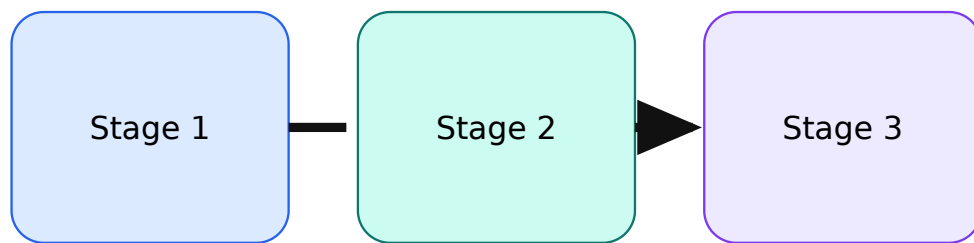


Figure 14. Exothermic energy profile. Labels and arrows identify the principal parts or process direction.

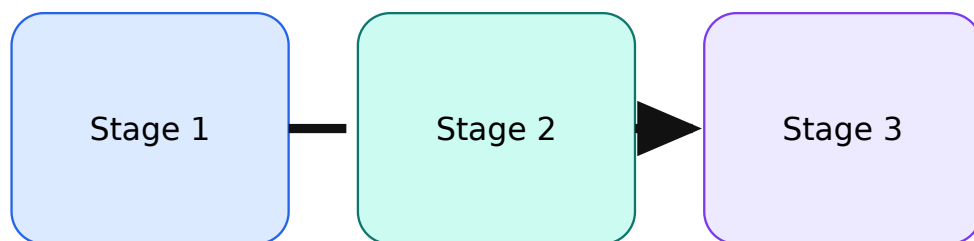


Figure 15. Standard-solution preparation. Labels and arrows identify the principal parts or process direction.

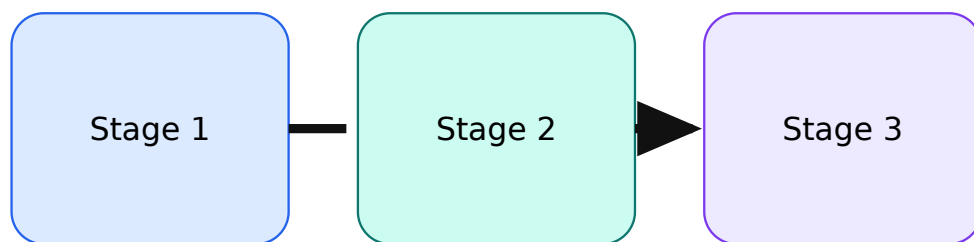


Figure 16. Volumetric flask use. Labels and arrows identify the principal parts or process direction.

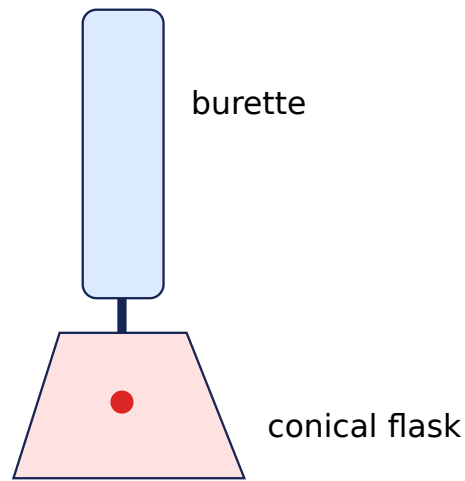


Figure 17. Titration setup. Labels and arrows identify the principal parts or process direction.

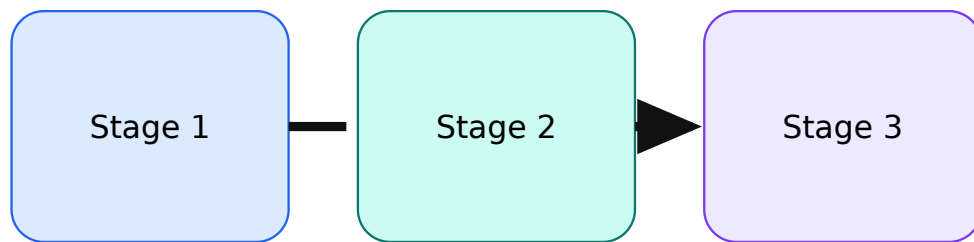


Figure 18. Indicator colour change. Labels and arrows identify the principal parts or process direction.

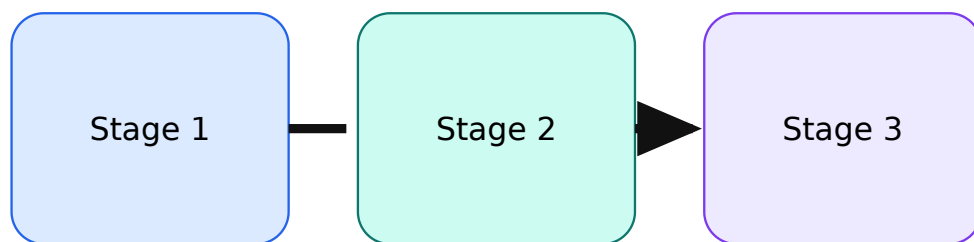


Figure 19. Water-hardness scale. Labels and arrows identify the principal parts or process direction.

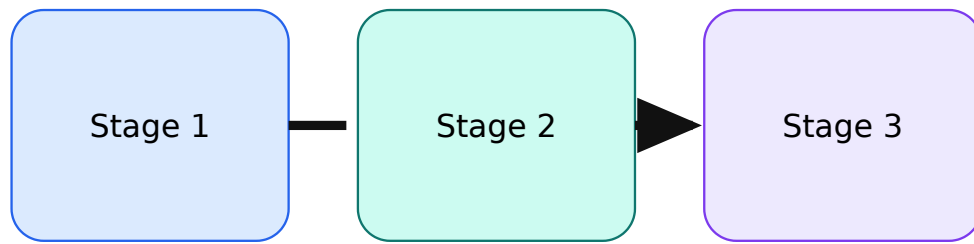


Figure 20. Boiler scale. Labels and arrows identify the principal parts or process direction.

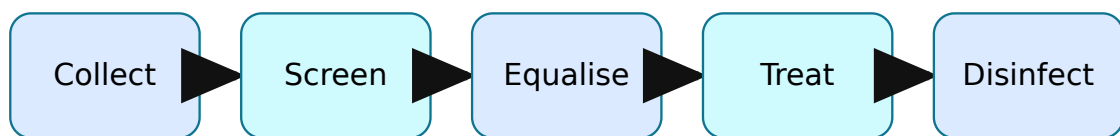


Figure 21. Effluent-treatment flow chart. Labels and arrows identify the principal parts or process direction.

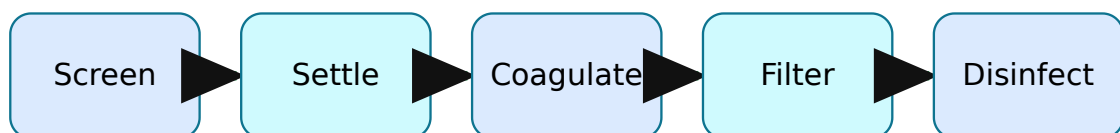


Figure 22. Municipal-water-treatment flow chart. Labels and arrows identify the principal parts or process direction.

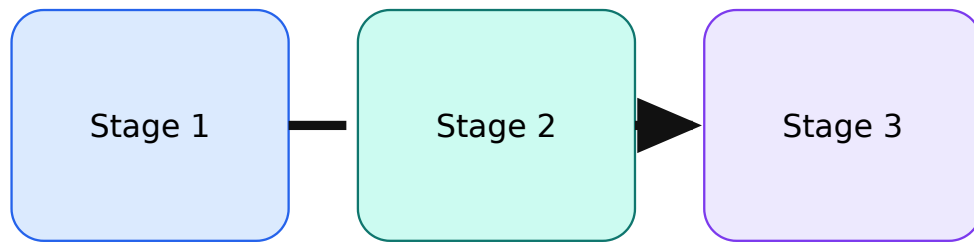


Figure 23. Chlorination. Labels and arrows identify the principal parts or process direction.

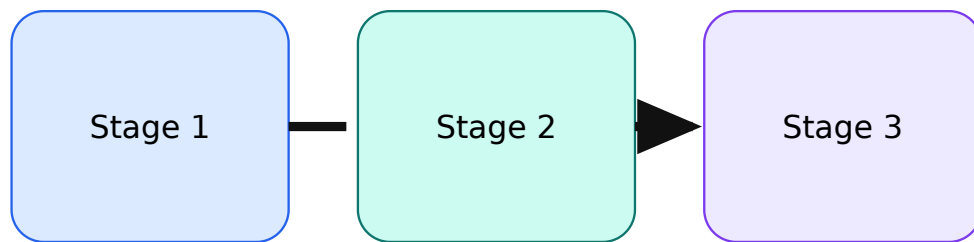


Figure 24. UV sterilisation. Labels and arrows identify the principal parts or process direction.

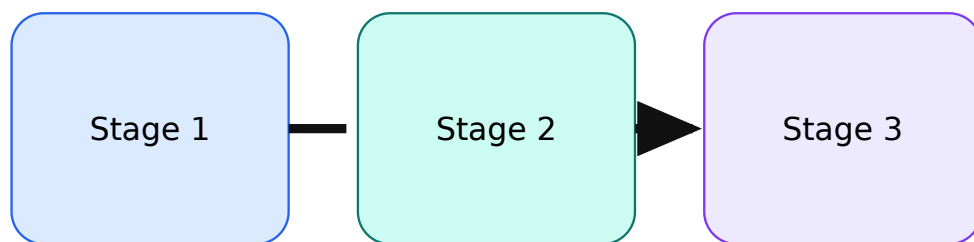


Figure 25. Ozonation. Labels and arrows identify the principal parts or process direction.

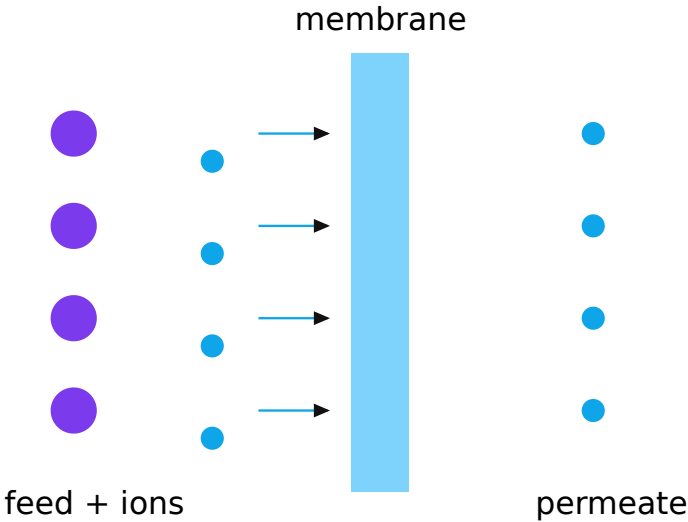


Figure 26. Reverse-osmosis membrane. Labels and arrows identify the principal parts or process direction.

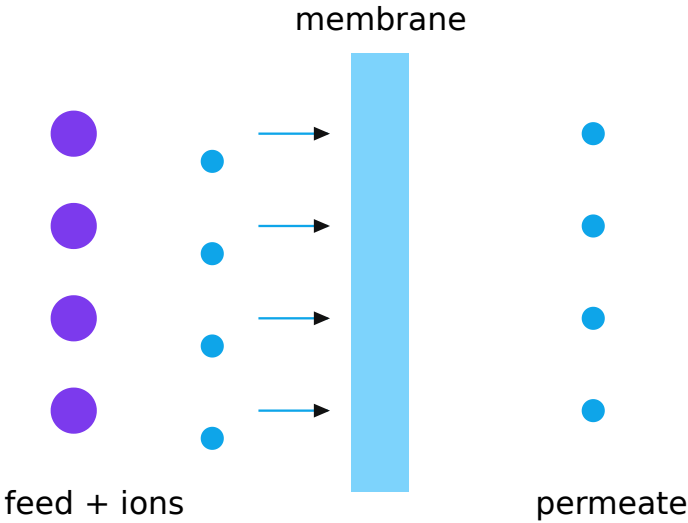


Figure 27. Nanofiltration membrane. Labels and arrows identify the principal parts or process direction.

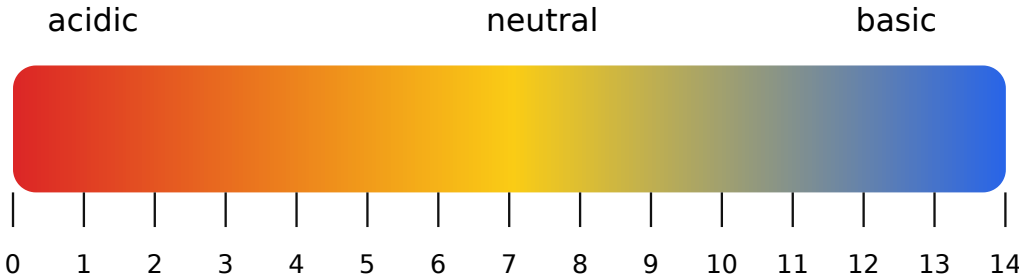


Figure 28. pH scale. Labels and arrows identify the principal parts or process direction.

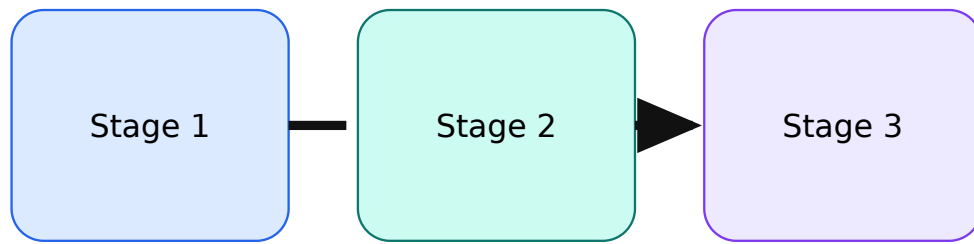


Figure 29. Metallic conduction. Labels and arrows identify the principal parts or process direction.

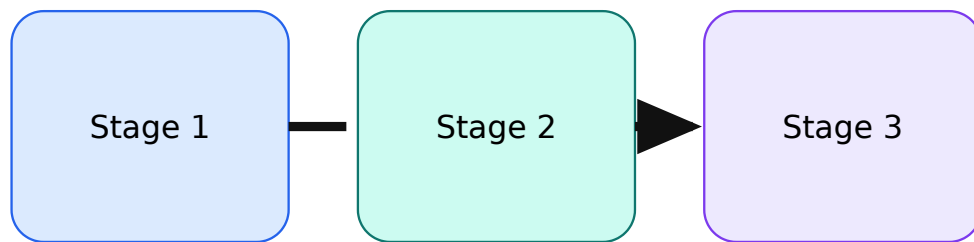


Figure 30. Electrolytic conduction. Labels and arrows identify the principal parts or process direction.

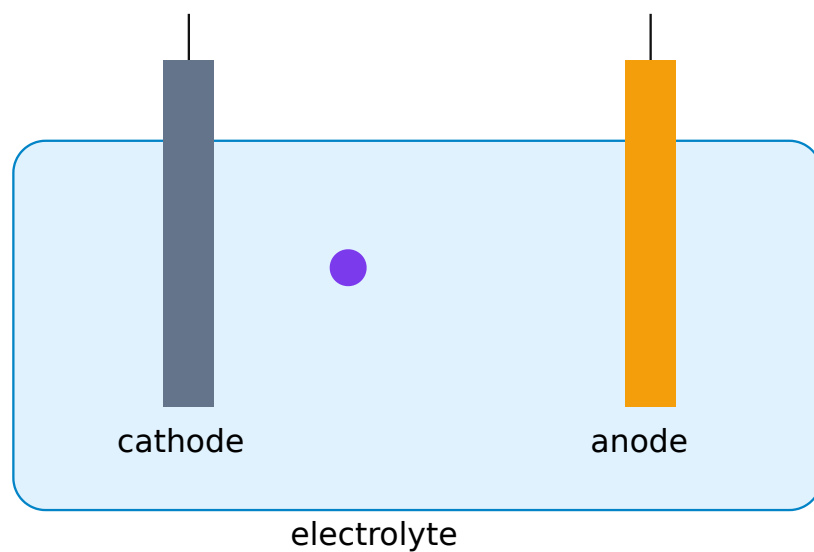


Figure 31. Electrolytic cell. Labels and arrows identify the principal parts or process direction.

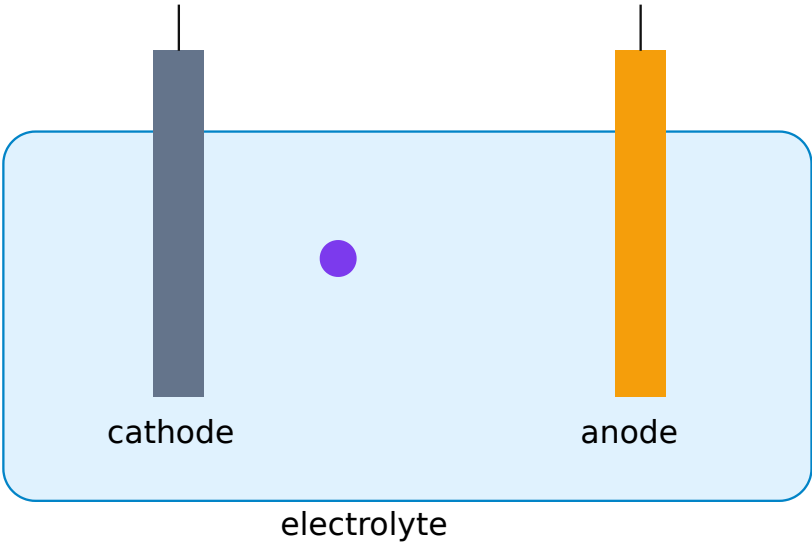


Figure 32. Electrolysis of water. Labels and arrows identify the principal parts or process direction.

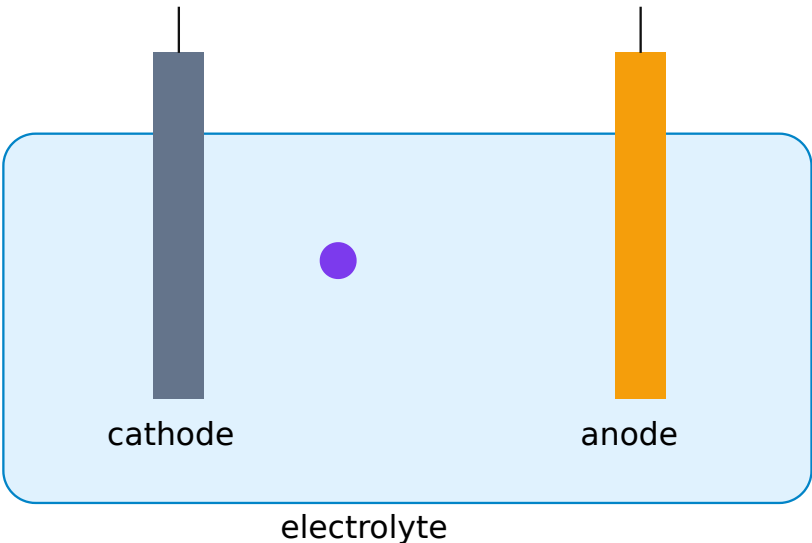


Figure 33. Nickel electroplating. Labels and arrows identify the principal parts or process direction.

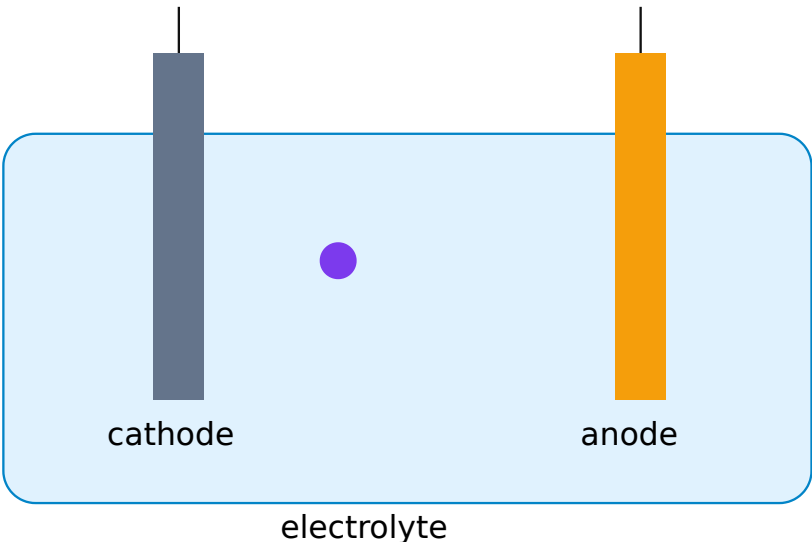


Figure 34. Electrolytic refining of copper. Labels and arrows identify the principal parts or process direction.

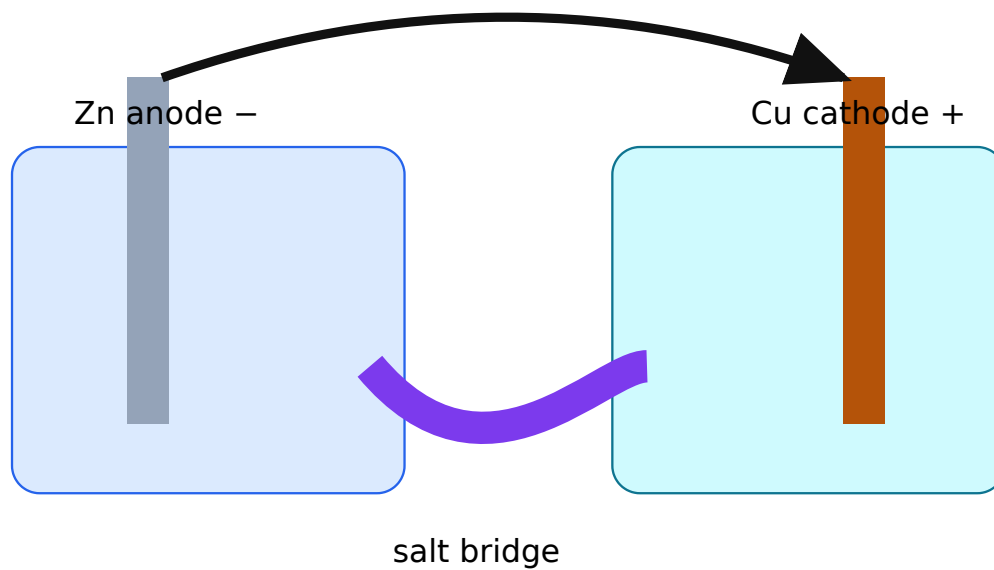


Figure 35. Daniell cell. Labels and arrows identify the principal parts or process direction.

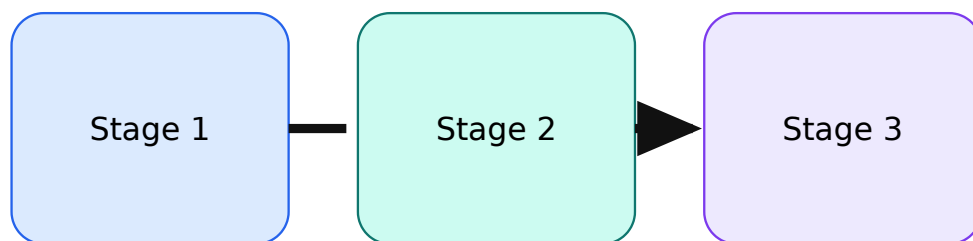


Figure 36. Salt bridge. Labels and arrows identify the principal parts or process direction.

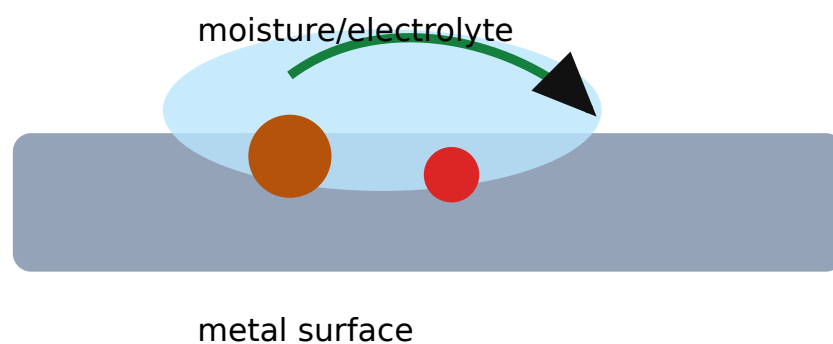


Figure 37. Corrosion cell. Labels and arrows identify the principal parts or process direction.

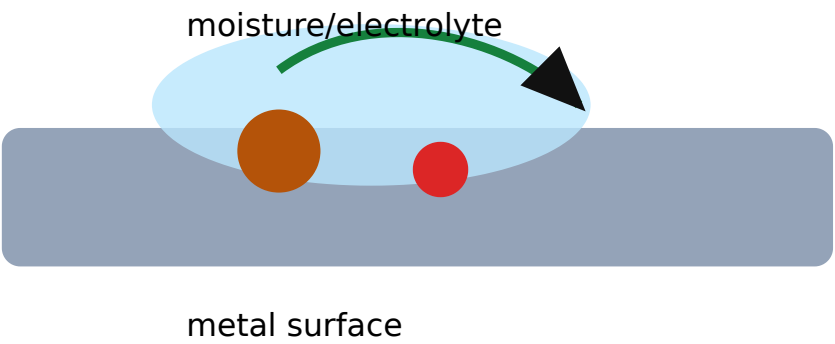


Figure 38. Sacrificial-anode protection. Labels and arrows identify the principal parts or process direction.

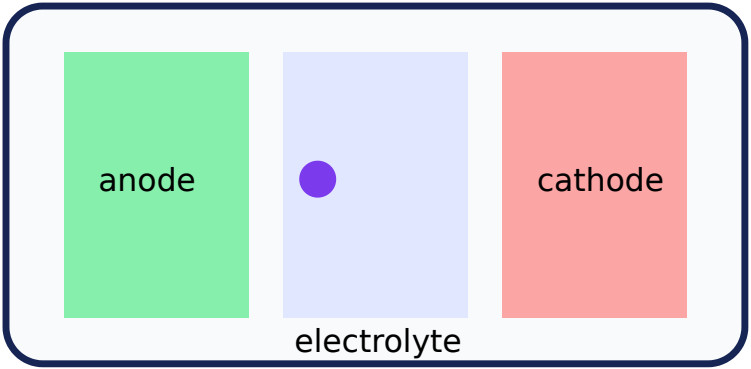


Figure 39. Lithium-ion battery. Labels and arrows identify the principal parts or process direction.

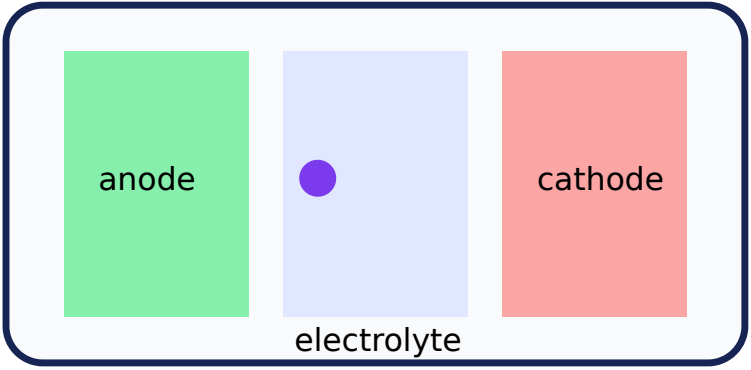


Figure 40. Sodium-ion battery. Labels and arrows identify the principal parts or process direction.

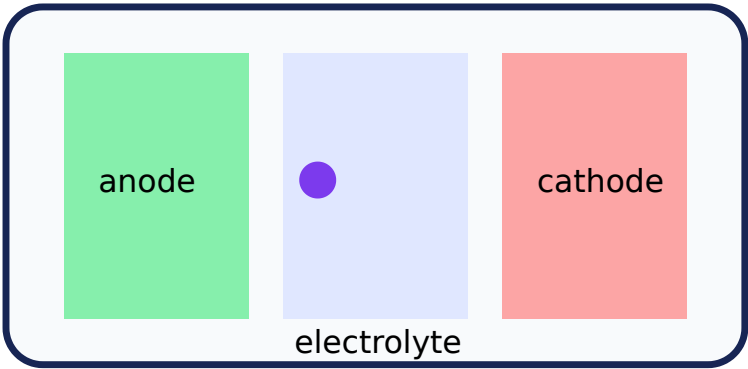


Figure 41. Solid-state battery. Labels and arrows identify the principal parts or process direction.

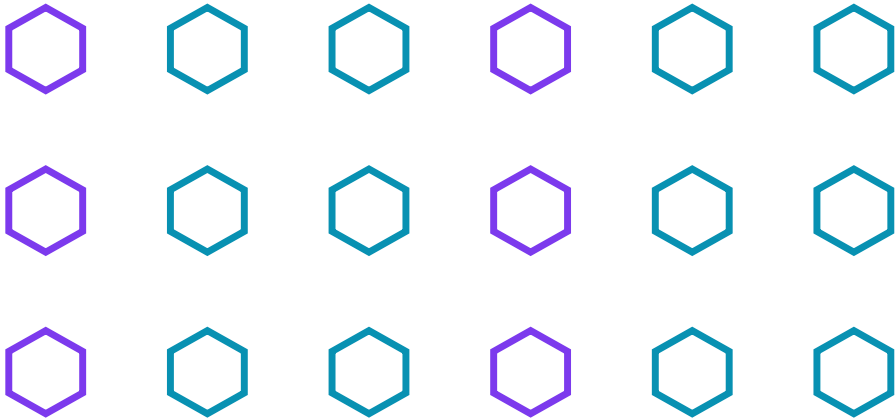


Figure 42. Alloy structure. Labels and arrows identify the principal parts or process direction.

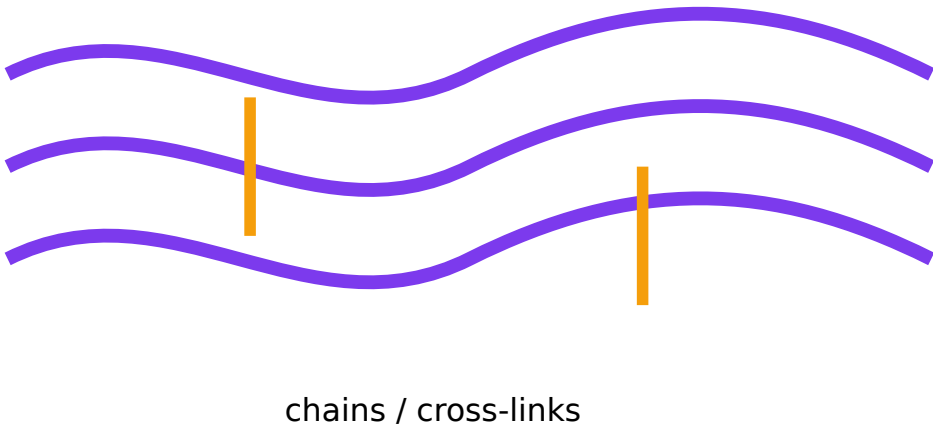


Figure 43. Polymer chains. Labels and arrows identify the principal parts or process direction.

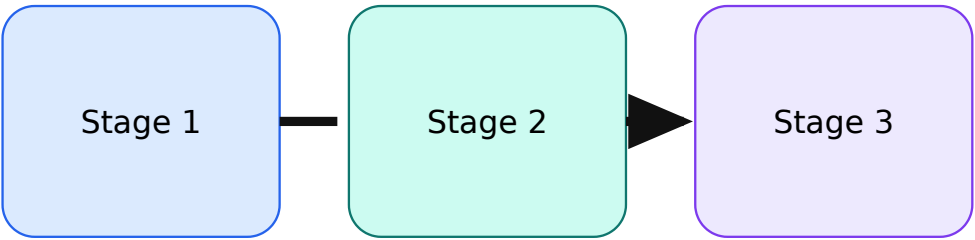


Figure 44. Thermoplastic and thermoset comparison. Labels and arrows identify the principal parts or process direction.

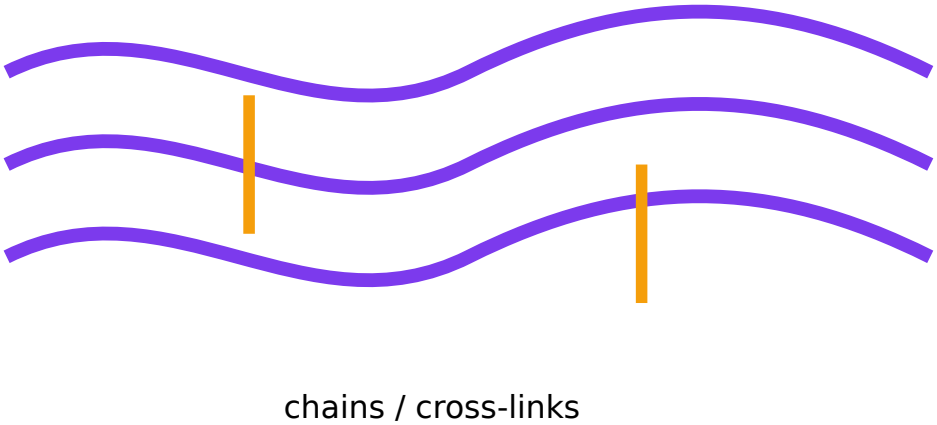


Figure 45. Rubber vulcanisation. Labels and arrows identify the principal parts or process direction.

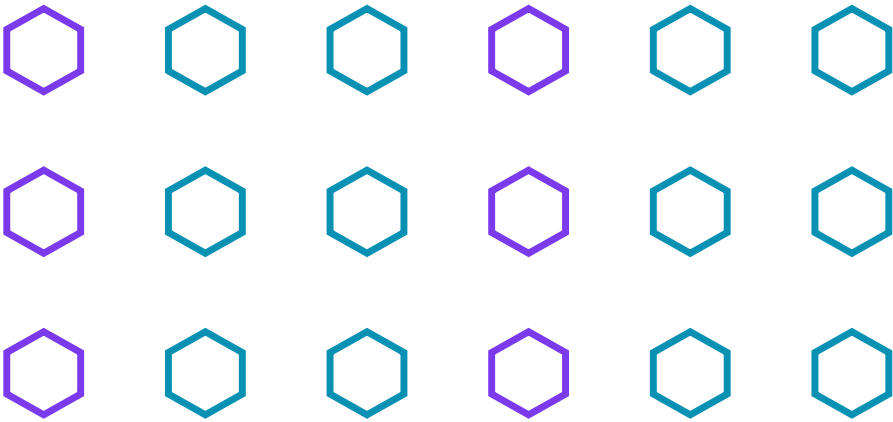


Figure 46. Carbon nanotube. Labels and arrows identify the principal parts or process direction.

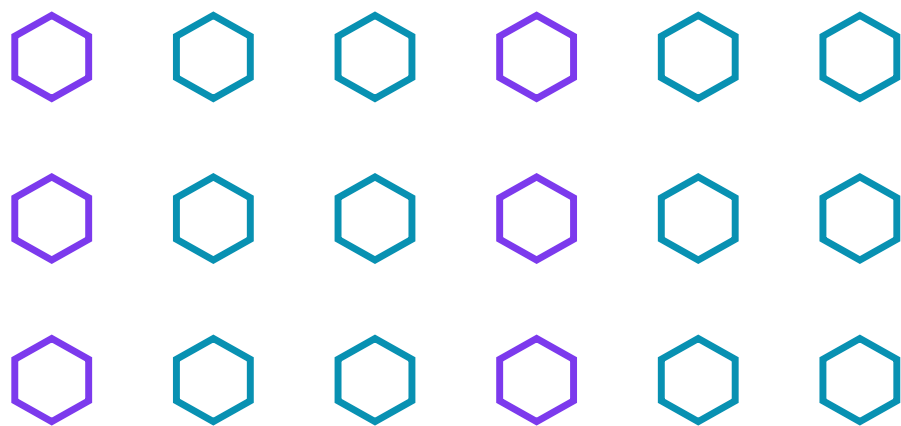


Figure 47. Fullerene. Labels and arrows identify the principal parts or process direction.

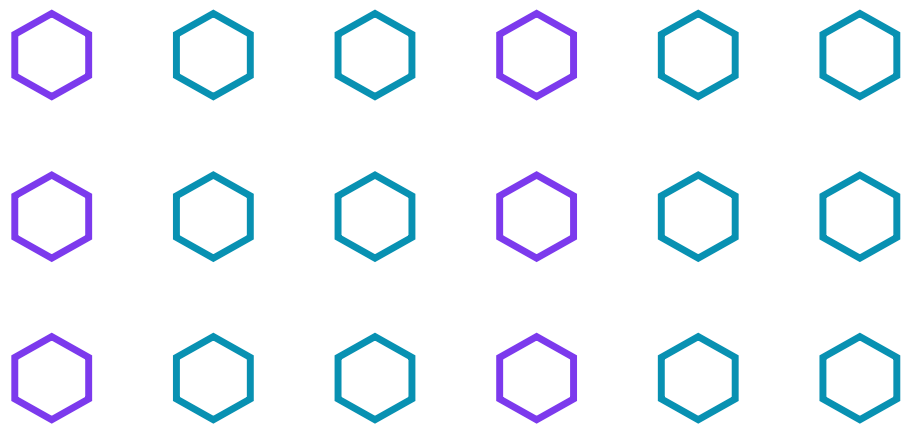


Figure 48. Graphene. Labels and arrows identify the principal parts or process direction.

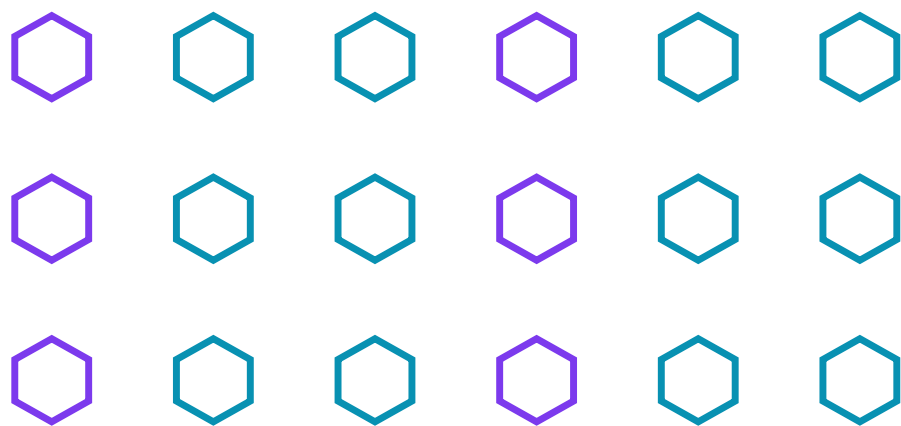


Figure 49. MOF lattice. Labels and arrows identify the principal parts or process direction.

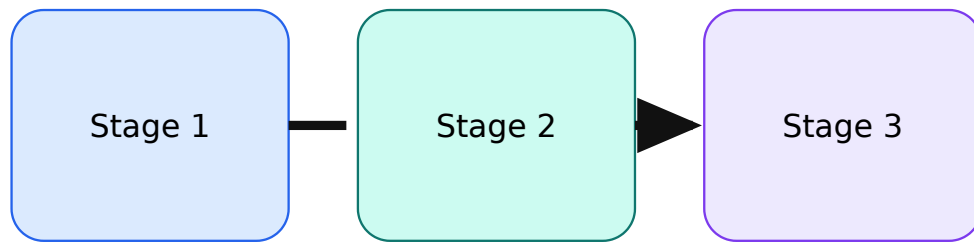


Figure 50. Biomaterial categories. Labels and arrows identify the principal parts or process direction.

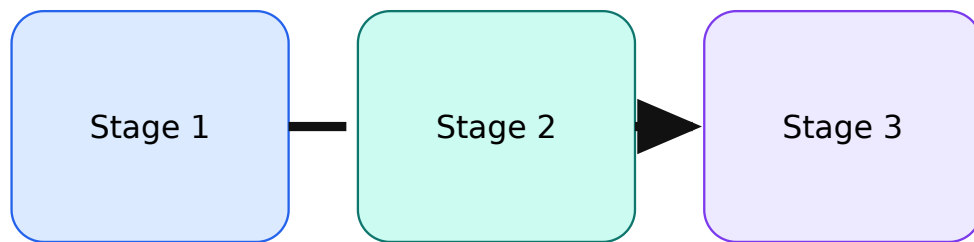


Figure 51. Biofuel cycle. Labels and arrows identify the principal parts or process direction.

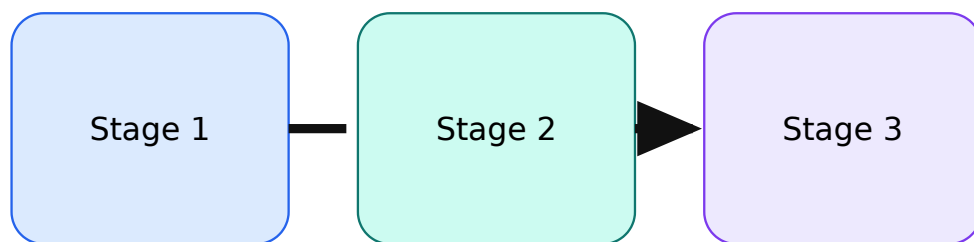
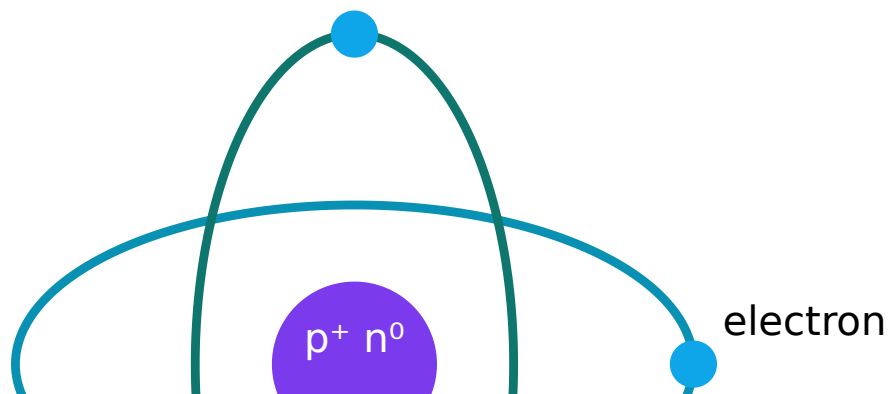


Figure 52. Green-chemistry principles. Labels and arrows identify the principal parts or process direction.

Motion that explains a process

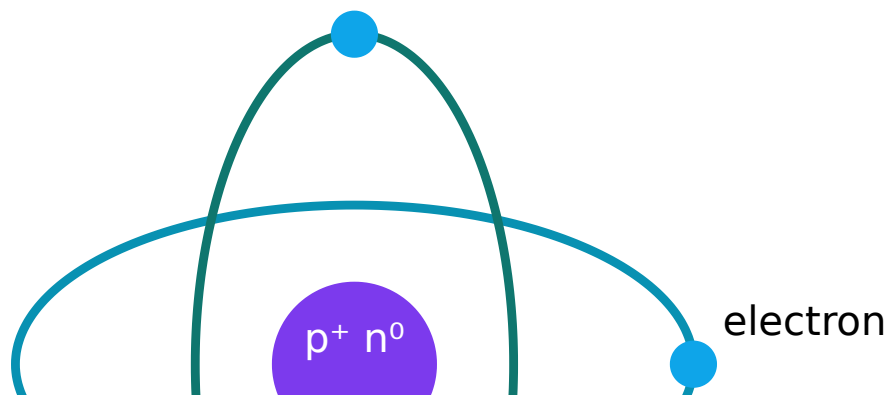
Animations start only when requested, pause outside the viewport and have static print fallbacks.

1. Atom construction



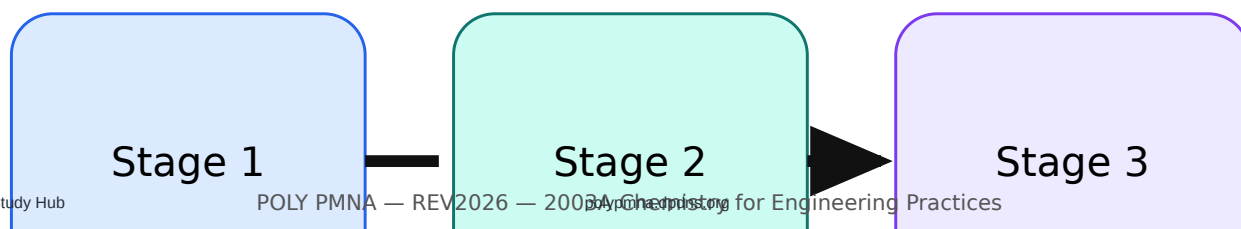
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

2. Electron-shell filling



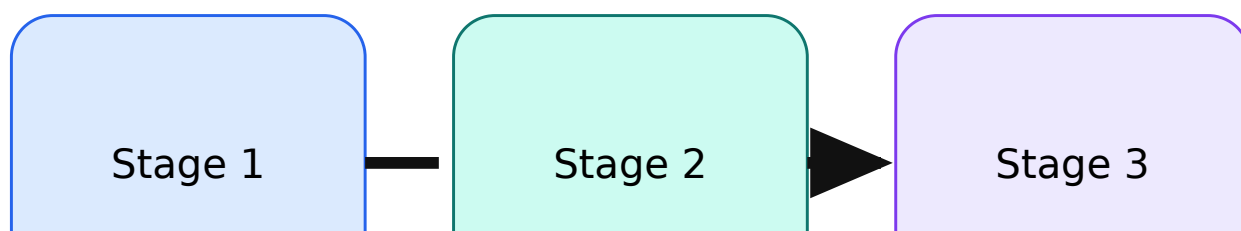
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

3. Solute dissolving



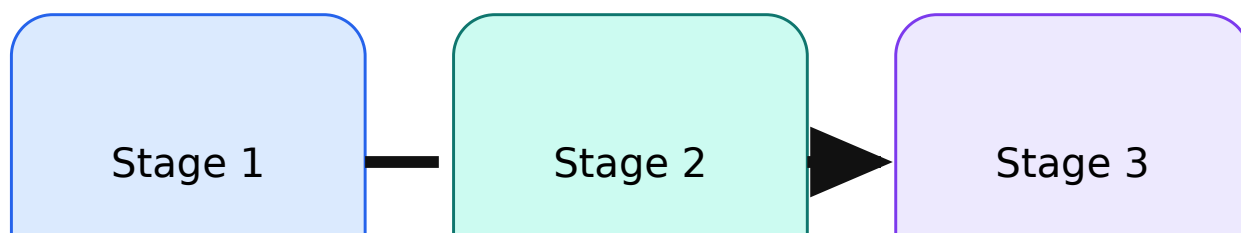
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

4. Standard-solution preparation



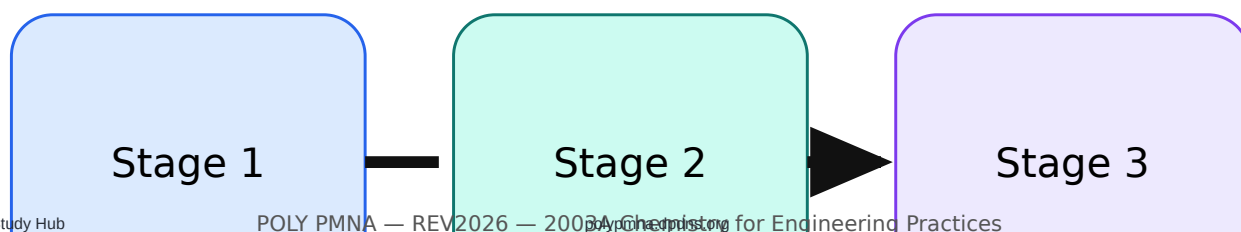
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

5. Endothermic temperature reduction



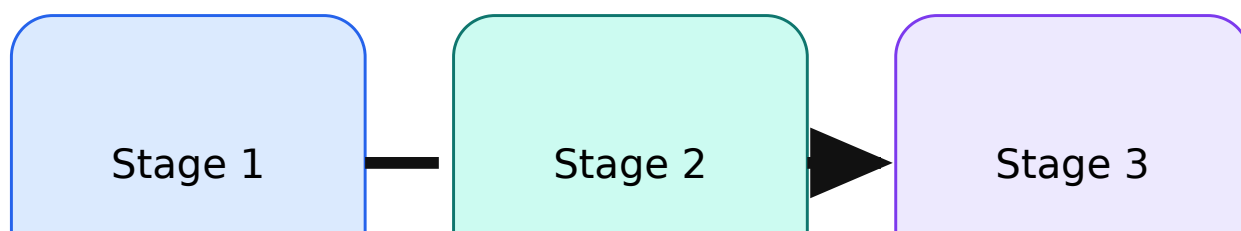
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

6. Exothermic temperature increase



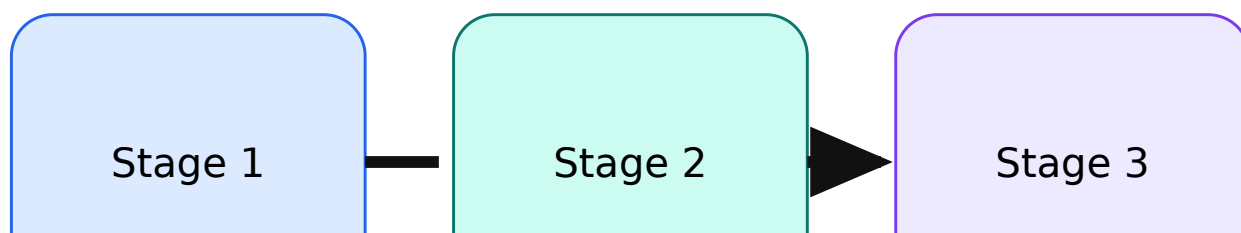
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

7. Matter and energy transfer in systems



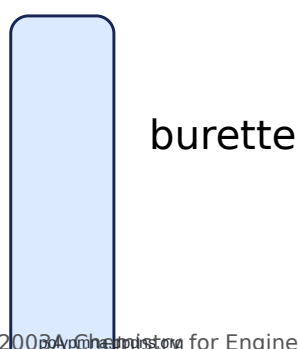
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

8. Thermodynamic-process graph



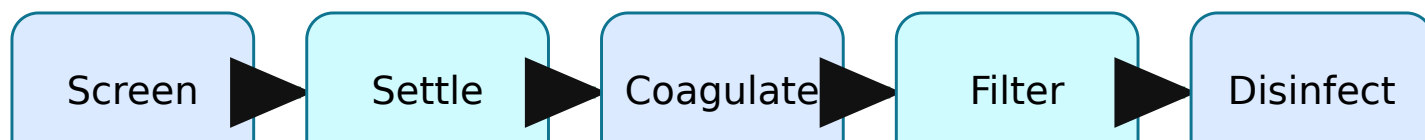
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

9. Titration and endpoint colour change



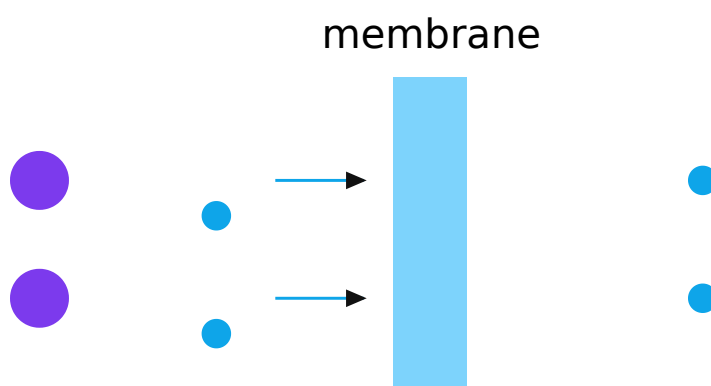
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

10. Municipal-water-treatment flow



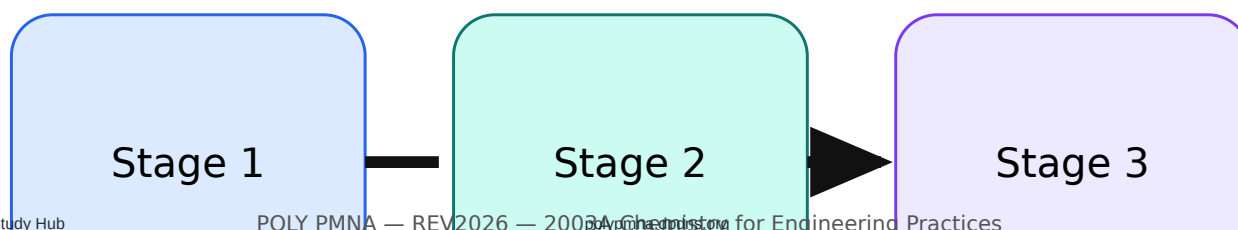
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

11. Reverse-osmosis separation



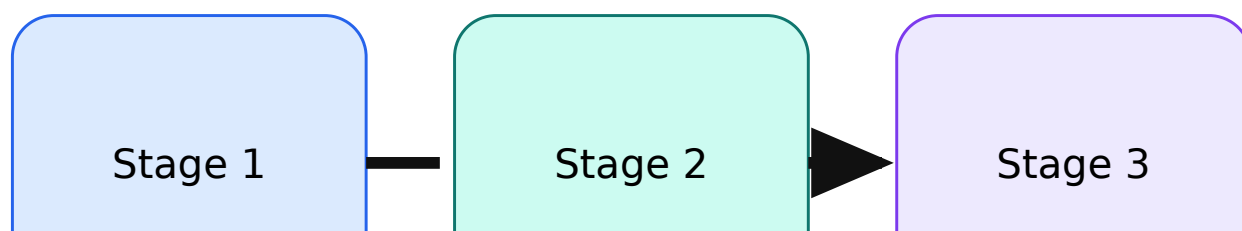
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

12. pH-scale response



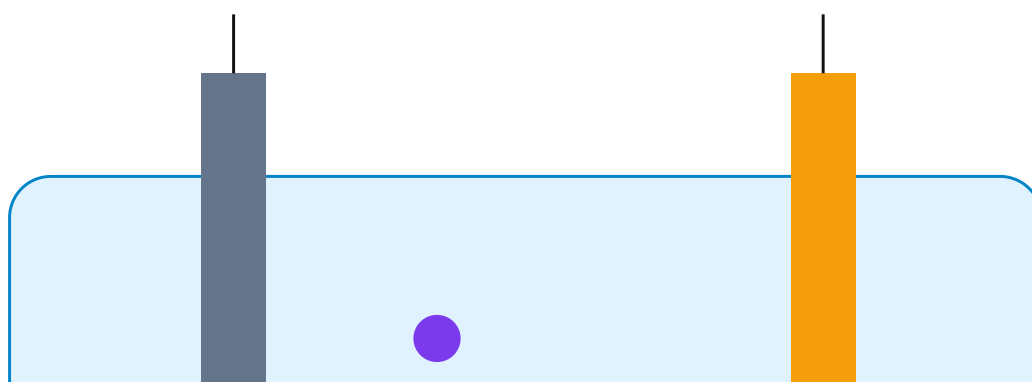
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

13. Ion movement in electrolyte



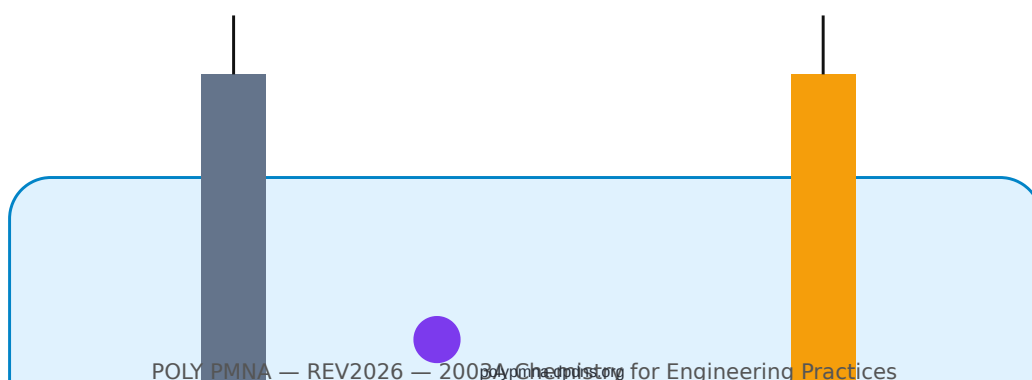
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

14. Electrolysis

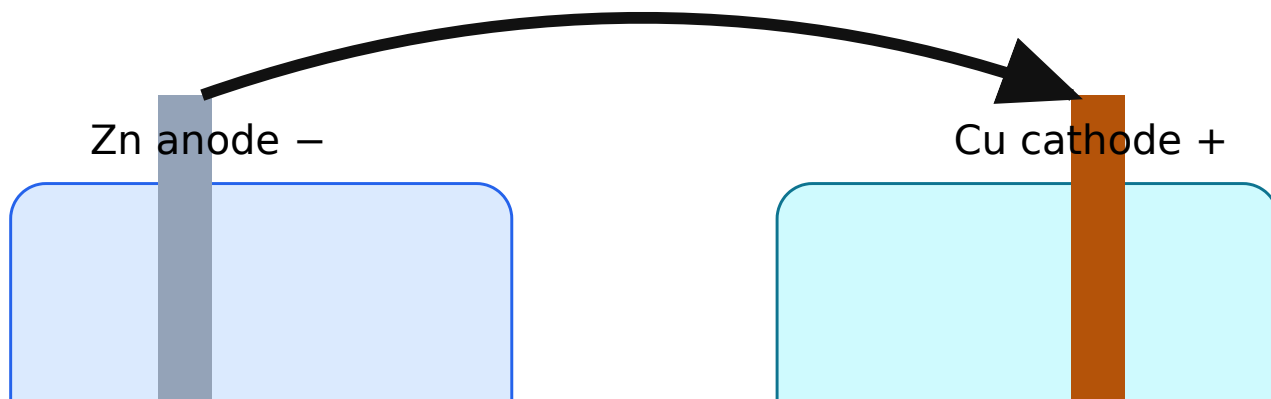


The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

15. Nickel electroplating

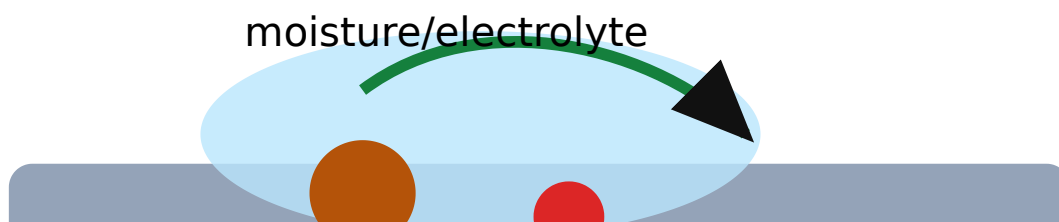


16. Daniell-cell electron flow



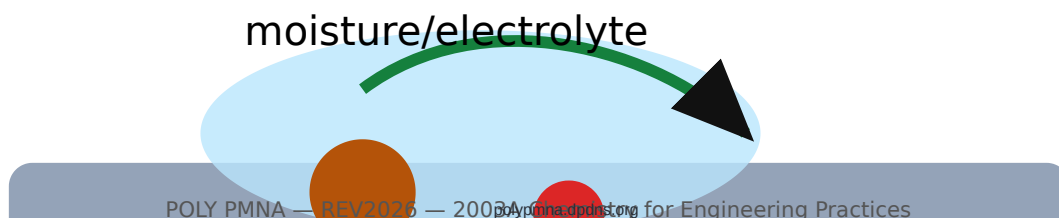
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

17. Corrosion formation



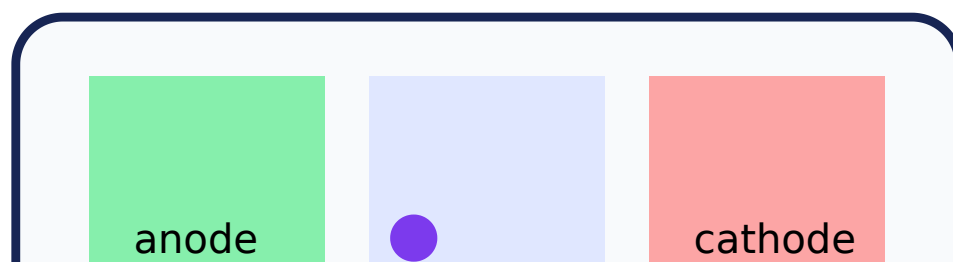
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

18. Sacrificial-anode protection



The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

19. Battery charging and discharging



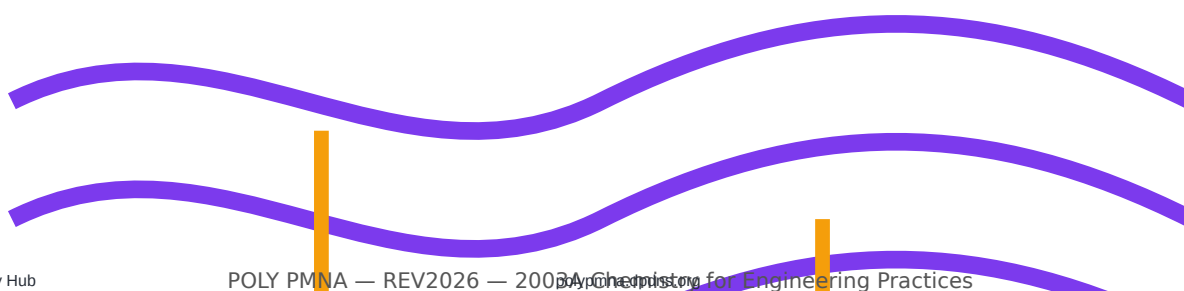
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

20. Polymerisation



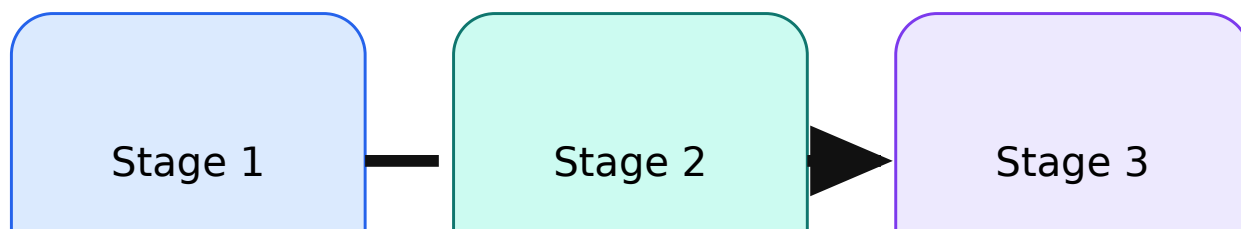
The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

21. Rubber vulcanisation



The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

22. Green-chemistry cycle



The moving marker indicates matter, ion, electron, heat or process progression. Pause and inspect the labelled static frame before proceeding.

SOLVED NUMERICAL EXAMPLES

Given → formula → substitution → verified answer

Example 1: Atomic particles

Given / required: For ^{27}Al ($Z=13$), find p , e and n .

Formula: $p=Z$; $e=Z$ for neutral atom; $n=A-Z$

Substitution and calculation: $p=13$; $e=13$; $n=27-13=14$

Final answer: 13 protons, 13 electrons, 14 neutrons

Verification: $13+14=27$ verifies mass number

Common mistake: Using 27 as electron count

Example 2: Ion particles

Given / required: For Ca^{2+} with $Z=20$ and $A=40$, find p, e, n .

Formula: $p=Z$; $e=Z-2$; $n=A-Z$

Substitution and calculation: $p=20$; $e=18$; $n=20$

Final answer: 20 p, 18 e, 20 n

Verification: Net charge +2

Common mistake: Neutral-atom electron rule applied to an ion

Example 3: Molarity

Given / required: 5.85 g NaCl (58.5 g mol^{-1}) in 500 mL.

Formula: $M=(m/M_r)/V(L)$

Substitution and calculation: $M=(5.85/58.5)/0.500=0.200$

Final answer: 0.200 mol L^{-1}

Verification: $0.2 \text{ mol/L} \times 0.5 \text{ L}=0.1 \text{ mol}=5.85 \text{ g}$

Common mistake: Using 500 as volume

Example 4: Required molar mass

Given / required: Prepare 250 mL 0.1 M NaCl.

Formula: $m=M M_r V$

Substitution and calculation: $m=0.1 \times 58.5 \times 0.250=1.4625$

Final answer: 1.46 g NaCl

Verification: Back-calculation gives 0.025 mol

Common mistake: Not converting 250 mL

Example 5: Normality

Given / required: 6.3 g oxalic acid dihydrate, Eq mass 63, in 1 L.

Formula: $N = (m/E)/V$

Substitution and calculation: $N = (6.3/63)/1 = 0.100$

Final answer: 0.100 N

Verification: 0.1 equivalent in 1 L

Common mistake: Using molar mass 126 as equivalent mass

Example 6: Required normal mass

Given / required: Prepare 250 mL 0.1 N Na_2CO_3 , Eq mass 53.

Formula: $m = N E V$

Substitution and calculation: $m = 0.1 \times 53 \times 0.250 = 1.325$

Final answer: 1.325 g

Verification: Expected value matches standard preparation record

Common mistake: Using 106 directly

Example 7: ppm mass basis

Given / required: 3 mg solute in 1.5 kg solution.

Formula: $\text{ppm} = \text{mass fraction} \times 10^6$

Substitution and calculation: $3 \text{ mg} / (1.5 \times 10^6 \text{ mg}) \times 10^6 = 2$

Final answer: 2 ppm

Verification: $2 \text{ mg/kg} \times 1.5 \text{ kg} = 3 \text{ mg}$

Common mistake: Using 1.5 as mg

Example 8: Dilution

Given / required: Volume of 1.0 M stock needed for 250 mL 0.10 M.

Formula: $C_1V_1=C_2V_2$

Substitution and calculation: $V_1=0.10\times 250/1.0=25$

Final answer: 25 mL stock, dilute to 250 mL

Verification: Moles before= $1\times 0.025=0.025$; after= $.1\times .25=.025$

Common mistake: Adding 250 mL water instead of making total volume 250 mL

Example 9: Titration

Given / required: 25.0 mL 0.100 N Na_2CO_3 requires 20.0 mL HCl.

Formula: $V_1N_1=V_2N_2$

Substitution and calculation: $N(\text{HCl})=25\times 0.100/20=0.125$

Final answer: 0.125 N HCl

Verification: Equivalent amount both sides=2.5 mEq

Common mistake: Reversing volumes

Example 10: pH

Given / required: Find pH for $[\text{H}^+]=2.0\times 10^{-3}$ M.

Formula: $\text{pH}=-\log[\text{H}^+]$

Substitution and calculation: $\text{pH}=-\log(2.0\times 10^{-3})=2.699$

Final answer: $\text{pH}\approx 2.70$

Verification: $10^{-2.7}\approx 2.0\times 10^{-3}$

Common mistake: Reporting pH 3 exactly

Example 11: pOH

Given / required: $[\text{OH}^-] = 1.0 \times 10^{-5} \text{ M}$.

Formula: $\text{pOH} = -\log[\text{OH}^-]$

Substitution and calculation: $\text{pOH} = 5.00$; $\text{pH} = 14 - 5 = 9.00$

Final answer: pOH 5.00, pH 9.00

Verification: Basic result is consistent

Common mistake: Calling solution acidic because $\text{pOH} < 7$

Example 12: Hydrogen ion

Given / required: Find $[\text{H}^+]$ at pH 4.30.

Formula: $[\text{H}^+] = 10^{-\text{pH}}$

Substitution and calculation: $10^{-4.30} = 5.01 \times 10^{-5}$

Final answer: $5.0 \times 10^{-5} \text{ mol L}^{-1}$

Verification: $-\log$ value returns 4.30

Common mistake: Using 4.30 mol/L

Example 13: Charge

Given / required: Current 1.5 A for 8 min.

Formula: $Q = It$

Substitution and calculation: $t = 480 \text{ s}$; $Q = 1.5 \times 480 = 720$

Final answer: 720 C

Verification: $720 \text{ C} / 480 \text{ s} = 1.5 \text{ A}$

Common mistake: Keeping time in minutes

Example 14: Metal deposition

Given / required: $Z=0.000329 \text{ g C}^{-1}$, $I=2 \text{ A}$, $t=10 \text{ min}$.

Formula: $m=ZIt$

Substitution and calculation: $t=600 \text{ s}$; $m=.000329 \times 2 \times 600=.3948$

Final answer: 0.395 g

Verification: $Q=1200 \text{ C}$ and $m/Z=1200 \text{ C}$

Common mistake: Using 10 s

Example 15: Electrolysis time

Given / required: Deposit 1.00 g when $Z=.000329 \text{ g/C}$ and $I=2 \text{ A}$.

Formula: $t=m/(ZI)$

Substitution and calculation: $t=1/(.000329 \times 2)=1519.8 \text{ s}$

Final answer: $1.52 \times 10^3 \text{ s} \approx 25.3 \text{ min}$

Verification: $ZIt \approx 1.00 \text{ g}$

Common mistake: Not converting final seconds

Example 16: Faraday second law

Given / required: Equal charge deposits Ag and Cu. $E_{\text{Ag}}=107.9$, $E_{\text{Cu}}=31.75$. If $\text{Cu}=0.500 \text{ g}$, Ag mass?

Formula: $m_{\text{Ag}}/m_{\text{Cu}}=E_{\text{Ag}}/E_{\text{Cu}}$

Substitution and calculation: $m_{\text{Ag}}=.500 \times 107.9/31.75=1.699$

Final answer: 1.70 g Ag

Verification: Mass ratio 3.40 matches equivalent ratio

Common mistake: Using atomic mass of Cu without valency

Example 17: Hardness

Given / required: 25.0 mL sample uses 12.5 mL 0.0100 M EDTA.

Formula: $\text{Hardness} = V \times M \times 100000 / \text{sample mL}$

Substitution and calculation: $12.5 \times .0100 \times 100000 / 25 = 500$

Final answer: 500 mg L⁻¹ as CaCO₃

Verification: Moles EDTA = .000125; CaCO₃ equivalent mass = .0125 g in 25 mL

Common mistake: Missing 100000 conversion factor

Example 18: System identification

Given / required: Steam enters and leaves a turbine while shaft work and heat cross boundary.

Formula: Check matter and energy transfer

Substitution and calculation: Matter and energy both cross

Final answer: Open system

Verification: Steady flow still permits matter transfer

Common mistake: Calling it closed because casing is closed

Example 19: Process identification

Given / required: Gas is heated in a rigid sealed vessel.

Formula: Rigid means V constant

Substitution and calculation: Volume does not change

Final answer: Isochoric process

Verification: Boundary work $P\Delta V$ is zero

Common mistake: Calling it isobaric

Example 20: Material selection

Given / required: Choose transparent impact-resistant guard: PMMA or PC.

Formula: Compare impact, clarity, temperature and cost

Substitution and calculation: PC offers much higher impact resistance; PMMA offers better surface hardness/optical weathering

Final answer: Select PC where impact safety dominates

Verification: Decision states trade-off rather than universal winner

Common mistake: Selecting only by transparency

PRACTICE QUESTIONS

Module-wise sample-question bank with answers

PRACTICE QUESTIONS. These are internally prepared learning questions, not official SITTR questions.

Module I — CO1

20 one-mark questions

M1 CO1 1 mark **1. Define or identify: Atoms and molecules.**

Model-answer points: A correct one-sentence definition or identification with essential condition.

M1 CO1 1 mark **2. Define or identify: Nucleus: compact centre containing protons and neutrons.**

Model-answer points: A correct one-sentence definition or identification with essential condition.

M1 CO1 1 mark **3. Define or identify: Electron: negatively charged particle occupying shells around the nucleus.**

Model-answer points: A correct one-sentence definition or identification with essential condition.

M1 CO1 1 mark **4. Define or identify: Solutions and standard solutions.**

Model-answer points: A correct one-sentence definition or identification with essential condition.

M1 CO1 1 mark **5. Define or identify: Aqueous solution: water is the solvent.**

Model-answer points: A correct one-sentence definition or identification with essential condition.

6. Define or identify: Concentrated and dilute are comparative, not exact, terms.

Model-answer points: A correct one-sentence definition or identification with essential condition.

7. Define or identify: Molarity and dilution.

Model-answer points: A correct one-sentence definition or identification with essential condition.

8. Define or identify: Mole links particle count to measurable mass.

Model-answer points: A correct one-sentence definition or identification with essential condition.

9. Define or identify: Molar mass is expressed in g mol⁻¹.

Model-answer points: A correct one-sentence definition or identification with essential condition.

10. Define or identify: Normality and equivalent mass.

Model-answer points: A correct one-sentence definition or identification with essential condition.

11. Define or identify: Normality unit: eq L⁻¹ or N.

Model-answer points: A correct one-sentence definition or identification with essential condition.

12. Define or identify: Oxalic acid dihydrate has equivalent mass 63 g eq⁻¹ in acid-base reaction.

Model-answer points: A correct one-sentence definition or identification with essential condition.

13. Define or identify: Parts per million.

Model-answer points: A correct one-sentence definition or identification with essential condition.

14. Define or identify: Exact mass form: mass solute/mass solution × 10⁶.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15. Define or identify: Useful for pollutants, dissolved ions and water quality.

Model-answer points: A correct one-sentence definition or identification with essential condition.

16. Define or identify: Thermodynamic system and surroundings.

Model-answer points: A correct one-sentence definition or identification with essential condition.

17. Define or identify: Open system exchanges matter and energy.

Model-answer points: A correct one-sentence definition or identification with essential condition.

18. Define or identify: Closed system exchanges energy but not matter.

Model-answer points: A correct one-sentence definition or identification with essential condition.

19. Define or identify: Thermodynamic processes.

Model-answer points: A correct one-sentence definition or identification with essential condition.

20. Define or identify: Isothermal: temperature constant.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15 three-mark questions**1. Explain Atoms and molecules using three technically distinct points and one example.**

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

2. Explain Solutions and standard solutions using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

3. Explain Molarity and dilution using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

4. Explain Normality and equivalent mass using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

5. Explain Parts per million using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

6. Explain Thermodynamic system and surroundings using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

7. Explain Thermodynamic processes using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

8. Explain Internal energy, enthalpy and entropy using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

9. Explain Atoms and molecules using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

10. Explain Solutions and standard solutions using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

11. Explain Molarity and dilution using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

12. Explain Normality and equivalent mass using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

13. Explain Parts per million using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

14. Explain Thermodynamic system and surroundings using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

15. Explain Thermodynamic processes using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

10 seven-mark questions

1. Develop a structured engineering note on Atoms and molecules with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

2. Develop a structured engineering note on Solutions and standard solutions with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

3. Develop a structured engineering note on Molarity and dilution with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

4. Develop a structured engineering note on Normality and equivalent mass with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

5. Develop a structured engineering note on Parts per million with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

6. Develop a structured engineering note on Thermodynamic system and surroundings with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

7. Develop a structured engineering note on Thermodynamic processes with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

8. Develop a structured engineering note on Internal energy, enthalpy and entropy with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

9. Develop a structured engineering note on Atoms and molecules with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

10. Develop a structured engineering note on Solutions and standard solutions with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

15 multiple-choice questions

M1

CO1

1 mark

1. Which option is most directly associated with Atoms and molecules?

- A. Atoms and molecules
- B. Solutions and standard solutions
- C. Molarity and dilution
- D. Normality and equivalent mass

Model-answer points: Answer: A. Atoms and molecules.

M1

CO1

1 mark

2. Which option is most directly associated with Nucleus: compact centre containing protons and neutrons?

- A. Nucleus: compact centre containing protons and neutrons
- B. Aqueous solution: water is the solvent
- C. Mole links particle count to measurable mass
- D. Normality unit: eq L⁻¹ or N

Model-answer points: Answer: A. Nucleus: compact centre containing protons and neutrons.

M1

CO1

1 mark

3. Which option is most directly associated with Electron: negatively charged particle occupying shells around the nucleus?

- A. Electron: negatively charged particle occupying shells around the nucleus
- B. Concentrated and dilute are comparative, not exact, terms
- C. Molar mass is expressed in g mol⁻¹
- D. Oxalic acid dihydrate has equivalent mass 63 g eq⁻¹ in acid-base reaction

Model-answer points: Answer: A. Electron: negatively charged particle occupying shells around the nucleus.

M1

CO1

1 mark

4. Which option is most directly associated with Solutions and standard solutions?

- A. Solutions and standard solutions
- B. Molarity and dilution
- C. Normality and equivalent mass
- D. Parts per million

Model-answer points: Answer: A. Solutions and standard solutions.

5. Which option is most directly associated with Aqueous solution: water is the solvent?

- A. Aqueous solution: water is the solvent**
- B. Mole links particle count to measurable mass**
- C. Normality unit: eq L⁻¹ or N**
- D. Exact mass form: mass solute/mass solution × 10⁶**

Model-answer points: Answer: A. Aqueous solution: water is the solvent.

6. Which option is most directly associated with Concentrated and dilute are comparative, not exact, terms?

- A. Concentrated and dilute are comparative, not exact, terms**
- B. Molar mass is expressed in g mol⁻¹**
- C. Oxalic acid dihydrate has equivalent mass 63 g eq⁻¹ in acid-base reaction**
- D. Useful for pollutants, dissolved ions and water quality**

Model-answer points: Answer: A. Concentrated and dilute are comparative, not exact, terms.

7. Which option is most directly associated with Molarity and dilution?

- A. Molarity and dilution**
- B. Normality and equivalent mass**
- C. Parts per million**
- D. Thermodynamic system and surroundings**

Model-answer points: Answer: A. Molarity and dilution.

8. Which option is most directly associated with Mole links particle count to measurable mass?

- A. Mole links particle count to measurable mass**
- B. Normality unit: eq L⁻¹ or N**
- C. Exact mass form: mass solute/mass solution × 10⁶**
- D. Open system exchanges matter and energy**

Model-answer points: Answer: A. Mole links particle count to measurable mass.

9. Which option is most directly associated with Molar mass is expressed in g mol⁻¹?

- A. Molar mass is expressed in g mol⁻¹**
- B. Oxalic acid dihydrate has equivalent mass 63 g eq⁻¹ in acid-base reaction**
- C. Useful for pollutants, dissolved ions and water quality**
- D. Closed system exchanges energy but not matter**

Model-answer points: Answer: A. Molar mass is expressed in g mol⁻¹.

10. Which option is most directly associated with Normality and equivalent mass?

- A. Normality and equivalent mass
- B. Parts per million
- C. Thermodynamic system and surroundings
- D. Thermodynamic processes

Model-answer points: Answer: A. Normality and equivalent mass.

11. Which option is most directly associated with Normality unit: eq L⁻¹ or N?

- A. Normality unit: eq L⁻¹ or N
- B. Exact mass form: mass solute/mass solution × 10⁶
- C. Open system exchanges matter and energy
- D. Isothermal: temperature constant

Model-answer points: Answer: A. Normality unit: eq L⁻¹ or N.

12. Which option is most directly associated with Oxalic acid dihydrate has equivalent mass 63 g eq⁻¹ in acid-base reaction?

- A. Oxalic acid dihydrate has equivalent mass 63 g eq⁻¹ in acid-base reaction
- B. Useful for pollutants, dissolved ions and water quality
- C. Closed system exchanges energy but not matter
- D. Isobaric: pressure constant

Model-answer points: Answer: A. Oxalic acid dihydrate has equivalent mass 63 g eq⁻¹ in acid-base reaction.

13. Which option is most directly associated with Parts per million?

- A. Parts per million
- B. Thermodynamic system and surroundings
- C. Thermodynamic processes
- D. Internal energy, enthalpy and entropy

Model-answer points: Answer: A. Parts per million.

14. Which option is most directly associated with Exact mass form: mass solute/mass solution × 10⁶?

- A. Exact mass form: mass solute/mass solution × 10⁶
- B. Open system exchanges matter and energy
- C. Isothermal: temperature constant
- D. Internal energy is a state function

Model-answer points: Answer: A. Exact mass form: mass solute/mass solution × 10⁶.

15. Which option is most directly associated with Useful for pollutants, dissolved ions and water quality?

- A. Useful for pollutants, dissolved ions and water quality
- B. Closed system exchanges energy but not matter
- C. Isobaric: pressure constant
- D. Enthalpy change is useful for reactions at constant pressure

Model-answer points: Answer: A. Useful for pollutants, dissolved ions and water quality.

10 fill-in-the-blank questions

1. The handbook topic that includes “Atoms and molecules” is _____.

Model-answer points: Atoms and molecules

2. The handbook topic that includes “Nucleus: compact centre containing protons and neutrons” is _____.

Model-answer points: Solutions and standard solutions

3. The handbook topic that includes “Electron: negatively charged particle occupying shells around the nucleus” is _____.

Model-answer points: Molarity and dilution

4. The handbook topic that includes “Solutions and standard solutions” is _____.

Model-answer points: Normality and equivalent mass

5. The handbook topic that includes “Aqueous solution: water is the solvent” is _____.

Model-answer points: Parts per million

6. The handbook topic that includes “Concentrated and dilute are comparative, not exact, terms” is _____.

Model-answer points: Thermodynamic system and surroundings

7. The handbook topic that includes “Molarity and dilution” is _____.

Model-answer points: Thermodynamic processes

8. The handbook topic that includes “Mole links particle count to measurable mass” is _____.

Model-answer points: Internal energy, enthalpy and entropy

9. The handbook topic that includes “Molar mass is expressed in g mol^{-1} ” is _____.

Model-answer points: Atoms and molecules

10. The handbook topic that includes “Normality and equivalent mass” is _____.

Model-answer points: Solutions and standard solutions

10 true-or-false questions

1. Atoms and molecules is included in Module 1.

Model-answer points: True.

2. All conditions and units may be ignored when applying Solutions and standard solutions.

Model-answer points: False. Conditions, units and assumptions are essential.

3. Electron: negatively charged particle occupying shells around the nucleus is included in Module 1.

Model-answer points: True.

4. All conditions and units may be ignored when applying Normality and equivalent mass.

Model-answer points: False. Conditions, units and assumptions are essential.

5. Aqueous solution: water is the solvent is included in Module 1.

Model-answer points: True.

6. All conditions and units may be ignored when applying Thermodynamic system and surroundings.

Model-answer points: False. Conditions, units and assumptions are essential.

7. Molarity and dilution is included in Module 1.

Model-answer points: True.

8. All conditions and units may be ignored when applying Internal energy, enthalpy and entropy.

Model-answer points: False. Conditions, units and assumptions are essential.

9. Molar mass is expressed in g mol^{-1} is included in Module 1.

Model-answer points: True.

10. All conditions and units may be ignored when applying Solutions and standard solutions.

Model-answer points: False. Conditions, units and assumptions are essential.

10 viva questions

1. What observation or measurement would confirm Atoms and molecules in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

2. What observation or measurement would confirm Solutions and standard solutions in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

3. What observation or measurement would confirm Molarity and dilution in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

4. What observation or measurement would confirm Normality and equivalent mass in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5. What observation or measurement would confirm Parts per million in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

6. What observation or measurement would confirm Thermodynamic system and surroundings in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

7. What observation or measurement would confirm Thermodynamic processes in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

8. What observation or measurement would confirm Internal energy, enthalpy and entropy in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

9. What observation or measurement would confirm Atoms and molecules in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

10. What observation or measurement would confirm Solutions and standard solutions in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5 numerical/application problems with solution cues

M1 CO1 7 marks **1. Calculate p, e and n for ^{35}Cl , Z=17.**

Model-answer points: p=17, e=17, n=18.

M1 CO1 7 marks **2. Find molarity of 4.9 g H_2SO_4 (98 g/mol) in 500 mL.**

Model-answer points: $M = (4.9/98)/0.5 = 0.100 \text{ M}$.

M1 CO1 7 marks **3. Mass needed for 250 mL 0.2 M NaOH (40 g/mol).**

Model-answer points: $m = .2 \times 40 \times .25 = 2.00 \text{ g}$.

M1 CO1 7 marks **4. Find normality of 5.3 g Na_2CO_3 (Eq 53) in 500 mL.**

Model-answer points: $N = (5.3/53)/.5 = 0.200 \text{ N}$.

M1 CO1 7 marks **5. Find ppm of 8 mg in 4 kg solution.**

Model-answer points: $8 \text{ mg}/4 \text{ kg} = 2 \text{ ppm}$.

5 diagram-based questions

M1 CO1 3 marks **1. Draw and label a diagram for Atoms and molecules; show direction of relevant flow or transfer.**

Model-answer points: Include all principal components, correct labels, arrows and a caption.

M1 CO1 3 marks **2. Draw and label a diagram for Solutions and standard solutions; show direction of relevant flow or transfer.**

Model-answer points: Include all principal components, correct labels, arrows and a caption.

3. Draw and label a diagram for Molarity and dilution; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

4. Draw and label a diagram for Normality and equivalent mass; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

5. Draw and label a diagram for Parts per million; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

5 engineering-application questions

1. Apply Molarity and dilution to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

2. Apply Normality and equivalent mass to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

3. Apply Parts per million to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

4. Apply Thermodynamic system and surroundings to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

5. Apply Thermodynamic processes to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

Module II — CO2

20 one-mark questions

M2

CO2

1 mark

1. Define or identify: Volumetric analysis and titration.

Model-answer points: A correct one-sentence definition or identification with essential condition.

M2

CO2

1 mark

2. Define or identify: End point: observed indicator colour change.

Model-answer points: A correct one-sentence definition or identification with essential condition.

M2

CO2

1 mark

3. Define or identify: Equivalence point: stoichiometric completion.

Model-answer points: A correct one-sentence definition or identification with essential condition.

M2

CO2

1 mark

4. Define or identify: Choice of acid-base indicator.

Model-answer points: A correct one-sentence definition or identification with essential condition.

M2

CO2

1 mark

5. Define or identify: Methyl orange: red in acid, yellow in alkali; transition about pH 3.1-4.4.

Model-answer points: A correct one-sentence definition or identification with essential condition.

M2

CO2

1 mark

6. Define or identify: Phenolphthalein: colourless in acid, pale pink in alkali; transition about pH 8.2-10.0.

Model-answer points: A correct one-sentence definition or identification with essential condition.

M2

CO2

1 mark

7. Define or identify: Principle $V_1N_1 = V_2N_2$.

Model-answer points: A correct one-sentence definition or identification with essential condition.

8. Define or identify: Define the identity of solution 1 and solution 2.

Model-answer points: A correct one-sentence definition or identification with essential condition.

9. Define or identify: Keep both volumes in the same unit.

Model-answer points: A correct one-sentence definition or identification with essential condition.

10. Define or identify: Water hardness.

Model-answer points: A correct one-sentence definition or identification with essential condition.

11. Define or identify: Hard water consumes soap and forms scum.

Model-answer points: A correct one-sentence definition or identification with essential condition.

12. Define or identify: Heating produces scale in boilers and heat exchangers.

Model-answer points: A correct one-sentence definition or identification with essential condition.

13. Define or identify: Industrial effluent treatment.

Model-answer points: A correct one-sentence definition or identification with essential condition.

14. Define or identify: Physical: screening, equalisation, sedimentation and filtration.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15. Define or identify: Chemical: neutralisation, precipitation, coagulation and oxidation.

Model-answer points: A correct one-sentence definition or identification with essential condition.

16. Define or identify: Potable water and municipal treatment.

Model-answer points: A correct one-sentence definition or identification with essential condition.

17. Define or identify: Screening removes large debris.

Model-answer points: A correct one-sentence definition or identification with essential condition.

18. Define or identify: Coagulation destabilises fine colloids.

Model-answer points: A correct one-sentence definition or identification with essential condition.

19. Define or identify: Reverse osmosis and nanofiltration.

Model-answer points: A correct one-sentence definition or identification with essential condition.

20. Define or identify: Feed, permeate and reject streams must be identified.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15 three-mark questions**1. Explain Volumetric analysis and titration using three technically distinct points and one example.**

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

2. Explain Choice of acid-base indicator using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

3. Explain Principle $V_1N_1 = V_2N_2$ using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

4. Explain Water hardness using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

5. Explain Industrial effluent treatment using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

6. Explain Potable water and municipal treatment using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

7. Explain Reverse osmosis and nanofiltration using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

8. Explain Arrhenius acids, bases and neutralisation using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

9. Explain pH, pOH and applications using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

10. Explain Volumetric analysis and titration using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

11. Explain Choice of acid-base indicator using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

12. Explain Principle $V_1N_1 = V_2N_2$ using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

13. Explain Water hardness using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

14. Explain Industrial effluent treatment using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

15. Explain Potable water and municipal treatment using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

10 seven-mark questions

1. Develop a structured engineering note on Volumetric analysis and titration with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

2. Develop a structured engineering note on Choice of acid-base indicator with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

3. Develop a structured engineering note on Principle $V_1N_1 = V_2N_2$ with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

4. Develop a structured engineering note on Water hardness with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

5. Develop a structured engineering note on Industrial effluent treatment with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

6. Develop a structured engineering note on Potable water and municipal treatment with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

7. Develop a structured engineering note on Reverse osmosis and nanofiltration with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

8. Develop a structured engineering note on Arrhenius acids, bases and neutralisation with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

9. Develop a structured engineering note on pH, pOH and applications with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

10. Develop a structured engineering note on Volumetric analysis and titration with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

M2

CO2

1 mark

1. Which option is most directly associated with Volumetric analysis and titration?

A. Volumetric analysis and titration

B. Choice of acid-base indicator

C. Principle $V_1N_1 = V_2N_2$

D. Water hardness

Model-answer points: Answer: A. Volumetric analysis and titration.

M2

CO2

1 mark

2. Which option is most directly associated with End point: observed indicator colour change?

A. End point: observed indicator colour change

B. Methyl orange: red in acid, yellow in alkali; transition about pH 3.1-4.4

C. Define the identity of solution 1 and solution 2

D. Hard water consumes soap and forms scum

Model-answer points: Answer: A. End point: observed indicator colour change.

M2

CO2

1 mark

3. Which option is most directly associated with Equivalence point: stoichiometric completion?

A. Equivalence point: stoichiometric completion

B. Phenolphthalein: colourless in acid, pale pink in alkali; transition about pH 8.2-10.0

C. Keep both volumes in the same unit

D. Heating produces scale in boilers and heat exchangers

Model-answer points: Answer: A. Equivalence point: stoichiometric completion.

M2

CO2

1 mark

4. Which option is most directly associated with Choice of acid-base indicator?

A. Choice of acid-base indicator

B. Principle $V_1N_1 = V_2N_2$

C. Water hardness

D. Industrial effluent treatment

Model-answer points: Answer: A. Choice of acid-base indicator.

M2

CO2

1 mark

5. Which option is most directly associated with Methyl orange: red in acid, yellow in alkali; transition about pH 3.1-4.4?

A. Methyl orange: red in acid, yellow in alkali; transition about pH 3.1-4.4

B. Define the identity of solution 1 and solution 2

C. Hard water consumes soap and forms scum

D. Physical: screening, equalisation, sedimentation and filtration

Model-answer points: Answer: A. Methyl orange: red in acid, yellow in alkali; transition about pH 3.1-4.4.

M2 Course CO2A

1 mark

6. Which option is most directly associated with Phenolphthalein: colourless in acid, pale pink in alkali; transition about pH 8.2–10.0?

- A. Phenolphthalein: colourless in acid, pale pink in alkali; transition about pH 8.2–10.0
- B. Keep both volumes in the same unit
- C. Heating produces scale in boilers and heat exchangers
- D. Chemical: neutralisation, precipitation, coagulation and oxidation

Model-answer points: Answer: A. Phenolphthalein: colourless in acid, pale pink in alkali; transition about pH 8.2–10.0.

M2

CO2

1 mark

7. Which option is most directly associated with Principle $V_1N_1 = V_2N_2$?

- A. Principle $V_1N_1 = V_2N_2$
- B. Water hardness
- C. Industrial effluent treatment
- D. Potable water and municipal treatment

Model-answer points: Answer: A. Principle $V_1N_1 = V_2N_2$.

M2

CO2

1 mark

8. Which option is most directly associated with Define the identity of solution 1 and solution 2?

- A. Define the identity of solution 1 and solution 2
- B. Hard water consumes soap and forms scum
- C. Physical: screening, equalisation, sedimentation and filtration
- D. Screening removes large debris

Model-answer points: Answer: A. Define the identity of solution 1 and solution 2.

M2

CO2

1 mark

9. Which option is most directly associated with Keep both volumes in the same unit?

- A. Keep both volumes in the same unit
- B. Heating produces scale in boilers and heat exchangers
- C. Chemical: neutralisation, precipitation, coagulation and oxidation
- D. Coagulation destabilises fine colloids

Model-answer points: Answer: A. Keep both volumes in the same unit.

M2

CO2

1 mark

10. Which option is most directly associated with Water hardness?

- A. Water hardness
- B. Industrial effluent treatment
- C. Potable water and municipal treatment
- D. Reverse osmosis and nanofiltration

Model-answer points: Answer: A. Water hardness.

11. Which option is most directly associated with Hard water consumes soap and forms scum?

- A. Hard water consumes soap and forms scum**
- B. Physical: screening, equalisation, sedimentation and filtration**
- C. Screening removes large debris**
- D. Feed, permeate and reject streams must be identified**

Model-answer points: Answer: A. Hard water consumes soap and forms scum.

12. Which option is most directly associated with Heating produces scale in boilers and heat exchangers?

- A. Heating produces scale in boilers and heat exchangers**
- B. Chemical: neutralisation, precipitation, coagulation and oxidation**
- C. Coagulation destabilises fine colloids**
- D. Pretreatment reduces fouling and scaling**

Model-answer points: Answer: A. Heating produces scale in boilers and heat exchangers.

13. Which option is most directly associated with Industrial effluent treatment?

- A. Industrial effluent treatment**
- B. Potable water and municipal treatment**
- C. Reverse osmosis and nanofiltration**
- D. Arrhenius acids, bases and neutralisation**

Model-answer points: Answer: A. Industrial effluent treatment.

14. Which option is most directly associated with Physical: screening, equalisation, sedimentation and filtration?

- A. Physical: screening, equalisation, sedimentation and filtration**
- B. Screening removes large debris**
- C. Feed, permeate and reject streams must be identified**
- D. HCl and HNO₃ are strong acids**

Model-answer points: Answer: A. Physical: screening, equalisation, sedimentation and filtration.

15. Which option is most directly associated with Chemical: neutralisation, precipitation, coagulation and oxidation?

- A. Chemical: neutralisation, precipitation, coagulation and oxidation**
- B. Coagulation destabilises fine colloids**
- C. Pretreatment reduces fouling and scaling**
- D. CH₃COOH is a weak acid**

Model-answer points: Answer: A. Chemical: neutralisation, precipitation, coagulation and oxidation.

1. The handbook topic that includes “Volumetric analysis and titration” is _____.

Model-answer points: Volumetric analysis and titration

2. The handbook topic that includes “End point: observed indicator colour change” is _____.

Model-answer points: Choice of acid-base indicator

3. The handbook topic that includes “Equivalence point: stoichiometric completion” is _____.

Model-answer points: Principle $V_1N_1 = V_2N_2$

4. The handbook topic that includes “Choice of acid-base indicator” is _____.

Model-answer points: Water hardness

5. The handbook topic that includes “Methyl orange: red in acid, yellow in alkali; transition about pH 3.1-4.4” is _____.

Model-answer points: Industrial effluent treatment

6. The handbook topic that includes “Phenolphthalein: colourless in acid, pale pink in alkali; transition about pH 8.2-10.0” is _____.

Model-answer points: Potable water and municipal treatment

7. The handbook topic that includes “Principle $V_1N_1 = V_2N_2$ ” is _____.

Model-answer points: Reverse osmosis and nanofiltration

8. The handbook topic that includes “Define the identity of solution 1 and solution 2” is _____.

Model-answer points: Arrhenius acids, bases and neutralisation

M2 Course CO2A

1 mark

9. The handbook topic that includes “Keep both volumes in the same unit” is _____.

Model-answer points: pH, pOH and applications

M2

CO2

1 mark

10. The handbook topic that includes “Water hardness” is _____.

Model-answer points: Volumetric analysis and titration

10 true-or-false questions

M2

CO2

1 mark

1. Volumetric analysis and titration is included in Module 2.

Model-answer points: True.

M2

CO2

1 mark

2. All conditions and units may be ignored when applying Choice of acid-base indicator.

Model-answer points: False. Conditions, units and assumptions are essential.

M2

CO2

1 mark

3. Equivalence point: stoichiometric completion is included in Module 2.

Model-answer points: True.

M2

CO2

1 mark

4. All conditions and units may be ignored when applying Water hardness.

Model-answer points: False. Conditions, units and assumptions are essential.

M2

CO2

1 mark

5. Methyl orange: red in acid, yellow in alkali; transition about pH 3.1-4.4 is included in Module 2.

Model-answer points: True.

M2

CO2

1 mark

6. All conditions and units may be ignored when applying Potable water and municipal treatment.

Model-answer points: False. Conditions, units and assumptions are essential.

M2

CO2

1 mark

7. Principle $V_1N_1 = V_2N_2$ is included in Module 2.

Model-answer points: True.

M2 Course CO2A

1 mark

8. All conditions and units may be ignored when applying Arrhenius acids, bases and neutralisation.

Model-answer points: False. Conditions, units and assumptions are essential.

M2

CO2

1 mark

9. Keep both volumes in the same unit is included in Module 2.

Model-answer points: True.

M2

CO2

1 mark

10. All conditions and units may be ignored when applying Volumetric analysis and titration.

Model-answer points: False. Conditions, units and assumptions are essential.

10 viva questions

M2

CO2

1 mark

1. What observation or measurement would confirm Volumetric analysis and titration in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

M2

CO2

1 mark

2. What observation or measurement would confirm Choice of acid-base indicator in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

M2

CO2

1 mark

3. What observation or measurement would confirm Principle $V_1N_1 = V_2N_2$ in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

M2

CO2

1 mark

4. What observation or measurement would confirm Water hardness in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5. What observation or measurement would confirm Industrial effluent treatment in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

6. What observation or measurement would confirm Potable water and municipal treatment in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

7. What observation or measurement would confirm Reverse osmosis and nanofiltration in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

8. What observation or measurement would confirm Arrhenius acids, bases and neutralisation in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

9. What observation or measurement would confirm pH, pOH and applications in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

10. What observation or measurement would confirm Volumetric analysis and titration in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5 numerical/application problems with solution cues

1. 25 mL 0.1 N acid neutralises 20 mL base; find base normality.

Model-answer points: $N = 25 \times .1 / 20 = .125 \text{ N}$.

2. Find pH of 1×10^{-4} M H^+ .**Model-answer points:** pH=4.**3. Find pOH and pH for 1×10^{-3} M OH^- .****Model-answer points:** pOH=3, pH=11.**4. 10 mL 1 M stock is diluted to 250 mL; find concentration.****Model-answer points:** $C_2 = 1 \times 10 / 250 = .04$ M.**5. 10 mL 0.01 M EDTA treats 20 mL water; hardness as $CaCO_3$.****Model-answer points:** $10 \times .01 \times 100000 / 20 = 500$ mg/L.

5 diagram-based questions

1. Draw and label a diagram for Volumetric analysis and titration; show direction of relevant flow or transfer.**Model-answer points:** Include all principal components, correct labels, arrows and a caption.**2. Draw and label a diagram for Choice of acid-base indicator; show direction of relevant flow or transfer.****Model-answer points:** Include all principal components, correct labels, arrows and a caption.**3. Draw and label a diagram for Principle $V_1N_1 = V_2N_2$; show direction of relevant flow or transfer.****Model-answer points:** Include all principal components, correct labels, arrows and a caption.**4. Draw and label a diagram for Water hardness; show direction of relevant flow or transfer.****Model-answer points:** Include all principal components, correct labels, arrows and a caption.

5. Draw and label a diagram for Industrial effluent treatment; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

5 engineering-application questions

1. Apply Principle $V_1N_1 = V_2N_2$ to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

2. Apply Water hardness to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

3. Apply Industrial effluent treatment to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

4. Apply Potable water and municipal treatment to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

5. Apply Reverse osmosis and nanofiltration to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

Module III — CO3

20 one-mark questions

1. Define or identify: Oxidation, reduction and redox agents.

Model-answer points: A correct one-sentence definition or identification with essential condition.

2. Define or identify: OIL RIG: Oxidation Is Loss, Reduction Is Gain.

Model-answer points: A correct one-sentence definition or identification with essential condition.

3. Define or identify: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is oxidation.

Model-answer points: A correct one-sentence definition or identification with essential condition.

4. Define or identify: Metallic and electrolytic conduction.

Model-answer points: A correct one-sentence definition or identification with essential condition.

5. Define or identify: Metallic conduction does not require chemical decomposition.

Model-answer points: A correct one-sentence definition or identification with essential condition.

6. Define or identify: Electrolytic conduction usually accompanies electrode reactions.

Model-answer points: A correct one-sentence definition or identification with essential condition.

7. Define or identify: Strong, weak and non-electrolytes.

Model-answer points: A correct one-sentence definition or identification with essential condition.

8. Define or identify: NaCl and HCl are strong electrolytes in water.

Model-answer points: A correct one-sentence definition or identification with essential condition.

9. Define or identify: Acetic acid is a weak electrolyte.

Model-answer points: A correct one-sentence definition or identification with essential condition.

10. Define or identify: Electrolysis and electrolytic cells.

Model-answer points: A correct one-sentence definition or identification with essential condition.

11. Define or identify: Cations move to the cathode.

Model-answer points: A correct one-sentence definition or identification with essential condition.

12. Define or identify: Anions move to the anode.

Model-answer points: A correct one-sentence definition or identification with essential condition.

13. Define or identify: Faraday's laws of electrolysis.

Model-answer points: A correct one-sentence definition or identification with essential condition.

14. Define or identify: $Q = It$.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15. Define or identify: Z is electrochemical equivalent in kg C^{-1} or g C^{-1} .

Model-answer points: A correct one-sentence definition or identification with essential condition.

16. Define or identify: Nickel electroplating and copper refining.

Model-answer points: A correct one-sentence definition or identification with essential condition.

17. Define or identify: Surface cleaning controls adhesion.

Model-answer points: A correct one-sentence definition or identification with essential condition.

18. Define or identify: Nickel anode: $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$.

Model-answer points: A correct one-sentence definition or identification with essential condition.

19. Define or identify: Daniell cell and salt bridge.

Model-answer points: A correct one-sentence definition or identification with essential condition.

20. Define or identify: Anode: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15 three-mark questions

Course 2003A

M3 CO3 3 marks 1. Explain Oxidation, reduction and redox agents using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

M3 CO3 3 marks 2. Explain Metallic and electrolytic conduction using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

M3 CO3 3 marks 3. Explain Strong, weak and non-electrolytes using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

M3 CO3 3 marks 4. Explain Electrolysis and electrolytic cells using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

M3 CO3 3 marks 5. Explain Faraday's laws of electrolysis using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

M3 CO3 3 marks 6. Explain Nickel electroplating and copper refining using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

M3 CO3 3 marks 7. Explain Daniell cell and salt bridge using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

8. Explain Primary, secondary and fuel cells using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

9. Explain Corrosion and factors affecting rate using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

10. Explain Barrier and cathodic protection using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

11. Explain Lithium-ion, sodium-ion and solid-state batteries using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

12. Explain Oxidation, reduction and redox agents using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

13. Explain Metallic and electrolytic conduction using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

14. Explain Strong, weak and non-electrolytes using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

15. Explain Electrolysis and electrolytic cells using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

10 seven-mark questions

1. Develop a structured engineering note on Oxidation, reduction and redox agents with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

2. Develop a structured engineering note on Metallic and electrolytic conduction with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

3. Develop a structured engineering note on Strong, weak and non-electrolytes with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

4. Develop a structured engineering note on Electrolysis and electrolytic cells with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

5. Develop a structured engineering note on Faraday's laws of electrolysis with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

6. Develop a structured engineering note on Nickel electroplating and copper refining with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

7. Develop a structured engineering note on Daniell cell and salt bridge with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

8. Develop a structured engineering note on Primary, secondary and fuel cells with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

9. Develop a structured engineering note on Corrosion and factors affecting rate with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

10. Develop a structured engineering note on Barrier and cathodic protection with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

15 multiple-choice questions

1. Which option is most directly associated with Oxidation, reduction and redox agents?

- A. Oxidation, reduction and redox agents**
- B. Metallic and electrolytic conduction**
- C. Strong, weak and non-electrolytes**
- D. Electrolysis and electrolytic cells**

Model-answer points: Answer: A. Oxidation, reduction and redox agents.

2. Which option is most directly associated with OIL RIG: Oxidation Is Loss,

Reduction Is Gain?

- A. OIL RIG: Oxidation Is Loss, Reduction Is Gain
- B. Metallic conduction does not require chemical decomposition
- C. NaCl and HCl are strong electrolytes in water
- D. Cations move to the cathode

Model-answer points: Answer: A. OIL RIG: Oxidation Is Loss, Reduction Is Gain.

3. Which option is most directly associated with $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is oxidation?

- A. $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is oxidation
- B. Electrolytic conduction usually accompanies electrode reactions
- C. Acetic acid is a weak electrolyte
- D. Anions move to the anode

Model-answer points: Answer: A. $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is oxidation.

4. Which option is most directly associated with Metallic and electrolytic conduction?

- A. Metallic and electrolytic conduction
- B. Strong, weak and non-electrolytes
- C. Electrolysis and electrolytic cells
- D. Faraday's laws of electrolysis

Model-answer points: Answer: A. Metallic and electrolytic conduction.

5. Which option is most directly associated with Metallic conduction does not require chemical decomposition?

- A. Metallic conduction does not require chemical decomposition
- B. NaCl and HCl are strong electrolytes in water
- C. Cations move to the cathode
- D. $Q = It$

Model-answer points: Answer: A. Metallic conduction does not require chemical decomposition.

6. Which option is most directly associated with Electrolytic conduction usually accompanies electrode reactions?

- A. Electrolytic conduction usually accompanies electrode reactions
- B. Acetic acid is a weak electrolyte
- C. Anions move to the anode
- D. Z is electrochemical equivalent in kg C^{-1} or g C^{-1}

Model-answer points: Answer: A. Electrolytic conduction usually accompanies electrode reactions.

7. Which option is most directly associated with Strong, weak and non-electrolytes?

- A. Strong, weak and non-electrolytes
- B. Electrolysis and electrolytic cells
- C. Faraday's laws of electrolysis
- D. Nickel electroplating and copper refining

Model-answer points: Answer: A. Strong, weak and non-electrolytes.

8. Which option is most directly associated with NaCl and HCl are strong electrolytes in water?

- A. NaCl and HCl are strong electrolytes in water
- B. Cations move to the cathode
- C. $Q = It$
- D. Surface cleaning controls adhesion

Model-answer points: Answer: A. NaCl and HCl are strong electrolytes in water.

9. Which option is most directly associated with Acetic acid is a weak electrolyte?

- A. Acetic acid is a weak electrolyte
- B. Anions move to the anode
- C. Z is electrochemical equivalent in kg C^{-1} or g C^{-1}
- D. Nickel anode: $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$

Model-answer points: Answer: A. Acetic acid is a weak electrolyte.

10. Which option is most directly associated with Electrolysis and electrolytic cells?

- A. Electrolysis and electrolytic cells
- B. Faraday's laws of electrolysis
- C. Nickel electroplating and copper refining
- D. Daniell cell and salt bridge

Model-answer points: Answer: A. Electrolysis and electrolytic cells.

11. Which option is most directly associated with Cations move to the cathode?

- A. Cations move to the cathode
- B. $Q = It$
- C. Surface cleaning controls adhesion
- D. Anode: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$

Model-answer points: Answer: A. Cations move to the cathode.

M3 Course

CO3

1 mark

12. Which option is most directly associated with Anions move to the anode?

- A. Anions move to the anode
- B. Z is electrochemical equivalent in kg C^{-1} or g C^{-1}
- C. Nickel anode: $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^{-}$
- D. Cathode: $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$

Model-answer points: Answer: A. Anions move to the anode.

M3

CO3

1 mark

13. Which option is most directly associated with Faraday's laws of electrolysis?

- A. Faraday's laws of electrolysis
- B. Nickel electroplating and copper refining
- C. Daniell cell and salt bridge
- D. Primary, secondary and fuel cells

Model-answer points: Answer: A. Faraday's laws of electrolysis.

M3

CO3

1 mark

14. Which option is most directly associated with $Q = It$?

- A. $Q = It$
- B. Surface cleaning controls adhesion
- C. Anode: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^{-}$
- D. Primary example: alkaline dry cell

Model-answer points: Answer: A. $Q = It$.

M3

CO3

1 mark

15. Which option is most directly associated with Z is electrochemical equivalent in kg C^{-1} or g C^{-1} ?

- A. Z is electrochemical equivalent in kg C^{-1} or g C^{-1}
- B. Nickel anode: $\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^{-}$
- C. Cathode: $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$
- D. Secondary example: lead-acid or lithium-ion battery

Model-answer points: Answer: A. Z is electrochemical equivalent in kg C^{-1} or g C^{-1} .

10 fill-in-the-blank questions

M3

CO3

1 mark

1. The handbook topic that includes "Oxidation, reduction and redox agents" is _____.**Model-answer points:** Oxidation, reduction and redox agents

M3

CO3

1 mark

2. The handbook topic that includes "OIL RIG: Oxidation Is Loss, Reduction Is Gain" is _____.**Model-answer points:** Metallic and electrolytic conduction

M3 course CO3A

1 mark

3. The handbook topic that includes “ $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is oxidation” is _____.

Model-answer points: Strong, weak and non-electrolytes

M3

CO3

1 mark

4. The handbook topic that includes “Metallic and electrolytic conduction” is _____.

Model-answer points: Electrolysis and electrolytic cells

M3

CO3

1 mark

5. The handbook topic that includes “Metallic conduction does not require chemical decomposition” is _____.

Model-answer points: Faraday’s laws of electrolysis

M3

CO3

1 mark

6. The handbook topic that includes “Electrolytic conduction usually accompanies electrode reactions” is _____.

Model-answer points: Nickel electroplating and copper refining

M3

CO3

1 mark

7. The handbook topic that includes “Strong, weak and non-electrolytes” is _____.

Model-answer points: Daniell cell and salt bridge

M3

CO3

1 mark

8. The handbook topic that includes “NaCl and HCl are strong electrolytes in water” is _____.

Model-answer points: Primary, secondary and fuel cells

M3

CO3

1 mark

9. The handbook topic that includes “Acetic acid is a weak electrolyte” is _____.

Model-answer points: Corrosion and factors affecting rate

M3

CO3

1 mark

10. The handbook topic that includes “Electrolysis and electrolytic cells” is _____.

Model-answer points: Barrier and cathodic protection

10 true-or-false questions

M3 Course CO3

1 mark

1. Oxidation, reduction and redox agents is included in Module 3.

Model-answer points: True.

M3 Course CO3

1 mark

2. All conditions and units may be ignored when applying Metallic and electrolytic conduction.

Model-answer points: False. Conditions, units and assumptions are essential.

M3 Course CO3

1 mark

3. $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ is oxidation is included in Module 3.

Model-answer points: True.

M3 Course CO3

1 mark

4. All conditions and units may be ignored when applying Electrolysis and electrolytic cells.

Model-answer points: False. Conditions, units and assumptions are essential.

M3 Course CO3

1 mark

5. Metallic conduction does not require chemical decomposition is included in Module 3.

Model-answer points: True.

M3 Course CO3

1 mark

6. All conditions and units may be ignored when applying Nickel electroplating and copper refining.

Model-answer points: False. Conditions, units and assumptions are essential.

M3 Course CO3

1 mark

7. Strong, weak and non-electrolytes is included in Module 3.

Model-answer points: True.

M3 Course CO3

1 mark

8. All conditions and units may be ignored when applying Primary, secondary and fuel cells.

Model-answer points: False. Conditions, units and assumptions are essential.

M3 Course CO3

1 mark

9. Acetic acid is a weak electrolyte is included in Module 3.

Model-answer points: True.

10. All conditions and units may be ignored when applying Barrier and cathodic protection.

Model-answer points: False. Conditions, units and assumptions are essential.

10 viva questions

1. What observation or measurement would confirm Oxidation, reduction and redox agents in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

2. What observation or measurement would confirm Metallic and electrolytic conduction in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

3. What observation or measurement would confirm Strong, weak and non-electrolytes in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

4. What observation or measurement would confirm Electrolysis and electrolytic cells in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5. What observation or measurement would confirm Faraday's laws of electrolysis in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

6. What observation or measurement would confirm Nickel electroplating and copper refining in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

7. What observation or measurement would confirm Daniell cell and salt bridge in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

8. What observation or measurement would confirm Primary, secondary and fuel cells in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

9. What observation or measurement would confirm Corrosion and factors affecting rate in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

10. What observation or measurement would confirm Barrier and cathodic protection in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5 numerical/application problems with solution cues

1. Calculate charge for 3 A during 5 min.

Model-answer points: $Q = 3 \times 300 = 900 \text{ C}$.

2. $Z = .000329 \text{ g/C}$, $I = 1.5 \text{ A}$, $t = 600 \text{ s}$; mass?

Model-answer points: $m = .296 \text{ g}$.

3. Time to deposit .5 g at $Z = .000329 \text{ g/C}$ and 2 A.

Model-answer points: $t = .5 / (.000329 \times 2) = 760 \text{ s}$.

4. Equal charge deposits 1 g Cu; $E_{\text{Ag}}/E_{\text{Cu}} = 3.40$; Ag mass?

Model-answer points: 3.40 g.

Model-answer points: $.34 - (-.76) = 1.10 \text{ V}$.

5 diagram-based questions

1. Draw and label a diagram for Oxidation, reduction and redox agents; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

2. Draw and label a diagram for Metallic and electrolytic conduction; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

3. Draw and label a diagram for Strong, weak and non-electrolytes; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

4. Draw and label a diagram for Electrolysis and electrolytic cells; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

5. Draw and label a diagram for Faraday's laws of electrolysis; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

5 engineering-application questions

1. Apply Strong, weak and non-electrolytes to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

2. Apply Electrolysis and electrolytic cells to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

3. Apply Faraday's laws of electrolysis to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

4. Apply Nickel electroplating and copper refining to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

5. Apply Daniell cell and salt bridge to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

Module IV — CO4

20 one-mark questions

1. Define or identify: Alloys and purpose of alloying.

Model-answer points: A correct one-sentence definition or identification with essential condition.

2. Define or identify: Stainless steel: iron with chromium and often nickel; corrosion-resistant equipment.

Model-answer points: A correct one-sentence definition or identification with essential condition.

3. Define or identify: Brass: copper-zinc; terminals, valves and fittings.

Model-answer points: A correct one-sentence definition or identification with essential condition.

4. Define or identify: Monomers, polymers and polymerisation.

Model-answer points: A correct one-sentence definition or identification with essential condition.

5. Define or identify: Addition polymerisation commonly opens a carbon-carbon double bond.

Model-answer points: A correct one-sentence definition or identification with essential condition.

6. Define or identify: Condensation polymerisation eliminates a small molecule such as water.

Model-answer points: A correct one-sentence definition or identification with essential condition.

7. Define or identify: Thermoplastics and thermosets.

Model-answer points: A correct one-sentence definition or identification with essential condition.

8. Define or identify: Thermoplastics are generally weldable and recyclable by remelting.

Model-answer points: A correct one-sentence definition or identification with essential condition.

9. Define or identify: Thermosets retain shape and heat resistance but are harder to recycle.

Model-answer points: A correct one-sentence definition or identification with essential condition.

10. Define or identify: Common engineering polymers.

Model-answer points: A correct one-sentence definition or identification with essential condition.

11. Define or identify: PE: cable insulation, tanks and films.

Model-answer points: A correct one-sentence definition or identification with essential condition.

12. Define or identify: PVC: conduit, cable sheath and pipes; plasticiser controls flexibility.

Model-answer points: A correct one-sentence definition or identification with essential condition.

13. Define or identify: Natural rubber, vulcanisation and synthetic rubber.

Model-answer points: A correct one-sentence definition or identification with essential condition.

14. Define or identify: Cross-links prevent permanent chain slippage.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15. Define or identify: Buna-S uses butadiene and styrene.

Model-answer points: A correct one-sentence definition or identification with essential condition.

16. Define or identify: Nanomaterials, carbon nanotubes, fullerenes and graphene.

Model-answer points: A correct one-sentence definition or identification with essential condition.

17. Define or identify: Carbon nanotube: rolled graphene-like cylinder.

Model-answer points: A correct one-sentence definition or identification with essential condition.

18. Define or identify: Fullerene: closed carbon cage such as C₆₀.

Model-answer points: A correct one-sentence definition or identification with essential condition.

19. Define or identify: Metal-organic frameworks.

Model-answer points: A correct one-sentence definition or identification with essential condition.

20. Define or identify: Metal nodes provide coordination sites.

Model-answer points: A correct one-sentence definition or identification with essential condition.

15 three-mark questions

1. Explain Alloys and purpose of alloying using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

2. Explain Monomers, polymers and polymerisation using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

3. Explain Thermoplastics and thermosets using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

4. Explain Common engineering polymers using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

5. Explain Natural rubber, vulcanisation and synthetic rubber using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

6. Explain Nanomaterials, carbon nanotubes, fullerenes and graphene using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

7. Explain Metal-organic frameworks using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

8. Explain Biomaterials and biofuels using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

9. Explain Green chemistry using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

10. Explain Alloys and purpose of alloying using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

11. Explain Monomers, polymers and polymerisation using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

12. Explain Thermoplastics and thermosets using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

13. Explain Common engineering polymers using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

14. Explain Natural rubber, vulcanisation and synthetic rubber using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

15. Explain Nanomaterials, carbon nanotubes, fullerenes and graphene using three technically distinct points and one example.

Model-answer points: Award one mark per correct point; accept a labelled example/condition as one point.

1. Develop a structured engineering note on Alloys and purpose of alloying with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

2. Develop a structured engineering note on Monomers, polymers and polymerisation with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

3. Develop a structured engineering note on Thermoplastics and thermosets with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

4. Develop a structured engineering note on Common engineering polymers with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

5. Develop a structured engineering note on Natural rubber, vulcanisation and synthetic rubber with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

6. Develop a structured engineering note on Nanomaterials, carbon nanotubes, fullerenes and graphene with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

7. Develop a structured engineering note on Metal-organic frameworks with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

8. Develop a structured engineering note on Biomaterials and biofuels with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

9. Develop a structured engineering note on Green chemistry with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

10. Develop a structured engineering note on Alloys and purpose of alloying with definition, principle, labelled diagram, application, limitation and safety/error point.

Model-answer points: Suggested distribution: definition 1, principle 2, diagram 1, application 1, limitation 1, safety/error 1.

15 multiple-choice questions

1. Which option is most directly associated with Alloys and purpose of alloying?

- A. Alloys and purpose of alloying
- B. Monomers, polymers and polymerisation
- C. Thermoplastics and thermosets
- D. Common engineering polymers

Model-answer points: Answer: A. Alloys and purpose of alloying.

2. Which option is most directly associated with Stainless steel: iron with chromium and often nickel; corrosion-resistant equipment?

- A. Stainless steel: iron with chromium and often nickel; corrosion-resistant equipment
- B. Addition polymerisation commonly opens a carbon-carbon double bond
- C. Thermoplastics are generally weldable and recyclable by remelting
- D. PE: cable insulation, tanks and films

Model-answer points: Answer: A. Stainless steel: iron with chromium and often nickel; corrosion-resistant equipment.

3. Which option is most directly associated with Brass: copper-zinc; terminals, valves and fittings?

- A. Brass: copper-zinc; terminals, valves and fittings**
- B. Condensation polymerisation eliminates a small molecule such as water**
- C. Thermosets retain shape and heat resistance but are harder to recycle**
- D. PVC: conduit, cable sheath and pipes; plasticiser controls flexibility**

Model-answer points: Answer: A. Brass: copper-zinc; terminals, valves and fittings.

4. Which option is most directly associated with Monomers, polymers and polymerisation?

- A. Monomers, polymers and polymerisation**
- B. Thermoplastics and thermosets**
- C. Common engineering polymers**
- D. Natural rubber, vulcanisation and synthetic rubber**

Model-answer points: Answer: A. Monomers, polymers and polymerisation.

5. Which option is most directly associated with Addition polymerisation commonly opens a carbon-carbon double bond?

- A. Addition polymerisation commonly opens a carbon-carbon double bond**
- B. Thermoplastics are generally weldable and recyclable by remelting**
- C. PE: cable insulation, tanks and films**
- D. Cross-links prevent permanent chain slippage**

Model-answer points: Answer: A. Addition polymerisation commonly opens a carbon-carbon double bond.

6. Which option is most directly associated with Condensation polymerisation eliminates a small molecule such as water?

- A. Condensation polymerisation eliminates a small molecule such as water**
- B. Thermosets retain shape and heat resistance but are harder to recycle**
- C. PVC: conduit, cable sheath and pipes; plasticiser controls flexibility**
- D. Buna-S uses butadiene and styrene**

Model-answer points: Answer: A. Condensation polymerisation eliminates a small molecule such as water.

7. Which option is most directly associated with Thermoplastics and thermosets?

- A. Thermoplastics and thermosets**
- B. Common engineering polymers**
- C. Natural rubber, vulcanisation and synthetic rubber**
- D. Nanomaterials, carbon nanotubes, fullerenes and graphene**

Model-answer points: Answer: A. Thermoplastics and thermosets.

8. Which option is most directly associated with Thermoplastics are generally weldable and recyclable by remelting?

- A. Thermoplastics are generally weldable and recyclable by remelting**
- B. PE: cable insulation, tanks and films**
- C. Cross-links prevent permanent chain slippage**
- D. Carbon nanotube: rolled graphene-like cylinder**

Model-answer points: Answer: A. Thermoplastics are generally weldable and recyclable by remelting.

9. Which option is most directly associated with Thermosets retain shape and heat resistance but are harder to recycle?

- A. Thermosets retain shape and heat resistance but are harder to recycle**
- B. PVC: conduit, cable sheath and pipes; plasticiser controls flexibility**
- C. Buna-S uses butadiene and styrene**
- D. Fullerene: closed carbon cage such as C₆₀**

Model-answer points: Answer: A. Thermosets retain shape and heat resistance but are harder to recycle.

10. Which option is most directly associated with Common engineering polymers?

- A. Common engineering polymers**
- B. Natural rubber, vulcanisation and synthetic rubber**
- C. Nanomaterials, carbon nanotubes, fullerenes and graphene**
- D. Metal-organic frameworks**

Model-answer points: Answer: A. Common engineering polymers.

11. Which option is most directly associated with PE: cable insulation, tanks and films?

- A. PE: cable insulation, tanks and films**
- B. Cross-links prevent permanent chain slippage**
- C. Carbon nanotube: rolled graphene-like cylinder**
- D. Metal nodes provide coordination sites**

Model-answer points: Answer: A. PE: cable insulation, tanks and films.

12. Which option is most directly associated with PVC: conduit, cable sheath and pipes; plasticiser controls flexibility?

- A. PVC: conduit, cable sheath and pipes; plasticiser controls flexibility**
- B. Buna-S uses butadiene and styrene**
- C. Fullerene: closed carbon cage such as C₆₀**
- D. Organic linkers define spacing and functionality**

Model-answer points: Answer: A. PVC: conduit, cable sheath and pipes; plasticiser controls flexibility.

13. Which option is most directly associated with Natural rubber, vulcanisation and synthetic rubber?

- A. Natural rubber, vulcanisation and synthetic rubber
- B. Nanomaterials, carbon nanotubes, fullerenes and graphene
- C. Metal-organic frameworks
- D. Biomaterials and biofuels

Model-answer points: Answer: A. Natural rubber, vulcanisation and synthetic rubber.

14. Which option is most directly associated with Cross-links prevent permanent chain slippage?

- A. Cross-links prevent permanent chain slippage
- B. Carbon nanotube: rolled graphene-like cylinder
- C. Metal nodes provide coordination sites
- D. Metallic implants: strength and toughness

Model-answer points: Answer: A. Cross-links prevent permanent chain slippage.

15. Which option is most directly associated with Buna-S uses butadiene and styrene?

- A. Buna-S uses butadiene and styrene
- B. Fullerene: closed carbon cage such as C₆₀
- C. Organic linkers define spacing and functionality
- D. Ceramics: hardness, wear resistance and bioactivity

Model-answer points: Answer: A. Buna-S uses butadiene and styrene.

10 fill-in-the-blank questions

1. The handbook topic that includes “Alloys and purpose of alloying” is _____.

Model-answer points: Alloys and purpose of alloying

2. The handbook topic that includes “Stainless steel: iron with chromium and often nickel; corrosion-resistant equipment” is _____.

Model-answer points: Monomers, polymers and polymerisation

3. The handbook topic that includes “Brass: copper-zinc; terminals, valves and fittings” is _____.

Model-answer points: Thermoplastics and thermosets

M4 Course CO4A

1 mark

4. The handbook topic that includes “Monomers, polymers and polymerisation” is _____.

Model-answer points: Common engineering polymers

M4 CO4

1 mark

5. The handbook topic that includes “Addition polymerisation commonly opens a carbon-carbon double bond” is _____.

Model-answer points: Natural rubber, vulcanisation and synthetic rubber

M4 CO4

1 mark

6. The handbook topic that includes “Condensation polymerisation eliminates a small molecule such as water” is _____.

Model-answer points: Nanomaterials, carbon nanotubes, fullerenes and graphene

M4 CO4

1 mark

7. The handbook topic that includes “Thermoplastics and thermosets” is _____.

Model-answer points: Metal-organic frameworks

M4 CO4

1 mark

8. The handbook topic that includes “Thermoplastics are generally weldable and recyclable by remelting” is _____.

Model-answer points: Biomaterials and biofuels

M4 CO4

1 mark

9. The handbook topic that includes “Thermosets retain shape and heat resistance but are harder to recycle” is _____.

Model-answer points: Green chemistry

M4 CO4

1 mark

10. The handbook topic that includes “Common engineering polymers” is _____.

Model-answer points: Alloys and purpose of alloying

10 true-or-false questions

M4 CO4

1 mark

1. Alloys and purpose of alloying is included in Module 4.

Model-answer points: True.

2. All conditions and units may be ignored when applying Monomers, polymers and polymerisation.

Model-answer points: False. Conditions, units and assumptions are essential.

3. Brass: copper-zinc; terminals, valves and fittings is included in Module 4.

Model-answer points: True.

4. All conditions and units may be ignored when applying Common engineering polymers.

Model-answer points: False. Conditions, units and assumptions are essential.

5. Addition polymerisation commonly opens a carbon-carbon double bond is included in Module 4.

Model-answer points: True.

6. All conditions and units may be ignored when applying Nanomaterials, carbon nanotubes, fullerenes and graphene.

Model-answer points: False. Conditions, units and assumptions are essential.

7. Thermoplastics and thermosets is included in Module 4.

Model-answer points: True.

8. All conditions and units may be ignored when applying Biomaterials and biofuels.

Model-answer points: False. Conditions, units and assumptions are essential.

9. Thermosets retain shape and heat resistance but are harder to recycle is included in Module 4.

Model-answer points: True.

10. All conditions and units may be ignored when applying Alloys and purpose of alloying.

Model-answer points: False. Conditions, units and assumptions are essential.

10 viva questions

1. What observation or measurement would confirm Alloys and purpose of alloying in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

2. What observation or measurement would confirm Monomers, polymers and polymerisation in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

3. What observation or measurement would confirm Thermoplastics and thermosets in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

4. What observation or measurement would confirm Common engineering polymers in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5. What observation or measurement would confirm Natural rubber, vulcanisation and synthetic rubber in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

6. What observation or measurement would confirm Nanomaterials, carbon nanotubes, fullerenes and graphene in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

7. What observation or measurement would confirm Metal-organic frameworks in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

8. What observation or measurement would confirm Biomaterials and biofuels in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

9. What observation or measurement would confirm Green chemistry in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

10. What observation or measurement would confirm Alloys and purpose of alloying in a laboratory or engineering system?

Model-answer points: State the expected measurable change, correct instrument/indicator and one source of error.

5 numerical/application problems with solution cues

1. A 2 kg polymer batch contains 5% additive; additive mass?

Model-answer points: $0.05 \times 2 = 0.10$ kg.

2. A brass sample is 60% Cu; Cu in 25 g?

Model-answer points: 15 g.

3. A coating loses 0.2 mm in 5 years; average rate?

Model-answer points: 0.04 mm/year.

4. Material is recycled at 80%; recovered from 500 kg?

Model-answer points: 400 kg.

Model-answer points: 65%.

5 diagram-based questions

1. Draw and label a diagram for Alloys and purpose of alloying; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

2. Draw and label a diagram for Monomers, polymers and polymerisation; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

3. Draw and label a diagram for Thermoplastics and thermosets; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

4. Draw and label a diagram for Common engineering polymers; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

5. Draw and label a diagram for Natural rubber, vulcanisation and synthetic rubber; show direction of relevant flow or transfer.

Model-answer points: Include all principal components, correct labels, arrows and a caption.

5 engineering-application questions

1. Apply Thermoplastics and thermosets to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

2. Apply Common engineering polymers to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

3. Apply Natural rubber, vulcanisation and synthetic rubber to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

4. Apply Nanomaterials, carbon nanotubes, fullerenes and graphene to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

5. Apply Metal-organic frameworks to an engineering maintenance, process-control, safety or material-selection situation.

Model-answer points: Define the problem, select the concept, state assumptions, apply it, and give a justified recommendation.

EXAMINATION PREPARATION

Practice papers, practical-test guidance and revision lists

PRACTICE MODEL PAPER — NOT AN OFFICIAL SITTTTR QUESTION PAPER.

Series Examination I practice

Time: 2 h · **Modules:** I-II · **Total:** 50

1. Four 1-mark definitions (4)
2. Six 3-mark questions, answer as instructed (18)
3. Four 7-mark questions with choice (28)

Series Examination II practice

Time: 2 h · **Modules:** III-IV · **Total:** 50

1. Four 1-mark questions (4)
2. Six 3-mark questions (18)

End-Semester practice model paper

Duration: 2.5 h · **Theory marks:** 60

Part A: two 1-mark questions from each module = 8 marks.

Part B: three 3-mark questions from each module; answer two from each = 24 marks.

Part C: one 7-mark set with choice from each module = 28 marks.

Part A

1. Define Atoms and molecules.
2. Define Solutions and standard solutions.
3. Define Volumetric analysis and titration.
4. Define Choice of acid-base indicator.
5. Define Oxidation, reduction and redox agents.
6. Define Metallic and electrolytic conduction.
7. Define Alloys and purpose of alloying.
8. Define Monomers, polymers and polymerisation.

Part B

Module 1: Explain Atoms and molecules; compare Solutions and standard solutions; solve/apply Molarity and dilution. Answer any two.

Module 2: Explain Volumetric analysis and titration; compare Choice of acid-base indicator; solve/apply Principle $V_1N_1 = V_2N_2$. Answer any two.

Module 3: Explain Oxidation, reduction and redox agents; compare Metallic and electrolytic conduction; solve/apply Strong, weak and non-electrolytes. Answer any two.

Module 4: Explain Alloys and purpose of alloying; compare Monomers, polymers and polymerisation; solve/apply Thermoplastics and thermosets. Answer any two.

Part C

Module 1: (a) Develop a complete answer on Thermodynamic processes OR (b) develop a complete answer on Internal energy, enthalpy and entropy.

Module 2: (a) Develop a complete answer on Arrhenius acids, bases and neutralisation OR (b) develop a complete answer on pH, pOH and applications.

Module 3: (a) Develop a complete answer on Barrier and cathodic protection OR (b) develop a complete answer on Lithium-ion, sodium-ion and solid-state batteries.

Module 4: (a) Develop a complete answer on Biomaterials and biofuels OR (b) develop a complete answer on Green chemistry.

Practical-test checklist

- Write-up/procedure: aim, diagram/flowchart, steps — 10

Course 2003A

- Setup and execution: correct setup, handling, accuracy — 10
- Observation and result: table, calculations/graphs, result — 8
- Viva voce: concepts and related theory — 8
- Record completion, neatness and certification — 4

Common mistakes

- Wrong hydrate or equivalent mass.
- mL not converted to L; minutes not converted to seconds.
- Burette funnel left in place; parallax in meniscus reading.
- Wrong electrode polarity or electron-flow arrow.
- Over-titration and non-concordant data accepted.
- pH meter not calibrated or electrode wiped harshly.
- Result stated without unit or basis.

Last-minute revision

Definitions: atom, molecule, standard solution, system, endpoint, hardness, electrolyte, electrolysis, corrosion, alloy, polymer, nanomaterial, biomaterial and green chemistry.

Equations: particle relations, molarity, normality, ppm, dilution, $V_1N_1=V_2N_2$, pH/pOH, $Q=It$ and Faraday laws.

Diagrams: thermodynamic systems, titration, water treatment, RO, electrolytic cell, Daniell cell, electroplating, corrosion protection, batteries, polymers, nanostructures and green-chemistry cycle.

SELF-LEARNING ACTIVITY CENTRE

All fifteen official weekly activity groups

Week 1 • Module 1

Official activity group: Atomic numbers and mass numbers of 20 elements; structures of five atoms; atoms and molecules in daily substances; simple atom model; record completion.

Objective: connect Module 1 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.
Course 2003A

Expected outcome: student can explain and apply the week's concepts and connect them to CO1.

Week 2 · Module 1

Official activity group: Concentration terms; molarity and normality problems; household-solution survey; solute and solvent identification; record completion.

Objective: connect Module 1 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO1.

Week 3 · Module 1

Official activity group: Thermodynamic-process report; daily-life examples; endothermic and exothermic observations; record completion.

Objective: connect Module 1 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO1.

Week 4 • Module 1

Official activity group: Spontaneous and non-spontaneous processes; standard-solution principle and procedure; record completion.

Objective: connect Module 1 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO1.

Week 5 • Module 2

Official activity group: Normality-equation problems; indicator-selection report; natural indicators; record completion.

Objective: connect Module 2 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO2.

Week 6 • Module 2

Official activity group: Water-sample collection; pH identification; soft and hard water; hard-water problems at home; record completion.

Objective: connect Module 2 concepts with observation, calculation, reporting and record completion.

Course 2003A

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO₂.

Week 7 • Module 2

Official activity group: Modern water filters; municipal-treatment stages and flow chart; muddy-water sedimentation and filtration; record completion.

Objective: connect Module 2 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO₂.

Week 8 • Module 2

Official activity group: Local water-purification methods; CO₁ and CO₂ assignment preparation; Lab Series Examination I preparation; record completion.

Objective: connect Module 2 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO2.

Week 9 · Module 3

Official activity group: Hydrogen economy and hydrogen fuel in automobiles; record completion.

Objective: connect Module 3 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO3.

Week 10 · Module 3

Official activity group: Oxidation-reduction reactions; Daniell-cell operation and diagram; record completion.

Objective: connect Module 3 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.
Course 2003A

Expected outcome: student can explain and apply the week's concepts and connect them to CO3.

Week 11 • Module 3

Official activity group: Battery recycling; record completion.

Objective: connect Module 3 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO3.

Week 12 • Module 3

Official activity group: EV battery types, advantages and disadvantages; record completion.

Objective: connect Module 3 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO3.

Week 13 • Module 4

Official activity group: Daily-life alloy articles; bioplastics and biodegradable polymers; reusable/recyclable polymers; nanomaterial presentation; record completion.

Objective: connect Module 4 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO4.

Week 14 · Module 4

Official activity group: Biomedical implant materials; biomaterial assignment; MOF applications; record completion.

Objective: connect Module 4 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week's concepts and connect them to CO4.

Week 15 · Module 4

Official activity group: Green-synthesis procedure; biofuel assignment; CO3 and CO4 preparation; Lab Series Examination II preparation; record completion.

Objective: connect Module 4 concepts with observation, calculation, reporting and record completion.

Resources: handbook, official syllabus, laboratory record, safe household observations where explicitly permitted, library or teacher-approved digital resources.

Safety: No chemical experiment at home unless the activity explicitly uses safe household materials and faculty has approved it. Never handle corrosive, toxic, metal-salt or resin chemicals outside the laboratory.

Instructions: define the task, collect verifiable information/data, use correct units and labels, analyse rather than copy, cite the source, and complete the record.

Observation/submission: dated table or labelled diagram plus 300–500 word report/presentation as applicable.

Evaluation checklist: accuracy, completeness, originality, safety, relevance, neatness and on-time submission.

Expected outcome: student can explain and apply the week’s concepts and connect them to CO4.

ENGINEERING APPLICATION CENTRE

Chemistry in real engineering work

Industrial-water treatment

Measure before selecting treatment. Separate suspended, dissolved, ionic, organic and microbial problems; then verify treatment by repeat analysis and compliant disposal.

Engineering practice

CO2

Effluent treatment

Measure before selecting treatment. Separate suspended, dissolved, ionic, organic and microbial problems; then verify treatment by repeat analysis and compliant disposal.

Engineering practice

CO2

Boiler-water scale and heat loss

Measure before selecting treatment. Separate suspended, dissolved, ionic, organic and microbial problems; then verify treatment by repeat analysis and compliant disposal.

Engineering practice

CO2

Corrosion of structural steel

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO3

Corrosion of escalator trusses

Inspect drainage, coating continuity, welds, crevices and contact between dissimilar metals. Specify surface preparation, compatible coating and maintenance interval; do not assume stainless appearance means immunity.

Engineering practice

CO3

Corrosion of brackets and fasteners

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO3

Anti-rust surface preparation

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO3

Electroplating of components

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO3

Battery selection and BMS limits

Compare chemistry, voltage, energy, current, cycle life, temperature range, charging method, fire behaviour, enclosure and recycling route. Use a BMS and manufacturer limits.

Engineering practice

CO3

EV battery chemistry

Compare chemistry, voltage, energy, current, cycle life, temperature range, charging method, fire behaviour, enclosure and recycling route. Use a BMS and manufacturer limits.

Engineering practice

CO3

Polymer insulation and cable sheaths

Select the exact grade using dielectric, mechanical, impact, UV, flame, chemical and service-temperature data. Verify ageing and disposal constraints.

PVC conduits

Select the exact grade using dielectric, mechanical, impact, UV, flame, chemical and service-temperature data. Verify ageing and disposal constraints.

Engineering practice

CO4

Nylon gears and guides

Select the exact grade using dielectric, mechanical, impact, UV, flame, chemical and service-temperature data. Verify ageing and disposal constraints.

Engineering practice

CO4

Polycarbonate enclosures

Select the exact grade using dielectric, mechanical, impact, UV, flame, chemical and service-temperature data. Verify ageing and disposal constraints.

Engineering practice

CO4

PMMA transparent covers

Select the exact grade using dielectric, mechanical, impact, UV, flame, chemical and service-temperature data. Verify ageing and disposal constraints.

Engineering practice

CO4

Brass terminals

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO4

Solder alloys

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO4

Stainless-steel equipment

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO4

Surface coatings

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO4

pH control

Measure before selecting treatment. Separate suspended, dissolved, ionic, organic and microbial problems; then verify treatment by repeat analysis and compliant disposal.

Engineering practice

CO2

Conductivity measurement

Measure before selecting treatment. Separate suspended, dissolved, ionic, organic and microbial problems; then verify treatment by repeat analysis and compliant disposal.

Engineering practice

CO3

Biomaterials in medical equipment

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO4

Sustainable-material selection

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO4

Green manufacturing

Translate chemistry into a specification: define environment, required property, hazard, inspection method, failure mode, maintenance need and end-of-life route.

Engineering practice

CO4

Textbooks, references and verification notes

Textbooks listed in the syllabus

1. Puri, Sharma & Pathania — Principles of Physical Chemistry.
2. P. L. Soni, O. P. Dharmarha and U. N. Dash — Textbook of Physical Chemistry.
3. A. I. Vogel — Vogel's Textbook of Quantitative Chemical Analysis.
4. Skoog, West and Holler — Fundamentals of Analytical Chemistry.
5. T. Pradeep — The Essentials: Understanding Nanoscience and Nanotechnology.
6. A. O. Thomas — Practical Chemistry.
7. Sudha Goel — Water and Waste Water Engineering.

Source discipline

Definitions, equations, experiments, hours, outcomes, activities and assessment structures are aligned to the supplied official syllabus. Expanded explanations, diagrams, examples and practice questions are original teaching material. Advanced battery, nanomaterial and thermodynamic details are explicitly treated as enrichment rather than mandatory syllabus derivation.

Official web resource listed: NPTEL course 104106119.